

The multiply iterated law of the iterated logarithm: game-theoretic foundations of sequential detection boundaries

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Abstract

Anytime-valid confidence sequences and e -processes are built almost universally from one recipe: average exponential test statistics over a prior on the tilting scale, then invoke Ville’s inequality on the resulting nonnegative supermartingale. The mixing prior sets the width of the detection boundary and is usually chosen by hand. We recast the recipe as a two-player game with information as currency. A Learner commits to the prior; Nature adaptively produces a mean-zero score process whose difficulty is priced by a cumulant-generating-function charge. The Learner’s mixture wealth is governed by a single pathwise Gibbs-variational identity,

$$\log Z_t = \int [\eta S_t - K_t(\eta)] \, d\pi_t(\eta) - \text{KL}(\pi_t \| \Pi), \quad \pi_t(d\eta) := \frac{e^{\eta S_t - K_t(\eta)}}{Z_t} \Pi(d\eta)$$

which holds along every realized path with no expectation operator; Ville’s inequality, the equalizer condition, the GROW characterization, and the saddlepoint formula are all specializations of it.

Three messages organize the rest. First, the law of the iterated logarithm (LIL) is the minimax boundary of this sequential-detection game, not arbitrary combinatorial slack. Second, the optimal prior is not a design choice but the forced *equalizer* strategy — the unique law that makes every boundary-crossing time equally costly for Nature — and it yields the sharp first iterated-log correction in closed form, with coefficient $3/2 = 1 + 1/2$ (one for the Erdős baseline, one half for the Laplace envelope around the saddle). Third, in the log-log scale chart the equalizer is exactly the Jeffreys prior on the scale-of-scales. The Erdős–Kolmogorov integral test is the criterion that selects it: the unique normalizability boundary at which the equalizer’s normalization integral is marginally divergent. The two-stage finite-time LIL proof, the Howard–Ramdas mixture and stitching constructions, and betting confidence sequences all read as instances of this equalizer principle. A companion empirical evaluation confirms the central identities and locates the Erdős threshold at the predicted value.

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1 Introduction

Sequential tests that remain valid at every stopping time — the anytime-valid confidence sequences and e -processes that drive modern nonparametric inference [13, 24, 33] — almost always rest on a single construction. Take an exponential test statistic at scale η , $M_t^\eta = \exp(\eta S_t - K_t(\eta))$ where K_t is the cumulant generating function of the score increments. Average it over a prior Π on η . The mixture $Z_t = \int M_t^\eta \Pi(d\eta)$ is a nonnegative supermartingale under any mean-zero alternative satisfying the CGF bound, and Ville’s inequality [27, 31] converts it into a time-uniform detection rule: $\mathbb{P}(\exists t : Z_t \geq 1/\alpha) \leq \alpha$.

The width of the resulting boundary is set by the prior. A heavy prior at small scales protects against late, slowly drifting alternatives but admits early false positives; a tight concentration at moderate scales tracks the central-limit regime closely but loses power on long horizons. The prior is usually treated as a design choice; the resulting boundary is read off and compared against classical reference lines.

The reference line for what is achievable in this regime is the law of the iterated logarithm. For a fair coin flipped N times the cumulative deviation $S_n = \xi_1 + \dots + \xi_n$ wanders past $\sqrt{n}$ infinitely often, but $\limsup_{n \rightarrow \infty} S_n / \sqrt{2n \log \log n} = 1$ almost surely [11, 16, 17]. The slow $\log \log n$ and the constant $\sqrt{2}$ together mark the exact boundary at which the supermartingale construction succeeds. Below this rate (e.g. $\sqrt{2t \log \log t}$ without correction) no choice of Π admits a normalizable equalizer; above it (e.g. adding $(3/2 + \varepsilon) \log \log \log t$ inside the radical) the prior is finite-mass and the detection rule closes the boundary.

This paper recasts the construction as a game. The Learner commits to a prior Π before any data arrive; Nature adaptively produces the score-increment distribution at each round, subject to a CGF budget. The prior is the Learner’s mixed strategy. The *equalizer* prior — the unique Π that makes every boundary-crossing time equally costly for Nature — is determined by a saddlepoint relation, and its normalizability is governed by the classical Erdős–Kolmogorov integral test. The iterated logarithm is not arbitrary combinatorial slack: it is the unique function that makes a normalization integral marginally divergent along the equalizer’s saddle.

The paper contributes along several complementary axes.

Exact minimax results. We prove that Balsubramani’s oblivious mixing law [1],

$$\Pi(d\lambda) = \frac{d\lambda}{|\lambda|(\log(1/|\lambda|))^2}, \quad 0 < |\lambda| \leq e^{-2}$$

is the unique minimax equalizer of a natural reduced scale-allocation game. The proof proceeds through four exact theorems: a finite-dimensional minimax coincidence-center theorem (Section 5), the continuum scale game and its value $V = 2$ (Section 6), the discrete shell version (Section 7), and a general equalizer theorem for arbitrary tax functions (Section 8).

Pathwise information-theoretic identity. The mixture wealth Z_t obeys an exact pathwise identity (Theorem 3.1) of Gibbs-variational form, which holds along the realized dynamics of the game without any expectation operator. Every subsequent identity — Ville’s inequality, the equalizer condition, the GROW characterization, the saddlepoint formula — is a specialization of this single identity to a particular event or constraint (Section 3). Carrying this identity into the saddle calculation is what allows the equalizer’s information-cost interpretation: the equalizer is the prior that makes the path’s information accumulation linear in time along the boundary.

Asymptotic equalizer analysis. We derive the equalizer density for a prescribed detection boundary $b(t) = \sqrt{2t h(t)}$ via Laplace’s method (Section 10), connect its normalizability to the Erdős integral test (Section 11), and show that the sharp first iterated-logarithm correction has coefficient $3/2$ (Section 12).

Four further structural results. Under the chart $\mu = \log \log(1/\lambda)$, the equalizer pushes forward to the rate-1 shifted exponential (Theorem 13.2; Section 13); for the LIL game with scale-family alternative, the GROW-optimal e -process equals the equalizer mixture (Theorem 14.1; Section 14); the finite-dimensional coincidence center converges to the continuum equalizer in 2-Wasserstein distance as the alphabet size $W \rightarrow \infty$ (Theorem 15.1; Section 15); and a finite- α shell-truncation tool produces hand-tractable anytime-valid confidence sequences (Proposition 16.1; Section 16).

Structural interpretation. The two-stage proof architecture of the finite-time LIL [1] is reinterpreted as two nested equalizer games (Section 18), and the adaptive betting strategies of modern confidence sequences are understood as sequential best-responses to the same game (Section 19). The LIL game itself is the exponentiated null-restricted shadow of a CGF-constrained repeated game over KL balls (Section 17).

Summary of contributions.

- C1. Exact game-theoretic derivation** of Balsubramani’s oblivious mixing law as the unique minimax equalizer of a reduced scale-allocation game (Theorems 5.1–8.1).
- C2. Pathwise Gibbs-variational identity** for the mixture wealth (Theorem 3.1), valid along realized paths without expectations, organizing every subsequent identity in the paper as a specialization of itself.
- C3. Unified statement** connecting the Erdős–Kolmogorov integral test to the normalizability of an asymptotic equalizer prior (Theorems 11.1–11.2).
- C4. Sharp $3/2$ theorem:** the first iterated-logarithm correction beyond $\sqrt{2t \log \log t}$ has the *exact* sharp coefficient $3/2$, and this constant emerges geometrically as the sum of the Laplace-envelope exponent and the Erdős baseline (Corollary 12.1 and Remark 12.2).
- C5. Closed-form description of the entire iterated-log hierarchy**, $(c_1^*, c_2^*, c_3^*, \dots) = (3/2, 1, 1, \dots)$ (Theorem 12.6), refuting a propagating-“ $+1/2$ ” conjecture; the threshold is universal across the Bernstein sub-exponential family (Corollary 12.4).
- C6. Jeffreys-iterated-log pushforward:** under $\mu = \log \log(1/\lambda)$, the Balsubramani equalizer is the rate-1 shifted exponential $2e^{-\mu}$ on $[\log 2, \infty)$ (Theorem 13.2).
- C7. GROW = equalizer:** for the LIL game with scale-family alternative, the GROW-optimal e -process equals the equalizer mixture (Theorem 14.1).
- C8. Wasserstein limit of the coincidence center:** the finite-dimensional minimax coincidence center converges in 2-Wasserstein distance to the continuum equalizer as $W \rightarrow \infty$ (Theorem 15.1).
- C9. Shell-truncation tool:** for any $\alpha \in (0, 1)$, truncating the shell mass at $j_{\max} = \lceil 2/\alpha \rceil$ produces an exact finite- α anytime-valid confidence sequence (Proposition 16.1).
- C10. Structural interpretation** of the two-stage finite-time LIL proof as two nested equalizer games (Section 18); of the Howard–Ramdas mixture and stitching constructions as oblivious instantiations of the same equalizer principle (Section 19); and of the Waudby-Smith–Ramdas betting confidence sequence as an adaptive sequential best-response that tracks the equalizer at each round (Section 19).
- C11. Explicit cross-link to a CGF-constrained repeated game:** the LIL scale-allocation game is the exponentiated, null-restricted shadow of a CGF-constrained repeated game over KL balls, and the same one-round identity drives both (Theorem 17.1; Section 17).

All logarithms are natural throughout. Throughout the paper, the symbols η and λ both denote the exponential tilting parameter (measurement scale); η is used in the sequential game and asymptotic analysis, while λ is used in the exact reduced game to match the notation of [1]. The correspondence is exact: $\lambda_t^* = \eta^*(t) = b(t)/t$ is the hindsight-optimal scale at time t , and the Balsubramani law $1/(|\lambda| \log^2(1/|\lambda|))$ on $[-e^{-2}, e^{-2}]$ is the image, under the identification $\lambda \leftrightarrow \eta$, of the normalized equalizer prior $\pi^*(\eta)$ of the sequential game. Below, $\log_k$ denotes the k -fold iterated logarithm and we write $f(t) \asymp g(t)$ for $f(t)/g(t)$ bounded above and below by positive constants as $t \rightarrow \infty$, $f(t) \sim g(t)$ for $f(t)/g(t) \rightarrow 1$, and $f(t) = o(g(t))$ for $f(t)/g(t) \rightarrow 0$. The cumulative intrinsic-time CGF is $K_t(\eta) = \sum_{s \leq t} \psi_s(\eta)$, and the single-scale wealth is $M_t^\eta = \exp(\eta S_t - K_t(\eta))$. The time- t posterior is $\pi_t(d\eta) = M_t^\eta \Pi(d\eta)/Z_t$.

Notation summary.

Symbol	Meaning
η, λ	Exponential tilting parameter / measurement scale (interchangeable)
$\eta^*(t), \lambda_t^*$	Saddlepoint scale at time t : $\eta^*(t) = b(t)/t$
Π, ν	Learner's prior over scales
π, f	Density of Π / density of ν
π^*, ν^*	Equalizer (minimax) prior
$F^*(\lambda)$	Equalizer CDF: $F^*(\lambda) = 2/\log(1/\lambda)$
S_t	Cumulative score (martingale)
$b(t)$	Detection boundary: $b(t) = \sqrt{2t h(t)}$
$h(t)$	Boundary-shape function: $h(t) \asymp \log \log t$ for LIL
$K_t(\eta)$	Cumulative CGF: $K_t(\eta) = \sum_{s \leq t} \psi_s(\eta)$
M_t^η	Single-scale wealth: $M_t^\eta = \exp(\eta S_t - K_t(\eta))$
π_t	Time- t posterior: $\pi_t(d\eta) = M_t^\eta \Pi(d\eta)/Z_t$
$Z_t, \tilde{Z}_t$	Oblivious / adaptive mixture wealth process
V	Game value (equals 2 for Balsubramani's interval $(0, e^{-2}]$)
$\log_k t$	k -fold iterated logarithm: $\log_1 t = \log t$, $\log_{k+1} t = \log \log_k t$

2 Related work

Anytime-valid testing and e -processes. The unified theory of time-uniform confidence sequences and e -processes [13]; the betting-based estimator framework [33]; and the game-theoretic-statistics survey [24] make the mixture-martingale mechanism explicit, positioning the oblivious-prior construction at the center of the wider safe anytime-valid inference program initiated in [27]. The GROW-optimal e -process framework [10] supplies the information-theoretic counterpart to the saddle-point view we use here: its growth-rate-optimal prior coincides with our equalizer prior under the scale-family alternative (Theorem 14.1). The unification of Robbins-style mixture martingales with adaptive “stitching” [13], and the self-normalized counterpart that adapts to the empirical variance [6], complete the modern picture. The contribution of the present paper to this line is to identify the specific mixing prior as the unique minimax equalizer of a reduced scale-allocation game, and to read the Erdős integral test as its normalizability criterion.

Recent betting CSes and safe testing. A companion survey [12] catalogues the mixture-and-stitching constructions whose equalizer reading we exploit. The Catoni-style robust extension of [32] carries this to confidence-sequence form for heavy-tailed data; the betting-CS view of that construction is the closest adaptive analogue of the present oblivious framework outside the sub-Gaussian regime. The reduction of sequential change-point detection to sequential estimation [29] and the systematic e -value combination framework [4] both exploit mixture martingales whose mixing priors are LIL-rate by design, the regime the equalizer characterizes. The safe-testing program [10] reframes classical hypothesis

testing in terms of e -values that compose with these mixture martingales; the likelihood-and-replicability tradition [23] places such constructions in their older statistical context.

Mixture martingales and sequential testing. The mixture-martingale construction of [15] revisits mixture martingales for best-arm identification and sequential tests; the implicit mixing prior there is a discrete-shell approximation of the continuum equalizer (cf. Theorem 7.1). An earlier line [2] uses the finite-time LIL directly as a sequential nonparametric test; the oblivious mixture there is the symmetric Balsubramani density of Corollary 6.4. Martingale methods for sequential estimation of functionals and divergences [21] admit the same equalizer interpretation when their log-optimal mixture is specialized to the LIL boundary.

Parameter-free online learning and universal portfolios. Coin-betting and parameter-free online learners [14, 22] achieve regret bounds at the LIL rate without hyperparameters; their potentials are instances of the adaptive wealth process, and the relevant rescaling of the Jeffreys prior is the same closest-to-Haar correction we identify in Section 19.5.

Small-sample evaluation in LLM benchmarks. Two recent applied lines test the LIL-rate regime in settings where the data is precisely small. [3] document that CLT-calibrated confidence intervals on small LLM-eval benchmarks dramatically underestimate uncertainty, recommending non-asymptotic frequentist or Bayesian alternatives; the LIL upper-confidence band derived here is exactly the non-asymptotic envelope they call for, and the equalizer characterization gives the sharp constant the LIL band charges. [20] build an applied SPRT-style adaptive stopping rule on chain-of-thought self-consistency that operates in the same small- n regime; their realized stopping time is the empirical face of the equalizer’s normalizability criterion, with the savings curves they report matching the LIL-rate bound at the relevant n . Both papers are concrete deployment evidence that the LIL boundary, rather than the CLT, is the operationally meaningful envelope for sequential LLM-evaluation.

Classical roots in random-walk LIL theory. The mixture-of-exponentials construction itself goes back to the boundary-crossing literature [5, 19, 25]; the Hartman–Wintner law [11] and the upper–lower-class test [8] established the upper-class characterization of LIL-rate boundaries, which an integral test [7] sharpened. The Komlós–Major–Tusnády strong approximation [18], together with its martingale extensions [26, 28], lifts these results from i.i.d. walks to martingales with controlled increments. The finite-time martingale LIL [1] is the proof we re-read game-theoretically.

The CGF-constrained game shadow. The LIL game studied here is the exponentiated null-restricted shadow of a CGF-constrained repeated game over KL balls. The Learner-side adaptive protocol of Section 19 draws on the intrinsic-time / Bayesian-update connection developed in that broader framework.

What is new here. Our equalizer characterization is incremental relative to these works in that the Balsubramani density has been widely used and, implicitly, understood to be minimax-like. What is new is

- (N1) the precise statement that this density is the *exact* minimax equalizer of a natural reduced game (Theorems 5.1–8.1), with explicit value $V = 2$;
- (N2) the identification of the Erdős integral test as the *normalizability* criterion of the asymptotic equalizer (Theorem 11.1);
- (N3) the pathwise Gibbs-variational identity for $\log Z_t$ (Theorem 3.1) that organizes every subsequent identity as a specialization of itself;
- (N4) the geometric decomposition $3/2 = 1 + 1/2$ of the sharp first-correction constant (Corollary 12.1, Remark 12.2);
- (N5) the closed-form description of the entire iterated-log hierarchy of Erdős thresholds, $(c_1^*, c_2^*, c_3^*, \dots) = (3/2, 1, 1, \dots)$ (Theorem 12.6; the Laplace “+1/2” contributes only at the first iterated-log layer, and is consumed by the leading $3/2 \log_3 t$ term via a saddlepoint substitution chain);

- (N6) the universality of the $3/2$ threshold across the Bernstein sub-exponential family (Corollary 12.4);
- (N7) the iterated-log-chart pushforward of the equalizer to the rate-1 shifted exponential (Theorem 13.2);
- (N8) the GROW = equalizer identity (Theorem 14.1);
- (N9) the W_2 convergence of the finite-dimensional coincidence center to the continuum equalizer (Theorem 15.1);
- (N10) the finite- α shell-truncation tool (Proposition 16.1); and
- (N11) the explicit cross-link to the CGF-constrained repeated game (Theorem 17.1; Section 17).

3 The pathwise information-theoretic identity

A single algebraic identity organizes every result that follows. It is exact, it holds along the realized dynamics of the game without any expectation operator, and every later identity — Ville’s inequality, the equalizer condition, the GROW characterization, the saddlepoint formula — is a specialization of it.

Theorem 3.1 (Pathwise information-theoretic identity for the mixture wealth). *Let Π be any probability measure on $(0, \infty)$ such that $Z_t > 0$ $\mathbb{P}$ -a.s. for all t . Define the time- t posterior*

$$\pi_t(d\eta) := \frac{e^{\eta S_t - K_t(\eta)}}{Z_t} \Pi(d\eta) = \frac{M_t^\eta}{Z_t} \Pi(d\eta) \quad (1)$$

Then, for every $\omega \in \Omega$ for which $Z_t(\omega) > 0$ and every $t \geq 0$, the identity

$$\log Z_t = \int_{(0, \infty)} [\eta S_t - K_t(\eta)] d\pi_t(\eta) - \text{KL}(\pi_t \| \Pi) \quad (2)$$

holds in $\mathbb{R} \cup \{-\infty\}$, with the convention that $\text{KL}(\pi_t \| \Pi) \in [0, \infty]$ and the integrand $\eta S_t - K_t(\eta)$ is π_t -integrable whenever $\text{KL}(\pi_t \| \Pi) < \infty$. Moreover, $\log Z_t$ is the supremum over all probability measures $\nu \ll \Pi$ of the right-hand side of (2) with ν in place of π_t , and the supremum is uniquely attained at $\nu = \pi_t$ (Gibbs variational principle, pathwise version).

Proof. By the definition (1), $\log \frac{d\pi_t}{d\Pi}(\eta) = \eta S_t - K_t(\eta) - \log Z_t$. Integrating against π_t gives

$$\begin{aligned} \text{KL}(\pi_t \| \Pi) &= \int \log \frac{d\pi_t}{d\Pi}(\eta) d\pi_t(\eta) \\ &= \int [\eta S_t - K_t(\eta)] d\pi_t(\eta) - \log Z_t, \end{aligned} \quad (3)$$

which is (2) after rearrangement. The Gibbs variational statement is the standard duality: for any $\nu \ll \Pi$ with $\text{KL}(\nu \| \Pi) < \infty$, write $J(\nu) := \int [\eta S_t - K_t(\eta)] d\nu(\eta) - \text{KL}(\nu \| \Pi)$. A direct calculation gives $J(\nu) = \log Z_t - \text{KL}(\nu \| \pi_t)$ (the free-energy identity), so $J(\nu) \leq \log Z_t$ with equality iff $\nu = \pi_t$. $\square$

Identity (2) holds *pathwise*: $S_t(\omega)$ is the actual realized value of the walk, no expectations have been taken, and the equation is true round by round. Three readings are immediate.

(a) Posterior accounting. The realized log-wealth at time t equals the posterior-mean per-bet log payoff, $\int [\eta S_t - K_t(\eta)] d\pi_t$, minus the information cost (in nats) the path has spent updating the prior to the posterior, $\text{KL}(\pi_t \| \Pi)$. Detection occurs at level α exactly when this difference exceeds $\log(1/\alpha)$. The Learner’s task is the resource-allocation problem of choosing Π so that the worst-case path is forced to spend the most information.

(b) E-value Gibbs duality. The variational form $\log Z_t = \sup_{\nu \ll \Pi} \left\{ \int [\eta S_t - K_t(\eta)] d\nu - \text{KL}(\nu \parallel \Pi) \right\}$ identifies $\log Z_t$ as the Donsker–Varadhan free energy of the collection of *instantaneous* per-bet log-payoffs $\eta S_t - K_t(\eta)$ under the prior Π . The supremum is achieved at the Gibbs measure $\pi_t \propto e^{\eta S_t - K_t(\eta)} \Pi$, and π_t tracks the time-evolving Bayesian update under Π .

(c) Exponentiated null restriction of the CGF-constrained game. A repeated CGF-constrained game over KL balls has an exact one-round identity $\langle p_t - \rho, c_t \rangle = \eta_t Q_t(c_t; p_t, \eta_t) + (\text{KL}(\rho \parallel p_t) - \text{KL}(\rho \parallel \tilde{p}_{t+1}))/\eta_t$, which, summed over rounds and exponentiated against a prior Π on the scale η , recovers (2). This is the structural identity that makes the LIL game and the CGF-constrained game one and the same up to the exponential map; the formalization is in Section 17.

The pathwise identity has the following immediate consequences.

Corollary 3.2 (Ville’s inequality, decomposed). *Under the assumptions of Theorem 3.1, the event $\{Z_t \geq 1/\alpha\}$ coincides with*

$$\left\{ \int [\eta S_t - K_t(\eta)] d\pi_t(\eta) \geq \log(1/\alpha) + \text{KL}(\pi_t \parallel \Pi) \right\}$$

Consequently $\mathbb{P}(\exists t : Z_t \geq 1/\alpha) \leq \alpha$ is the assertion that the path-wise information cost of any sample path suffices to certify a posterior-mean per-bet log payoff exceeding $\log(1/\alpha)$ with probability at most α .

Corollary 3.3 (Equalizer condition restated information-theoretically). *Suppose Π is such that $Z_t|_{S_t=b(t)} = C$ is constant in t (the equalizer condition). Then along the boundary $\{S_t = b(t)\}$, the identity (2) reads*

$$\log C = \int [\eta b(t) - K_t(\eta)] d\pi_t(\eta) - \text{KL}(\pi_t \parallel \Pi)$$

i.e. the posterior-mean per-bet log payoff minus the prior-to-posterior information cost is constant in t along the boundary. The equalizer is the unique Π for which the path’s accounting balance is independent of when the boundary is reached.

Remark 3.4 (Pathwise vs. stochastic identities). The identity (2) is purely algebraic — it follows from the definition of π_t as a Bayes update. The supermartingale property of Z_t , in contrast, is a stochastic statement that involves $\mathbb{E}[\cdot | \mathcal{F}_{t-1}]$ and the CGF hypothesis on ξ_t . In this paper we use (2) to *define* the equalizer (a deterministic property of Π along the boundary) and the supermartingale property only to translate the boundary inequality $S_t \geq b(t)$ into a probability statement via Ville. This division of labor is what makes the intrinsic-time treatment so clean: the algebra of $\log Z_t$ is unchanged whether we use the variance proxy $\eta^2 t/2$ or the exact ψ_t .

3.1 Pathwise increments: the per-round IT update

Differencing (2) between rounds $t-1$ and t gives the per-round version. Define $\Delta_t := \log Z_t - \log Z_{t-1}$, the round- t log-payoff of the mixture, and $\text{KL}_t := \text{KL}(\pi_t \parallel \pi_{t-1})$, the round- t posterior-update information cost.

Proposition 3.5 (Per-round IT update). *For every round $t \geq 1$,*

$$\Delta_t = \int [\eta \xi_t - \psi_t(\eta)] d\pi_t(\eta) - \text{KL}_t \tag{4}$$

Proof. Subtract (2) at $t-1$ from (2) at t . The integrand $[\eta S_t - K_t(\eta)] - [\eta S_{t-1} - K_{t-1}(\eta)]$ equals $\eta \xi_t - \psi_t(\eta)$. The KL terms combine via the chain rule for KL divergence: $\text{KL}(\pi_t \parallel \Pi) - \text{KL}(\pi_{t-1} \parallel \Pi) = \text{KL}(\pi_t \parallel \pi_{t-1})$, valid since $\pi_t \ll \pi_{t-1}$. $\square$

The per-round identity (4) is the LIL-game analogue of the standard one-round identity in a CGF-constrained repeated game over KL balls. The integrand $\eta \xi_t - \psi_t(\eta)$ is the centered per-bet log payoff (a true martingale increment, by definition of ψ_t); the KL_t term is the information cost of one Bayes update; and Δ_t is the realized log-payoff after both effects. The supermartingale property $\mathbb{E}[\Delta_t | \mathcal{F}_{t-1}] \leq 0$ is now visibly the statement $\int \mathbb{E}[\eta \xi_t - \psi_t(\eta) | \mathcal{F}_{t-1}] d\pi_t(\eta) \leq \text{KL}_t$, i.e. the prospective expected log-payoff is bounded by the information cost of the update; the inequality is tight when the CGF constraint is tight, recovering the martingale property exactly under $\mathbb{E}[e^{\eta \xi_t} | \mathcal{F}_{t-1}] = e^{\psi_t(\eta)}$.

4 The sequential detection game

4.1 Protocol

Fix a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, P)$. Let $(S_t)_{t \geq 1}$ be a real-valued score process with $S_0 = 0$ and increments $\xi_t = S_t - S_{t-1}$.

Definition 4.1 (Sequential detection game). The game has two players and runs in discrete rounds $t = 1, 2, \dots$

- (i) **Learner** (oblivious; commits before any data): chooses a probability measure Π on $(0, \infty)$, the *prior over measurement scales*.
- (ii) **Nature** (adaptive): at each round t , chooses the conditional distribution of the increment $\xi_t = S_t - S_{t-1}$ given $\mathcal{F}_{t-1}$, subject to the martingale-difference constraint $\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$. The conditional cumulant-generating function $\psi_t(\eta) := \log \mathbb{E}[e^{\eta \xi_t} \mid \mathcal{F}_{t-1}]$ is required to be finite on a neighborhood of the origin.

The Learner's payoff is the indicator that the mixture wealth process Z_t defined in (6) below stays under $1/\alpha$ for all t ; equivalently, that S_t never crosses the implicit boundary $b(t)$ given by $Z_t|_{S_t=b(t)} = 1/\alpha$. The Learner seeks the smallest such boundary; Nature seeks to cross it.

For each scale $\eta > 0$, write $K_t(\eta) := \sum_{s=1}^t \psi_s(\eta)$ for the cumulative CGF cost. In the Hoeffding regime where $|\xi_t| \leq 1$ a.s., Hoeffding's lemma gives $\psi_t(\eta) \leq \eta^2/2$, so $K_t(\eta) \leq \eta^2 t/2$. For a Rademacher walk ($\xi_t \in \{-1, +1\}$ equiprobably), $\psi_t(\eta) = \log \cosh(\eta)$, so $K_t(\eta) = t \log \cosh(\eta)$; when η is small, $\log \cosh(\eta) = \frac{1}{2}\eta^2 + O(\eta^4)$, so $K_t(\eta) \approx \eta^2 t/2$. The Hoeffding bound is thus tight at leading order in the LIL regime.

4.2 The mixture wealth process

For each fixed $\eta > 0$, the exponential process

$$M_t^\eta := \exp(\eta S_t - K_t(\eta)) \quad (5)$$

is a nonnegative supermartingale under the null $\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$. Indeed, by the tower property,

$$\mathbb{E}[M_t^\eta \mid \mathcal{F}_{t-1}] = M_{t-1}^\eta \cdot e^{-\psi_t(\eta)} \cdot \mathbb{E}[e^{\eta \xi_t} \mid \mathcal{F}_{t-1}] = M_{t-1}^\eta$$

Equality holds throughout when the CGF is computed exactly, so M_t^η is in fact a martingale; it becomes a strict supermartingale if $\psi_t(\eta)$ is replaced by an upper bound such as $\eta^2/2$. The Learner's mixture wealth process is

$$Z_t := \int_0^\infty M_t^\eta \Pi(d\eta) = \int_0^\infty \exp(\eta S_t - K_t(\eta)) \Pi(d\eta) \quad (6)$$

Since mixtures of nonnegative supermartingales are nonnegative supermartingales, Ville's inequality [27, 31] gives:

$$\mathbb{P}(\exists t \geq 1 : Z_t \geq 1/\alpha) \leq \alpha \quad (7)$$

The detection boundary $b(t)$ is implicitly defined by $Z_t = 1/\alpha$ when $S_t = b(t)$. The Learner wants $b(t)$ as small as possible (tight concentration); Nature wants to cross $b(t)$.

4.3 What “equalizer” means here, and why it forces a unique prior

The intuition. The Learner has fixed a prior Π once and for all; she now has to pay for whatever Nature does. Nature's best move, given the Learner's commitment, is to *wait* until a time t at which the boundary $b(t)$ is least well defended by Π

— specifically, until a time when the integrand $\exp(\eta b(t) - \eta^2 t/2)$, viewed as a function of η , is sharply peaked at a value of η that lies in a region where Π puts little mass. By choosing such a t , Nature can force a crossing without paying its expected CGF charge.

The Learner’s defense against this is the *equalizer*: choose Π so that the integrand peaks at *every* time t produce the *same* mixture wealth $Z_t|_{S_t=b(t)}$. With every crossing time equally costly to Nature, no waiting move is strictly preferred. The min-max strategy reduces to choosing the prior that exactly equalizes Z_t along the boundary, and that constraint turns out to determine the prior *uniquely* (up to normalization): the Laplace approximation in Section 4.4 below shows that the equalizer condition is one equation per time t , and inverting it gives the density at the saddlepoint $\eta^*(t)$.

Tiny worked example: Gaussian random walk + linear boundary. Specialize momentarily to $\xi_t \stackrel{\text{iid}}{\sim} N(0, 1)$ and ask Nature to cross the linear boundary $b(t) = \lambda_0 t$ for a fixed $\lambda_0 > 0$. The saddlepoint at this boundary is $\eta^*(t) = \lambda_0$, constant in t . The two-point prior $\Pi = \frac{1}{2}\delta_{\lambda_0} + \frac{1}{2}\delta_{-\lambda_0}$ yields the wealth process $Y_t = \cosh(\lambda_0 S_t)e^{-\lambda_0^2 t/2}$, and along $|S_t| = \lambda_0 t$ one finds $Y_t = \frac{1}{2}(e^{\lambda_0^2 t/2} + e^{-\lambda_0^2 t/2})e^{-\lambda_0^2 t/2} \sim \frac{1}{2}e^{\lambda_0^2 t/2}$, which depends on t — hence *not* an equalizer along the whole linear boundary. But this is the right strategy for the *LLN equalizer* sub-game of Section 18: there, Nature is restricted to crossing $|S_t| = \lambda_0 t$ at *all* t *uniformly*, and the two-point prior makes Y_t blow up as soon as a crossing occurs, no matter when. This trades sharpness for uniformity, and is what the first stage of Balsubramani’s proof actually needs.

The LIL game asks more. For the LIL boundary $b(t) = \sqrt{2t h(t)}$ with $h(t) \asymp \log \log t$, the hindsight-optimal scale $\eta^*(t) = b(t)/t \rightarrow 0$ depends on t . Now Nature can choose t to make $\eta^*(t)$ land anywhere in $(0, \eta_{\max}]$, and the Learner must hedge across all of these landing scales simultaneously. The equalizer condition becomes a constraint on the *shape* of Π on this interval — specifically, that $\pi(\eta^*(t)) \cdot \sqrt{t} \cdot e^{-h(t)}$ be constant in t ((10) below). The reduced game of Sections 6–8 extracts this constraint into a clean $\sup_\nu \inf_\lambda F_\nu(\lambda) \log(1/\lambda)$ statement, whose unique exact solution is the Balsubramani density.

From i.i.d. to martingale. The protocol of Definition 4.1 already allows Nature to choose the conditional distribution of ξ_t adaptively, so all constructions in this paper are stated for martingale-difference sequences. The role of the i.i.d. assumption is purely to invoke the classical Erdős–Kolmogorov–Feller integral test (Theorem 11.2), whose original statement is iid; the extension to bounded-increment martingales goes through the Komlós–Major–Tusnády strong approximation [18] and its martingale lifts [26, 28]. The *game-theoretic* achievability and converse statements (Proposition 4.3, Corollary 11.3) are direct consequences of the Laplace analysis + normalizability and do *not* require this passage — they hold verbatim for any martingale satisfying the CGF-finiteness condition of Definition 4.1. This is the sense in which the game-theoretic reading is sharper than the classical statement.

4.4 Saddlepoint evaluation along the boundary

We work throughout this subsection in the *Gaussian surrogate regime* $K_t(\eta) = \eta^2 t/2$, which is a tight upper bound in the Hoeffding setting $|\xi_t| \leq 1$ and an equality for i.i.d. Gaussian increments. For sub-Gaussian martingales with a Hoeffding-type CGF bound $\psi_t(\eta) \leq \sigma^2 \eta^2/2$, the same analysis applies after rescaling t by σ^2 ; the LIL boundaries we derive are therefore sharp up to this overall variance scaling. (The non-Gaussian corrections reappear in the saddlepoint curvature as subleading $O((\eta^*)^2)$ terms; they do not affect the leading $\sqrt{t \log \log t}$ rate. See Example 4.2 below for explicit Rademacher calculations.)

Evaluate the mixture (6) at $S_t = b(t)$ by Laplace’s method. The exponent $f(\eta) = \eta b(t) - \eta^2 t/2$ is maximized at the saddlepoint

$$\eta^*(t) = \frac{b(t)}{t} \tag{8}$$

with value $f(\eta^*) = b(t)^2/(2t)$ and curvature $f''(\eta^*) = -t$. The Laplace approximation gives

$$Z_t|_{S_t=b(t)} = \pi(\eta^*(t)) \cdot \sqrt{\frac{2\pi}{t}} \cdot \exp\left(\frac{b(t)^2}{2t}\right) \cdot (1 + o(1)) \quad (9)$$

where π is the density of Π and the $o(1)$ remainder holds under the regularity conditions (H1)–(H3) of Lemma 10.1 below. For non-Gaussian ψ_t , replace $\eta^2 t/2$ by $K_t(\eta)$ and the saddlepoint by the solution of $K'_t(\eta) = b(t)$; under the LIL-regime condition $b(t)/t \rightarrow 0$, the two saddlepoints agree to leading order in η (cf. Example 4.2).

Example 4.2 (Rademacher walk). For a Rademacher walk, $K_t(\eta) = t \log \cosh(\eta)$, so the exponent $\eta b(t) - t \log \cosh(\eta)$ is maximized at $\eta^*(t) = \tanh^{-1}(b(t)/t)$. In the LIL regime $b(t)/t \rightarrow 0$, Taylor expansion gives $\tanh^{-1}(x) = x + x^3/3 + O(x^5)$, so $\eta^*(t) \approx b(t)/t$ in agreement with the Hoeffding saddlepoint (8). The exact Laplace curvature at the saddlepoint is $t/\cosh^2(\eta^*)$, which also agrees with the Hoeffding curvature t to leading order when $\eta^* \rightarrow 0$. Thus the Hoeffding and Rademacher analyses produce the same LIL-regime equalizer density, differing only in subleading corrections that vanish as $t \rightarrow \infty$.

4.5 The equalizer density

In a zero-sum game, the minimax strategy is an *equalizer*: it makes the opponent indifferent among all pure strategies. Here, Nature's pure strategies include choosing *when* to attempt a boundary crossing. The equalizer prior makes Z_t constant along the boundary $\{S_t = b(t)\}$ for all t .

Setting $Z_t|_{S_t=b(t)} = C$ and solving for π :

$$\pi(\eta^*(t)) = C' \cdot \sqrt{t} \cdot \exp\left(-\frac{b(t)^2}{2t}\right) \quad (10)$$

This is the *equalizer density*: the unique prior (up to normalization) that distributes detection power uniformly across time.

Proposition 4.3 (Equalizer characterization). *The prior Π is an asymptotic equalizer strategy for boundary $b(t)$ (in the sense of Lemma 10.1 below) if and only if its density satisfies (10). The boundary is achievable by an oblivious Learner strategy — meaning the corresponding mixture wealth process is a well-defined nonnegative supermartingale with finite total mass — if and only if the equalizer density is normalizable: $\int_0^\infty \pi(\eta) d\eta < \infty$.*

Proof. The proposition has two assertions; we prove each in turn.

(I) *Equalizer characterization.* Suppose Π has continuous strictly positive density π near $\eta^*(t)$ for all large t . By Lemma 10.1, the Laplace approximation gives

$$Z_t|_{S_t=b(t)} = \sqrt{2\pi/t} \pi(\eta^*(t)) e^{h(t)} (1 + o(1))$$

Setting the LHS equal to a constant C and solving for π yields (10) with $C' := C/\sqrt{2\pi}$; conversely, substituting (10) into the Laplace expression makes $Z_t|_{S_t=b(t)} \rightarrow C$, so the equalizer condition holds asymptotically. This is an exact equivalence at the level of densities up to subleading $o(1)$ terms.

(II) *Achievability equivalence.* We treat the two directions separately.

(II.if) *Normalizability $\Rightarrow$ valid Learner strategy.* Given a normalizable density π satisfying (10), set $\Pi(d\eta) := \pi(\eta) d\eta / \int \pi$, so that Π is a probability measure on $(0, \infty)$. Each single-scale process M_t^η in (5) is a nonnegative supermartingale with $M_0^\eta = 1$, and the family $\{M_t^\eta\}_\eta$ is jointly measurable in (η, t, ω) . Tonelli's theorem then gives the supermartingale property of the mixture: $\mathbb{E}[Z_t | \mathcal{F}_{t-1}] \leq Z_{t-1}$. Hence $Z_0 = 1$ and $\mathbb{P}(\exists t : Z_t \geq 1/\alpha) \leq \alpha$ by Ville's inequality (7). Combined with Laplace, $Z_t|_{S_t=b(t)} \rightarrow C \in (0, \infty)$, so for any $\alpha < C^{-1}$ the event $\{S_t \geq b(t)\}$ implies $\{Z_t \geq 1/\alpha\}$ for all sufficiently large t , and the Learner controls the boundary crossing probability by α .

(II.only-if) *Non-normalizability $\Rightarrow$ no oblivious strategy.* Suppose, toward a contradiction, that the equalizer density π for boundary b has $\int \pi = \infty$, yet some probability measure $\tilde{\Pi}$ on $(0, \infty)$ produces a mixture $\tilde{Z}_t$ that satisfies $\tilde{Z}_t|_{S_t=b(t)} \geq$

$c > 0$ for all sufficiently large t . Restricting to a neighborhood $N_t = [\eta^*(t)/2, 2\eta^*(t)]$ of the saddlepoint and applying the Laplace argument of Lemma 10.1 to the truncated measure $\tilde{\Pi}|_{N_t}$,

$$\tilde{Z}_t|_{S_t=b(t)} \leq \tilde{\pi}(\eta^*(t)) \sqrt{2\pi/t} e^{h(t)} (1 + o(1)) + (\text{tail outside } N_t)$$

The tail outside N_t contributes at most a constant since the exponent in the integrand decays as $\exp(-t(\eta - \eta^*(t))^2/2)$. Hence $\tilde{\pi}(\eta^*(t)) \gtrsim \sqrt{t} e^{-h(t)}$ for all sufficiently large t , matching the equalizer density along the saddle curve $\{\eta^*(t) : t \geq t_0\}$. Changing variables $\eta = \eta^*(t)$ with Jacobian $|\mathrm{d}\eta^*/\mathrm{d}t| \asymp \sqrt{h(t)}/t^{3/2}$, the integral of $\tilde{\pi}$ over the saddle range $(\eta^*(\infty), \eta^*(t_0)]$ is bounded below by $\int_{t_0}^{\infty} \sqrt{t} e^{-h(t)} \cdot \sqrt{h(t)}/t^{3/2} \mathrm{d}t = \int_{t_0}^{\infty} \sqrt{h(t)} e^{-h(t)}/t \mathrm{d}t$, the Erdős integral, which diverges precisely by the non-normalizability hypothesis. Hence $\int \tilde{\pi} = \infty$, contradicting that $\tilde{\Pi}$ is a probability measure. The classical statement that an i.i.d. Gaussian random walk crosses $b(t)$ i.o. when π is non-normalizable is then a special case of Theorem 11.2; for our argument we only need the normalizability obstruction, which is internal to the Laplace analysis. The i.o. statement is classical [13, 25]. $\square$

5 The finite-dimensional minimax coincidence center

The finite-dimensional theorem is the exact prototype for the continuum story. Let Δ_K denote the probability simplex on $[K] := \{1, \dots, K\}$, and let $\pi_1, \dots, \pi_W \in \Delta_K$ be strictly positive priors. For $\alpha \in \Delta_W$, define the geometric pool

$$p_\alpha(i) := \frac{\prod_{w=1}^W \pi_w(i)^{\alpha_w}}{\sum_{j=1}^K \prod_{w=1}^W \pi_w(j)^{\alpha_w}}, \quad i \in [K]$$

and the coincidence functional

$$C_\alpha(\pi_{1:W}) := -\log \sum_{i=1}^K \prod_{w=1}^W \pi_w(i)^{\alpha_w}$$

In words: the next theorem says that the optimal mixing weights α^ over a finite collection of priors $\{\pi_w\}$ produce a geometric pool $p^* = p_{\alpha^*}$ whose reverse KL-divergence to each actively used component is equal — no actively pooled prior is reverse-closer to p^* than any other. This is the equalizer condition in finite dimensions.*

Theorem 5.1 (Finite-dimensional minimax coincidence center). *Let $\pi_1, \dots, \pi_W \in \Delta_K$ be strictly positive. Then*

$$\max_{\alpha \in \Delta_W} C_\alpha(\pi_{1:W}) = \min_{p \in \Delta_K} \max_{w \in [W]} \mathrm{KL}(p \| \pi_w)$$

If α^ maximizes the left-hand side and $p^* := p_{\alpha^*}$, then p^* attains the minimum on the right. Moreover:*

$$\begin{aligned} \alpha_w^* > 0 &\implies \mathrm{KL}(p^* \| \pi_w) = R^* \\ \alpha_w^* = 0 &\implies \mathrm{KL}(p^* \| \pi_w) \leq R^* \end{aligned}$$

where $R^ := \min_{p \in \Delta_K} \max_{w \in [W]} \mathrm{KL}(p \| \pi_w)$. The active reverse-KL constraints are equalized.*

Proof. Fix $\alpha \in \Delta_W$ and define

$$F(p, \alpha) := \sum_{w=1}^W \alpha_w \mathrm{KL}(p \| \pi_w), \quad p \in \Delta_K$$

Because every π_w is strictly positive, F is finite and continuous, strictly convex in p (relative entropy is strictly convex in its first argument whenever all second-argument supports contain $\mathrm{supp}(p)$), and linear in α . A Gibbs calculation gives,

for every $p \in \Delta_K$,

$$\begin{aligned}
F(p, \alpha) &= \sum_{i=1}^K p(i) \log p(i) - \sum_{w=1}^W \alpha_w \sum_{i=1}^K p(i) \log \pi_w(i) \\
&= \sum_{i=1}^K p(i) \log \frac{p(i)}{\prod_{w=1}^W \pi_w(i)^{\alpha_w}} \\
&= \sum_{i=1}^K p(i) \log \frac{p(i)}{p_\alpha(i)} - \log \sum_{j=1}^K \prod_{w=1}^W \pi_w(j)^{\alpha_w} \\
&= \text{KL}(p \| p_\alpha) + C_\alpha(\pi_{1:W})
\end{aligned}$$

Therefore $\min_{p \in \Delta_K} F(p, \alpha) = C_\alpha(\pi_{1:W})$, attained uniquely at $p = p_\alpha$ by strict convexity.

Apply Sion's minimax theorem [30] to F on the compact convex sets $\Delta_W \times \Delta_K$:

$$\begin{aligned}
\max_{\alpha \in \Delta_W} C_\alpha(\pi_{1:W}) &= \max_{\alpha} \min_p F(p, \alpha) = \min_p \max_{\alpha} F(p, \alpha) \\
&= \min_p \max_w \text{KL}(p \| \pi_w)
\end{aligned}$$

since maximizing a linear form $\alpha \mapsto \sum_w \alpha_w \text{KL}(p \| \pi_w)$ over the simplex selects the largest coordinate. This proves the identity and that $p^* = p_{\alpha^*}$ attains the right-hand-side minimum.

For the equalizer statement, differentiate C_α directly. Using the chain rule and the identity $\sum_i p_\alpha(i) \log \pi_w(i) = -H(p_\alpha) - \text{KL}(p_\alpha \| \pi_w)$,

$$\frac{\partial C_\alpha}{\partial \alpha_w} = - \frac{\sum_i (\prod_{w'} \pi_{w'}(i)^{\alpha_{w'}}) \log \pi_w(i)}{\sum_j \prod_{w'} \pi_{w'}(j)^{\alpha_{w'}}} = - \sum_i p_\alpha(i) \log \pi_w(i) = H(p_\alpha) + \text{KL}(p_\alpha \| \pi_w)$$

where $H(p) = -\sum_i p(i) \log p(i)$ is Shannon entropy. The map $\alpha \mapsto C_\alpha$ is concave (as the minimum of affine functions of α for each fixed p). The KKT conditions for maximizing a concave function over the simplex Δ_W state: there exists a constant $\mu \in \mathbb{R}$ (the Lagrange multiplier for $\sum_w \alpha_w = 1$) such that

$$\left. \frac{\partial C_\alpha}{\partial \alpha_w} \right|_{\alpha^*} \begin{cases} = \mu & \text{if } \alpha_w^* > 0, \\ \leq \mu & \text{if } \alpha_w^* = 0. \end{cases}$$

Because $H(p_{\alpha^*})$ does not depend on w , subtracting it from both sides shows that all active reverse-KL values $\text{KL}(p^* \| \pi_w)$ share a common value, R_{act}^* say, and inactive values cannot exceed it. To identify R_{act}^* with the minimax value $R^* = \min_p \max_w \text{KL}(p \| \pi_w)$, evaluate

$$C_{\alpha^*} = F(p^*, \alpha^*) = \sum_w \alpha_w^* \text{KL}(p^* \| \pi_w) = \sum_{w: \alpha_w^* > 0} \alpha_w^* \cdot R_{\text{act}}^* = R_{\text{act}}^*$$

using the active-set equalization in the third equality and $\sum_{w: \alpha_w^* > 0} \alpha_w^* = 1$ in the fourth. The minimax identity proved above gives $C_{\alpha^*} = R^*$, so $R_{\text{act}}^* = R^*$, as claimed. $\square$

6 The exact continuum scale-allocation game

We now define the continuum game whose solution is exactly Balsubramani's prior.

6.1 The one-sided game

Fix the interval $I := (0, e^{-2}]$. The Learner chooses a probability measure ν on I . Nature chooses a target scale $\lambda \in I$. The Learner's payoff is

$$G(\nu, \lambda) := F_\nu(\lambda) \cdot \log(1/\lambda), \quad F_\nu(\lambda) := \nu((0, \lambda])$$

Here $F_\nu(\lambda)$ is the cumulative prior mass below λ , and $\log(1/\lambda)$ is the *scale tax* — the information-theoretic cost of operating at scale λ , inherited from the exponential tilting that defines the wealth process. Small λ values correspond to late times (when $\lambda_t^* \approx \sqrt{2h(t)/t}$ is small), and Nature can exploit them by waiting.

The game value is $V := \sup_\nu \inf_{\lambda \in I} G(\nu, \lambda)$.

In words: the Learner spreads probability mass across scales $\lambda \in (0, e^{-2}]$, and Nature picks the weakest scale; the next theorem says the Learner's best mass-spreading strategy is the one that puts exactly $F^(\lambda) = 2/\log(1/\lambda)$ of its mass below scale λ , with tail logarithmic in $1/\lambda$.*

Theorem 6.1 (Exact minimax of the continuum scale game). *The value of the game is $V = 2$. The unique equalizer strategy is the probability law ν^* with CDF*

$$F^*(\lambda) = \frac{2}{\log(1/\lambda)}, \quad 0 < \lambda \leq e^{-2}$$

and density

$$\nu^*(d\lambda) = \frac{2 d\lambda}{\lambda(\log(1/\lambda))^2}$$

For this law, $G(\nu^*, \lambda) = 2$ for every $\lambda \in (0, e^{-2}]$.

Proof. Upper bound. For any probability law ν on I , we have $F_\nu(e^{-2}) = 1$. Therefore

$$\inf_\lambda G(\nu, \lambda) \leq G(\nu, e^{-2}) = 1 \cdot \log(e^2) = 2$$

Taking the supremum over ν gives $V \leq 2$.

Lower bound. Define $F^*(\lambda) := 2/\log(1/\lambda)$. This is increasing on $(0, e^{-2}]$, satisfies $\lim_{\lambda \downarrow 0} F^*(\lambda) = 0$, and has $F^*(e^{-2}) = 2/2 = 1$. So it is a valid CDF on I . For this law,

$$G(\nu^*, \lambda) = F^*(\lambda) \log(1/\lambda) = 2 \quad \forall \lambda \in I$$

Hence $\inf_\lambda G(\nu^*, \lambda) = 2$, so $V \geq 2$.

Uniqueness of the equalizer. An equalizer by definition satisfies $G(\nu, \lambda) = c$ for some constant c and all $\lambda \in I$, so $F_\nu(\lambda) = c/\log(1/\lambda)$. Evaluating at $\lambda = e^{-2}$ forces $1 = c/2$, hence $c = 2$ and $F_\nu \equiv F^*$, so $\nu = \nu^*$. $\square$

Remark 6.2 (Minimax versus equalizer). Theorem 6.1 asserts uniqueness within the *equalizer* class. The full minimax set is in general strictly larger: any law ν with $F_\nu(\lambda) \log(1/\lambda) \geq 2$ for all $\lambda \in I$ and $F_\nu(e^{-2}) = 1$ (equivalently, with CDF pointwise above F^* , so that ν is stochastically dominated by ν^*) attains $\inf_\lambda G(\nu, \lambda) = 2$ and is therefore minimax. Conversely, every minimax law must satisfy $F_\nu \geq F^*$ pointwise: if $F_\nu(\lambda_0) < F^*(\lambda_0) = 2/\log(1/\lambda_0)$ at any $\lambda_0 \in I$, then $G(\nu, \lambda_0) = F_\nu(\lambda_0) \log(1/\lambda_0) < 2$, contradicting $\inf_\lambda G(\nu, \lambda) \geq V = 2$. Hence:

- The equalizer ν^* is the *pointwise-smallest CDF* among all minimax laws (every other minimax law puts strictly more mass below every scale λ).
- All minimax laws share the same tail behavior: $F_\nu(\lambda) \geq 2/\log(1/\lambda)$ for every λ , so they all allocate at least $2/\log(1/\lambda)$ of their mass below scale λ .
- The equalizer is the unique minimax law that makes Nature indifferent across all pure strategies. Among non-equalizer minimax laws, Nature has a strict best response: she plays any λ at which $F_\nu \log(1/\lambda) > 2$.

The selection of the equalizer is thus a proper refinement of minimax, justified by indifference rather than by saddle-point optimality alone. This is also why the equalizer is the natural mixing prior for the method-of-mixtures construction: it is the strategy that does not waste mass at any scale.

Remark 6.3. The endpoint e^{-2} is not arbitrary: it is the truncation used in the earlier construction [1]. If the interval were $(0, e^{-a}]$, the value would be a and the equalizer CDF would be $a/\log(1/\lambda)$.

Why $\eta_{\max} = e^{-2}$? (IT-flavored truncation.) The pathwise identity (2) gives an information-theoretic reading of the truncation. At the equalizer prior $\pi^*(\eta) = 2/(\eta \log^2(1/\eta))$, the per-round IT identity (4) says $\Delta_t = \int [\eta \xi_t - \psi_t(\eta)] d\pi_t - \text{KL}_t$. For the boundary scale $\eta = \eta_{\max} = e^{-2}$, the posterior-prior KL is $\text{KL}(\pi_t \| \Pi^*) = \log(1/\eta_{\max}) = 2$ at the equalizer's saddle, exactly matching the value $V = 2$. A larger truncation would require a larger budget and break the equalizer's $F^*(\eta_{\max}) = 1$ normalization. The choice $\eta_{\max} = e^{-2}$ is the largest truncation compatible with a posterior-budget of V nats per round.

6.2 The signed game

Balsubramani's law is symmetric in sign. Let $I_{\pm} := [-e^{-2}, e^{-2}] \setminus \{0\}$. The Learner chooses a probability measure $\tilde{\nu}$ on $I_{\pm}$. Nature chooses a sign $\sigma \in \{+, -\}$ and a magnitude $\lambda \in (0, e^{-2}]$, with payoff $\tilde{G}(\tilde{\nu}, \sigma, \lambda) := \tilde{\nu}(\sigma(0, \lambda)) \cdot \log(1/\lambda)$.

Corollary 6.4 (Symmetric exact minimax law). *The unique symmetric minimax law is*

$$\tilde{\nu}^*(d\lambda) = \frac{d\lambda}{|\lambda|(\log(1/|\lambda|))^2}, \quad 0 < |\lambda| \leq e^{-2}$$

Proof. Write any law as $\tilde{\nu} = p\nu_+ + (1-p)\nu_-$ where $\nu_{\pm}$ are probability laws on the positive/negative half-lines and $p \in [0, 1]$. For Nature's choice (σ, λ) , $\tilde{\nu}(\sigma(0, \lambda)) = p F_{\nu_+}(\lambda)$ if $\sigma = +$ and $(1-p) F_{\nu_-}(\lambda)$ if $\sigma = -$; Nature plays the side giving the smaller product. Hence

$$\inf_{\sigma, \lambda} \tilde{G}(\tilde{\nu}, \sigma, \lambda) = \min \left\{ p \cdot \inf_{\lambda} F_{\nu_+}(\lambda) \log(1/\lambda), (1-p) \cdot \inf_{\lambda} F_{\nu_-}(\lambda) \log(1/\lambda) \right\}$$

By Theorem 6.1 the one-sided suprema equal 2 and are attained uniquely by the one-sided Balsubramani CDF $F^*(\lambda) = 2/\log(1/\lambda)$ on each side. Maximizing $\min\{2p, 2(1-p)\}$ over p forces $p = 1/2$, giving signed game value 1. The unique minimax law is therefore symmetric with each conditional equal to the one-sided equalizer. The resulting density is

$$\tilde{\nu}^*(d\lambda) = \frac{1}{2} \cdot \frac{2 d\lambda}{|\lambda|(\log(1/|\lambda|))^2} = \frac{d\lambda}{|\lambda|(\log(1/|\lambda|))^2}$$

supported on $I_{\pm}$, as claimed. □

7 The discrete shell game

The continuum law has a transparent discrete cousin. For $j = 2, 3, \dots$, define the shell $I_j := (e^{-(j+1)}, e^{-j}]$. The Learner assigns weights $w_j \geq 0$ with $\sum_{j \geq 2} w_j = 1$. Define tail masses $T_j := \sum_{i \geq j} w_i$. Nature chooses a shell index $j \geq 2$, and the payoff is

$$G_{\text{sh}}(w, j) := j \cdot T_j$$

The factor j is the shell version of $\log(1/\lambda)$, since $j \asymp \log(1/\lambda)$ on shell I_j .

In words: when scales are bucketed into geometric shells, the equalizer assigns the j -th tail-mass exactly $2/j$, so that the cost $j \cdot T_j$ is the same constant 2 at every shell. The induced individual shell weights $w_j^ = 2/(j(j+1))$ are a Stirling-style telescoping decomposition of unity.*

Theorem 7.1 (Exact shell equalizer). *The shell game has value $V_{\text{sh}} = 2$. Its unique equalizer is*

$$T_j^* = \frac{2}{j}, \quad w_j^* = T_j^* - T_{j+1}^* = \frac{2}{j(j+1)}, \quad j \geq 2$$

In particular, $G_{\text{sh}}(w^, j) = 2$ for every $j \geq 2$.*

Proof. For any probability weights w , we have $T_2 = 1$, so $\inf_{j \geq 2} jT_j \leq 2T_2 = 2$, giving $V_{\text{sh}} \leq 2$.

Define $T_j^* := 2/j$. Then $T_2^* = 1$, $T_j^* \downarrow 0$, and $w_j^* = 2/j - 2/(j+1) = 2/(j(j+1)) \geq 0$ with $\sum_{j \geq 2} w_j^* = T_2^* = 1$. Moreover, $jT_j^* = 2$ for all $j \geq 2$, so $V_{\text{sh}} \geq 2$.

Uniqueness: if $jT_j \equiv c$ then evaluating at $j = 2$ gives $c = 2$, forcing $T_j = 2/j$. $\square$

Corollary 7.2 (Continuum density from shell weights). *The shell weights satisfy $w_j^* \sim 2/j^2$. Since $j \asymp \log(1/\lambda)$ on shell I_j and $\mathrm{d}j \asymp -\mathrm{d}\lambda/\lambda$, the corresponding continuum density is $f(\lambda) \asymp 1/(\lambda(\log(1/\lambda))^2)$.*

Proof. By Theorem 7.1, $w_j^* = 2/(j(j+1)) = 2/j^2 + O(1/j^3)$, so $w_j^* \sim 2/j^2$ as $j \rightarrow \infty$. Within shell $I_j = (e^{-(j+1)}, e^{-j}]$ the variable λ satisfies $\log(1/\lambda) \in [j, j+1)$, hence $j = \log(1/\lambda) + O(1)$. Differentiating $j = -\log \lambda + O(1)$ gives $\mathrm{d}j = -\mathrm{d}\lambda/\lambda$, so the continuum density that pushes forward to the shell weights is $f(\lambda) = w_j^* |\mathrm{d}j/\mathrm{d}\lambda| \asymp (2/j^2)(1/\lambda) \asymp 1/(\lambda \log^2(1/\lambda))$, matching the Balsubramani density of Theorem 6.1 up to the constant factor 2. $\square$

Remark 7.3. The family $\{M_t^\eta : \eta > 0\}$ is a scale family. The right Haar measure for the multiplicative group on $(0, \infty)$ is $\mathrm{d}\eta/\eta$, which is also the Jeffreys prior for $\log \eta$. But $\mathrm{d}\eta/\eta$ has infinite mass. The equalizer density $1/(\eta(\log(1/\eta))^2)$ is the closest normalizable distribution: it equals $\mathrm{d}\eta/\eta$ weighted by $1/\log^2(1/\eta)$, the gentlest correction that yields finite total mass. This is why this particular density appears universally in LIL constructions.

8 A general equalizer theorem

The previous theorems admit a completely general form.

In words: replace the specific scale tax $\log(1/\lambda)$ by any continuous, strictly decreasing tax function $L(\lambda)$ that diverges at zero. The minimax value is then the tax at the right endpoint, and the equalizer puts CDF $F^ = L_0/L$ — the inverse-tax. The Balsubramani density of Theorem 6.1 is the special case $L = \log(1/\cdot)$ on $(0, e^{-2}]$.*

Theorem 8.1 (General continuum equalizer). *Let $L : (0, \lambda_{\max}] \rightarrow (0, \infty)$ be continuous, strictly decreasing, with $L(\lambda_{\max}) =: L_0 < \infty$ and $\lim_{\lambda \downarrow 0} L(\lambda) = \infty$. For a probability law ν on $(0, \lambda_{\max}]$, define*

$$G_L(\nu, \lambda) := F_\nu(\lambda) L(\lambda)$$

Then

$$\sup_\nu \inf_{\lambda \in (0, \lambda_{\max}]} G_L(\nu, \lambda) = L_0$$

The unique equalizer has CDF

$$F^*(\lambda) = \frac{L_0}{L(\lambda)}$$

If L is differentiable, the density is $f^(\lambda) = -L_0 L'(\lambda)/L(\lambda)^2$.*

Proof. For any ν , $\inf_\lambda G_L(\nu, \lambda) \leq G_L(\nu, \lambda_{\max}) = L_0$, so the value is at most L_0 .

Define $F^*(\lambda) := L_0/L(\lambda)$. Since L is decreasing and diverges at the origin, F^* is increasing, tends to 0 at the origin, and satisfies $F^*(\lambda_{\max}) = 1$. For this law, $G_L(\nu^*, \lambda) = L_0$ for all λ , so the value is at least L_0 . Uniqueness follows as before. $\square$

Example 8.2 (Recovering Balsubramani). Take $\lambda_{\max} = e^{-2}$ and $L(\lambda) = \log(1/\lambda)$. Then $L_0 = 2$, and Theorem 8.1 recovers $F^*(\lambda) = 2/\log(1/\lambda)$.

Example 8.3 (Heavier tail protection). With $L(\lambda) = \log(1/\lambda)[\log \log(e/\lambda)]^\beta$ for $\beta > 0$, the equalizer density acquires an additional iterated-logarithm factor, encoding extra protection for extremely small scales.

The general shell version holds as well.

Theorem 8.4 (General shell equalizer). *Let $(a_j)_{j \geq j_0}$ be positive and increasing with $a_{j_0} < \infty$ and $a_j \rightarrow \infty$. The Learner chooses weights $w_j \geq 0$ summing to 1; Nature chooses $j \geq j_0$; the payoff is $a_j T_j$. Then the value is a_{j_0} , with unique equalizer $T_j^* = a_{j_0}/a_j$.*

Proof. **Upper bound.** For any probability weights w , $T_{j_0} = 1$, so $\inf_{j \geq j_0} a_j T_j \leq a_{j_0}$.

Lower bound. Define $T_j^* := a_{j_0}/a_j$. Monotonicity of a_j gives T_j^* decreasing in j , $T_{j_0}^* = 1$, and $T_j^* \rightarrow 0$. The induced weights $w_j^* = T_j^* - T_{j+1}^* = a_{j_0}(1/a_j - 1/a_{j+1}) \geq 0$ sum to $\sum_{j \geq j_0} w_j^* = T_{j_0}^* = 1$. For this strategy, $a_j T_j^* = a_{j_0}$ for every $j \geq j_0$, so $\inf_j a_j T_j^* = a_{j_0}$.

Uniqueness. If $a_j T_j \equiv c$ for all $j \geq j_0$, then $j = j_0$ gives $c = a_{j_0}$, forcing $T_j = a_{j_0}/a_j$. $\square$

9 How the reduced game appears inside Balsubramani's proof

The reduced scale-allocation game whose minimax equalizer is the Balsubramani density is not an artefact of the present formulation. It is exactly the subproblem the method-of-mixtures proof exposes when the constant-scale exponential supermartingale is factorized around the hindsight-optimal scale.

9.1 Factorization around the hindsight-optimal scale

A constant-scale exponential supermartingale has the form

$$K_t^\lambda = \exp(\lambda M_t - k\lambda^2 U_t)$$

where M_t is the signed score, U_t is the cumulative variance proxy (e.g. $U_t = t$ in the Hoeffding-bounded i.i.d. regime), and $k > 0$ is the sub-Gaussian-CGF constant of the increments ($k = 1/2$ for unit-variance Gaussian, $k = 1/2$ for Rademacher under Hoeffding's lemma, and k in general absorbs the constant in the upper bound $\psi_t(\lambda) \leq k\lambda^2$ on the conditional log-MGF). With this normalization, the exponent $\lambda M_t - k\lambda^2 U_t$ is maximized at $\lambda_t^* := |M_t|/(2kU_t)$, with value $M_t^2/(4kU_t)$.

Lemma 9.1 (Exact factorization). *For every nonnegative measure ν on $(0, e^{-2}]$,*

$$K_t^\nu = \exp\left(\frac{M_t^2}{4kU_t}\right) \int_0^{e^{-2}} \exp(-kU_t(\lambda - \lambda_t^*)^2) \nu(d\lambda)$$

Proof. Complete the square: $\lambda|M_t| - k\lambda^2 U_t = M_t^2/(4kU_t) - kU_t(\lambda - \lambda_t^*)^2$. $\square$

After peeling off the universal factor $\exp(M_t^2/(4kU_t))$, the remaining problem is one of allocating mass near the relevant scale λ_t^* . The reduced game of Section 6 captures the most robust version of that allocation problem.

9.2 The reciprocal-log term

The CDF of the minimax equalizer at the effective scale is

$$F^*(\lambda_t^*) = \frac{2}{\log(2kU_t/|M_t|)}$$

This is exactly the decisive reciprocal-log term in the earlier proof [1]. The continuum game is not invented after the fact to match the density; it isolates the exact quantity that the proof uses.

Proposition 9.2 (Game-theoretic justification). *Balsubramani's oblivious mixing law is the unique exact minimax equalizer of the reduced continuum scale-allocation game. Because the key reciprocal-log term in the finite-time LIL proof is the cumulative mass of this law up to the effective hindsight-optimal scale, the law has a rigorous minimax justification as the natural mixed strategy for the method-of-mixtures construction.*

Proof. The first sentence is Theorem 6.1 (one-sided) together with Corollary 6.4 (signed): the unique CDF on $(0, e^{-2}]$ that equalizes $F_\nu(\lambda) \log(1/\lambda)$ is $F^*(\lambda) = 2/\log(1/\lambda)$, with corresponding density $2/(\lambda(\log(1/\lambda))^2)$.

The second sentence requires identifying the reciprocal-log expression that appears in the proof of [1, Theorem 3] with F^* evaluated at the effective hindsight-optimal scale. Lemma 9.1 factorizes the constant-scale exponential K_t^λ into a universal envelope $\exp(M_t^2/(4kU_t))$ multiplied by a Gaussian-in- λ residual centered at $\lambda_t^* = |M_t|/(2kU_t)$. Integrating the residual against $\tilde{\nu}^*$ truncated to $(0, e^{-2}]$ and using the Laplace approximation on the residual gives

$$\int_0^{e^{-2}} e^{-kU_t(\lambda-\lambda_t^*)^2} \tilde{\nu}^*(d\lambda) \asymp \tilde{\nu}^*((0, \lambda_t^*]) = F^*(\lambda_t^*)$$

with equality up to lower-order Laplace-curvature corrections that [1] absorbs into a multiplicative constant. By Theorem 6.1, $F^*(\lambda_t^*) = 2/\log(1/\lambda_t^*) = 2/\log(2kU_t/|M_t|)$, which is the decisive reciprocal-log factor in [1, (2.6)]. This identification is direct: no constant-matching is required beyond rewriting $\log(1/\lambda_t^*)$ in terms of $U_t/|M_t|$. $\square$

10 Asymptotic equalizers for prescribed boundaries

We now derive the asymptotic equalizer for a prescribed boundary, connecting the exact reduced game to the classical Erdős theory.

10.1 Laplace approximation along the boundary

Consider the mixture process

$$Z_t = \int_0^{\eta_{\max}} \exp\left(\eta S_t - \frac{\eta^2 t}{2}\right) \pi(\eta) d\eta$$

tracking the boundary $b(t) = \sqrt{2t h(t)}$ for a positive, eventually nondecreasing function h satisfying $h(t) \rightarrow \infty$ and $th'(t)/h(t) \rightarrow 0$ as $t \rightarrow \infty$ (so h is slowly varying in the sense of Karamata). The saddlepoint is $\eta^*(t) = b(t)/t = \sqrt{2h(t)}/t$; note that $\eta^*(t) \rightarrow 0$ and $t\eta^*(t)^2 = 2h(t) \rightarrow \infty$.

Lemma 10.1 (Laplace approximation). *Assume (H1) $h(t) \rightarrow \infty$; (H2) $th'(t)/h(t) \rightarrow 0$; and (H3) π is continuous and strictly positive in a neighborhood of $\eta^*(t)$ for all large t , with π bounded on $(0, \eta_{\max}]$. Then along $S_t = b(t)$,*

$$Z_t \sim \sqrt{\frac{2\pi}{t}} \pi(\eta^*(t)) e^{h(t)} \quad (t \rightarrow \infty)$$

Proof. Assumptions (H1) and (H2) together imply $h(t)/t \rightarrow 0$ as $t \rightarrow \infty$: under slow variation (H2), h grows slower than any positive power of t , so the Karamata representation $h(t) = \exp(\int_1^t \varepsilon(s)/s ds)$ with $\varepsilon(s) \rightarrow 0$ gives $\log h(t) = o(\log t)$, hence $h(t) = o(t^\gamma)$ for every $\gamma > 0$, and in particular $h(t)/t \rightarrow 0$. Therefore $\eta^*(t) = \sqrt{2h(t)}/t \rightarrow 0$, and for all large t the saddlepoint lies inside the integration interval $(0, \eta_{\max})$.

The exponent $\phi_t(\eta) := \eta b(t) - \eta^2 t/2$ satisfies the exact completion of the square

$$\phi_t(\eta) = h(t) - \frac{t}{2}(\eta - \eta^*(t))^2$$

since $\phi_t(\eta^*(t)) = b(t)^2/(2t) = h(t)$. Therefore

$$Z_t = e^{h(t)} \int_0^{\eta_{\max}} e^{-t(\eta - \eta^*(t))^2/2} \pi(\eta) \, d\eta$$

Substitute $u = \sqrt{t}(\eta - \eta^*(t))$, with $d\eta = du/\sqrt{t}$, to get

$$Z_t = \frac{e^{h(t)}}{\sqrt{t}} \int_{-\sqrt{t}\eta^*(t)}^{\sqrt{t}(\eta_{\max} - \eta^*(t))} e^{-u^2/2} \pi\left(\eta^*(t) + \frac{u}{\sqrt{t}}\right) du$$

Assumption (H1) gives $\sqrt{t}\eta^*(t) = \sqrt{2h(t)} \rightarrow \infty$, and since $\eta^*(t) \rightarrow 0$ (shown above) while $\eta_{\max}$ is fixed, the upper limit $\sqrt{t}(\eta_{\max} - \eta^*(t)) \rightarrow +\infty$ as well. By (H3), for every fixed u , $\pi(\eta^*(t) + u/\sqrt{t}) \rightarrow \pi(\eta^*(t))$ in the sense that $\pi(\eta^*(t) + u/\sqrt{t})/\pi(\eta^*(t)) \rightarrow 1$ uniformly on compact sets of u ; outside a large compact set the Gaussian factor $e^{-u^2/2}$ is exponentially small and π is bounded, so the tails contribute $o(1)$ relative to the Gaussian mass. Dominated convergence therefore yields

$$\int e^{-u^2/2} \pi(\eta^*(t) + u/\sqrt{t}) \, du \sim \pi(\eta^*(t)) \int_{\mathbb{R}} e^{-u^2/2} \, du = \pi(\eta^*(t)) \sqrt{2\pi}$$

Multiplying by $e^{h(t)}/\sqrt{t}$ gives the claimed asymptotic. $\square$

Corollary 10.2 (Asymptotic equalizer density). *The density that asymptotically equalizes the mixture wealth along $S_t = b(t)$ satisfies*

$$\pi(\eta^*(t)) \asymp \sqrt{t} e^{-h(t)}$$

10.2 Intrinsic-time version of the equalizer density

The argument above used the Gaussian surrogate $\eta^2 t/2$. The same calculation in intrinsic time, with $K_t(\eta)$ in place of $\eta^2 t/2$, yields a sharper statement for general sub- ψ processes. For a prescribed boundary $b(t)$, evaluate Z_t along $\{S_t = b(t)\}$ by Laplace's method on the exponent $f_t(\eta) = \eta b(t) - K_t(\eta)$. The saddlepoint equation is

$$K'_t(\eta^*(t)) = b(t) \tag{11}$$

and the Laplace expansion at $\eta^*(t)$ gives

$$Z_t|_{S_t=b(t)} = \pi(\eta^*(t)) \cdot \sqrt{\frac{2\pi}{K''_t(\eta^*(t))}} \cdot \exp(\eta^*(t) b(t) - K_t(\eta^*(t))) \cdot (1 + o(1)) \tag{12}$$

where the $o(1)$ remainder is uniform under continuity of π at the saddle and curvature growth $K''_t(\eta^*) \rightarrow \infty$. The equalizer condition $Z_t|_{S_t=b(t)} = C$ becomes

$$\pi(\eta^*(t)) = C' \cdot \sqrt{K''_t(\eta^*(t))} \cdot e^{-h_t}, \quad C' = C/\sqrt{2\pi} \tag{13}$$

where $h_t := K_t^*(b(t)) = \sup_{\eta} [\eta b(t) - K_t(\eta)]$ is the Cramér transform of the increments evaluated at the boundary. This is the universal equalizer-density formula in intrinsic time, valid for any sub- ψ process whose CGF is finite on a neighborhood of zero.

Corollary 10.3 (Variance relaxation). *For sub-Gaussian increments with the bound $\psi_t(\eta) \leq \eta^2/2$, $K''_t(\eta) = t$ identically and the equalizer (13) specializes to $\pi(\eta^*(t)) = C' \sqrt{t} e^{-h(t)}$ with $h(t) = b(t)^2/(2t)$, matching Corollary 10.2. The boundary $b(t) = \sqrt{2t} h(t)$ produces the LIL rate $\sqrt{2t} \log \log t$ when $h(t)$ tracks $\log \log t$.*

Remark 10.4 (The intrinsic-time view of the boundary). The role of “ t ” in $\sqrt{2t \log \log t}$ is played here by $K''_t(\eta^*(t))$, the curvature of the cumulant generating function at the saddle. For sub-Gaussian increments, $K''_t(\eta) = t$ identically and

the curvature is just the time elapsed; for general martingales, $K_t''(\eta^*)$ is the natural intrinsic clock at the saddle. We will see that the rate of LIL boundaries is universally $\sqrt{2K_t''(\eta^*) \log \log K_t''(\eta^*)}$ in this clock.

11 The Erdős integral test as normalizability

11.1 Change of variables

Write $b(t) = \sqrt{2t h(t)}$ with $\eta^*(t) = \sqrt{2h(t)/t}$. Differentiating:

$$|(\eta^*)'(t)| \sim \frac{\eta^*(t)}{2t} = \frac{\sqrt{h(t)}}{\sqrt{2} t^{3/2}}$$

Change variables from η to t in the normalization integral. Substituting $\pi(\eta^*(t)) \asymp \sqrt{t} e^{-h(t)}$:

$$\int_0^{\eta_{\max}} \pi(\eta) d\eta \asymp \int_2^\infty \frac{\sqrt{h(t)}}{t} e^{-h(t)} dt \quad (14)$$

In words: the change of variable $\eta \leftrightarrow t$ on the saddlepoint trajectory turns the normalization integral over the prior into a one-dimensional integral over time. The next theorem says: a valid Learner strategy for boundary b exists if and only if this time integral converges.

Theorem 11.1 (Normalizability criterion). *Under the assumptions of Lemma 10.1, an asymptotic equalizer density for boundary $b(t) = \sqrt{2t h(t)}$ is normalizable if and only if*

$$\int_2^\infty \frac{\sqrt{h(t)}}{t} e^{-h(t)} dt < \infty$$

Proof. Differentiate $\eta^*(t) = \sqrt{2h(t)/t}$. The assumption $th'(t)/h(t) \rightarrow 0$ gives $|(\eta^*)'(t)| \sim \eta^*(t)/(2t) = \sqrt{h(t)}/(\sqrt{2} t^{3/2})$. Substituting $\pi(\eta^*(t)) \asymp \sqrt{t} e^{-h(t)}$ and $|d\eta| \asymp \sqrt{h(t)} t^{-3/2} dt$ into $\int \pi(\eta) d\eta$ yields the t -integral (14). $\square$

11.2 Connection to the classical integral test

The integral (14) is exactly the Erdős–Kolmogorov integral. The classical result states:

Theorem 11.2 (Erdős–Kolmogorov–Feller integral test, [7, 8]). *Let $(S_t)_{t \geq 1}$ be a random walk with i.i.d. mean-zero, unit-variance increments, and let $b(t) = \sqrt{2t h(t)}$ for $h : [1, \infty) \rightarrow (0, \infty)$ eventually nondecreasing and slowly varying in the sense that $th'(t)/h(t) \rightarrow 0$. Then*

$$\mathbb{P}(S_t > b(t) \text{ i.o.}) = \begin{cases} 0 & \text{if } \int_2^\infty \frac{\sqrt{h(t)}}{t} e^{-h(t)} dt < \infty, \\ 1 & \text{if it diverges.} \end{cases}$$

The dichotomy extends to martingale difference sequences with uniformly bounded increments via Strassen-type invariance and martingale strong-approximation results [26, 28], which lift the Komlós–Major–Tusnády iid approximation [18] to dependent settings (after an overall variance rescaling of t if the conditional variances are not unit).

Scope of the martingale extension. The game-theoretic converse stated in Corollary 11.3 below uses only the *oblivious-strategy* direction of Theorem 11.2 — namely that no normalizable oblivious prior can keep Z_t above $1/\alpha$ along $S_t = b(t)$ when the equalizer prior is non-normalizable. This direction does not require the iid $\rightarrow$ martingale strong-approximation

passage; it is a direct consequence of the Laplace-equalizer Lemma 10.1 combined with normalizability (Theorem 11.1). The martingale strengthening of Theorem 11.2 is invoked only when one wants to translate “no oblivious strategy” into a probabilistic statement about i.o.-crossings of an arbitrary bounded-increment martingale, as stated explicitly in Corollary 12.1(ii).

The game-theoretic reading is now immediate: *the integral converges if and only if the equalizer prior is normalizable.*

Corollary 11.3 (Game-theoretic interpretation). *The boundary $b(t) = \sqrt{2t h(t)}$ is achievable by the Learner if and only if the Erdős–Kolmogorov integral converges. Equivalently, the value of the oblivious detection game at boundary b is finite if and only if the equalizer density for b has finite total mass.*

Proof. Combine three equivalences. By Proposition 4.3, boundary $b(t)$ is achievable by some oblivious Learner strategy iff the asymptotic equalizer density π for b is normalizable. By Theorem 11.1, normalizability of π is equivalent to convergence of the Erdős integral $\int_0^\infty t^{-1} \sqrt{h(t)} e^{-h(t)} dt$. Composing the two, achievability $\Leftrightarrow$ Erdős-integral convergence. The contrapositive is the conventional statement: divergence $\Leftrightarrow$ no normalizable oblivious strategy $\Leftrightarrow$ Nature can force boundary crossings, which by Theorem 11.2 is also $\Leftrightarrow \mathbb{P}(S_t > b(t) \text{ i.o.}) = 1$ in the i.i.d. setting (and the martingale extension noted in Theorem 11.2 for bounded increments). $\square$

11.3 The multiply iterated logarithms as the normalizability threshold

Write $\log_k$ for the k -fold iterated logarithm: $\log_1 t = \log t$, $\log_2 t = \log \log t$, etc.

The LIL boundary is critical. At $h(t) = \log_2 t$:

$$\int_0^\infty \frac{\sqrt{\log_2 t}}{t \log t} dt \geq \int_0^\infty \frac{dt}{t \log t} = \infty$$

The equalizer prior has infinite mass. Nature crosses $b(t) = \sqrt{2t \log_2 t}$ infinitely often.

The full Erdős hierarchy. The boundary

$$h(t) = \log_2 t + \log_3 t + \cdots + \log_{k-1} t + (1 + \varepsilon) \log_k t$$

gives

$$\int_0^\infty \frac{dt}{t \log t \log_2 t \cdots \log_{k-1} t \cdot (\log_k t)^{1/2+\varepsilon}}$$

which converges if and only if $\varepsilon > 0$. Each iterated logarithm absorbs one more layer of the integral’s divergence.

Remark 11.4. The game interpretation is vivid. Nature can always choose to “wait” until late times at which a given iterated logarithm $\log_k t$ is large, thereby exploiting the Learner’s prior at those times. The Learner must spread prior mass to cover these late-time contingencies. Each additional $\log_k$ correction is the cost of hedging against Nature’s ability to wait until the $(k-1)$ -st iterated logarithm itself grows large. The hierarchy terminates (the prior becomes normalizable) precisely when a factor of $(1 + \varepsilon)$ is inserted, giving the Learner enough surplus mass to cover all remaining delays.

12 The 3/2 correction

The Erdős hierarchy of Section 11 shows that each iterated logarithm absorbs one layer of integral divergence. But the *first* correction beyond $\sqrt{2t \log \log t}$ has a sharp coefficient that is not immediately obvious: it is 3/2, not 1. The shift arises because the $\sqrt{h(t)}$ factor in the numerator of the Erdős integral contributes a half-power of $\log \log t$.

In words: the smallest constant c for which $h(t) = \log \log t + c \log \log \log t$ produces an achievable boundary is $c = 3/2$, not

$c = 1$. The extra half on top of the 1 from the classical Erdős baseline is the cost of the Gaussian fluctuation envelope around the saddlepoint — Remark 12.2 reads it geometrically as $3/2 = 1 + 1/2$.

Corollary 12.1 (The sharp $3/2$ threshold). *Let $h(t) = \log \log t + c \log \log \log t + O(1)$. The normalizability criterion of Theorem 11.1 holds if and only if $c > 3/2$. Correspondingly:*

- (i) *for $c > 3/2$, the equalizer prior for $b(t) = \sqrt{2th(t)}$ has finite mass, so a valid oblivious Learner strategy exists and $\mathbb{P}(S_t > b(t) \text{ i.o.}) = 0$;*
- (ii) *for $c \leq 3/2$, the equalizer prior has infinite mass and Theorem 11.2 gives $\mathbb{P}(S_t > b(t) \text{ i.o.}) = 1$.*

Proof. Write $h(t) = \log \log t + c \log \log \log t + O(1)$. The decay factor is an *exact* product, not merely an order relation: ignoring the $O(1)$ term,

$$e^{-h(t)} = e^{-\log \log t} \cdot e^{-c \log \log \log t} = \frac{1}{\log t} \cdot \frac{1}{(\log \log t)^c},$$

an equality that holds verbatim and carries the entire convergence/divergence behavior. The only genuinely asymptotic factor is the Laplace numerator $\sqrt{h(t)} = \sqrt{\log \log t} (1 + o(1))$, since $h(t)/\log \log t = 1 + c \log \log \log t / \log \log t + o(1) \rightarrow 1$. Multiplying the exact decay by this numerator, the Erdős integrand $\sqrt{h(t)} e^{-h(t)}/t$ satisfies

$$\frac{\sqrt{h(t)} e^{-h(t)}}{t} = \frac{\sqrt{\log \log t} (1 + o(1))}{t \log t (\log \log t)^c} = \frac{1 + o(1)}{t \log t (\log \log t)^{c-1/2}}$$

Substitute $u = \log \log t$, so $du = dt/(t \log t)$ and the integral over $t \geq e^e$ becomes $\int_1^\infty u^{-(c-1/2)} (1 + o(1)) du$, which converges if and only if $c - 1/2 > 1$, i.e., $c > 3/2$ (the $1 + o(1)$ multiplicative factor and the $O(1)$ additive term in h each perturb the integrand by a bounded factor, which cannot change convergence). The exponent $c - 1/2$ is thus produced by exactly one half-power of $\log \log t$ in the numerator, the signature of the Laplace envelope. Statements (i)–(ii) then follow by combining Theorem 11.1 (normalizability $\Leftrightarrow$ integral convergence), Proposition 4.3 (normalizability $\Leftrightarrow$ existence of a valid oblivious Learner strategy), and Theorem 11.2 (divergence of the Erdős integral $\Leftrightarrow$ infinite crossings). $\square$

Remark 12.2. The decomposition $3/2 = 1 + 1/2$ traces to two factors of distinct character, one exact and one asymptotic. The Erdős baseline 1 is the *exact* decay $e^{-h(t)} = (\log t)^{-1} (\log \log t)^{-c}$ together with the exact measure $dt/(t \log t) = du$ under $u = \log \log t$; this produces the bare $\int u^{-c} du$ that diverges at $c = 1$. The extra $1/2$ is the *asymptotic* Laplace numerator $\sqrt{h(t)} = \sqrt{\log \log t} (1 + o(1))$, which contributes exactly one half-power of $\log \log t$ and shifts the divergence threshold from $c = 1$ to $c = 3/2$. The numerator itself is the product of the curvature prefactor $\sqrt{t}$ in the asymptotic equalizer density $\pi(\eta^*(t)) \asymp \sqrt{t} e^{-h(t)}$ (Corollary 10.2) and the Jacobian $|d\eta| \asymp \sqrt{h(t)} t^{-3/2} dt$ of the change of variable $t \leftrightarrow \eta^*(t)$, whose product is $\sqrt{h(t)}/t$. This is the geometric reason the first iterated-logarithm correction cannot be tuned away: it is the cost of integrating the Gaussian-fluctuation envelope of the saddlepoint along the boundary, and no normalization of the prior can absorb it. In the $1 + 1/2$ split, the 1 is what is fixed by the exact decay of the equalizer mass and the $1/2$ is the irreducible price of the saddle’s Gaussian width.

IT explanation of the $1 + 1/2$ split. In the pathwise IT identity (2), the prior-to-posterior KL cost $\text{KL}(\pi_t \parallel \Pi)$ along the boundary has two leading-order contributions:

- the *direct* cost of allocating mass at scale $\eta^*(t) \sim t^{-1/2} \sqrt{\log_2 t}$, which scales as the prior tax $L(\eta^*) \sim \frac{1}{2} \log t$ — this is the Erdős “1” baseline;
- the *curvature* cost of localizing the posterior to a saddle width $\sim 1/\sqrt{K_t''(\eta^*)} = 1/\sqrt{t}$, contributing an additional $\frac{1}{2} \log K_t''(\eta^*) \sim \frac{1}{2} \log t$ via the Laplace prefactor — this is the “ $1/2$ ” Laplace envelope.

The total KL budget at the LIL boundary therefore grows as $\log_2 t + \frac{3}{2} \log_3 t + O(1)$, exactly the rate of Corollary 12.1.

Remark 12.3. The law of the iterated logarithm is the $c = 0$ case: the boundary $\sqrt{2t \log \log t}$ is crossed infinitely often. The first correction that makes it finitely crossable requires adding $\frac{3}{2} \log \log \log t$ inside the square root. The boundary

$$b(t) = \sqrt{2t(\log \log t + \frac{3}{2} \log \log \log t + O(1))}$$

is the sharpest achievable detection boundary.

Corollary 12.4 (The $3/2$ threshold extends to the Bernstein sub-exponential family). *Let the increments have a Bernstein-type CGF $\psi_c(\eta) = \eta^2/(2(1 - c\eta))$ for some $c \geq 0$ (with $c = 0$ recovering the sub-Gaussian case). Let $h(t) = \log \log t + c' \log \log \log t + O(1)$. Then the normalizability criterion of Theorem 11.1 – and hence the conclusion of Corollary 12.1 – continues to hold with the same threshold $c' = 3/2$, independently of the Bernstein parameter c .*

Proof sketch. The saddlepoint equation $K'_t(\eta) = b(t)$ with the Bernstein CGF yields an admissible saddle $\eta^*(t)$ that lies inside the domain $\eta < 1/c$ at all sufficiently large t along the LIL boundary $b(t) = \sqrt{2th(t)}$, because $\eta^*(t) = O(\sqrt{\log \log t/t}) \rightarrow 0$. The Laplace prefactor along the boundary acquires a multiplicative correction $(1 - c\eta^*(t))^{-3/2}$ relative to the Gaussian case. At the LIL saddle this factor equals $1 + \frac{3}{2}c\eta^*(t) + O((c\eta^*)^2) = 1 + O(c\sqrt{\log \log t/t}) \rightarrow 1$ as $t \rightarrow \infty$. The asymptotic equivalence $e^{-h(t)} \asymp 1/(\log t \cdot (\log \log t)^{c'})$ used in the proof of Corollary 12.1 is therefore preserved up to a $1 + o(1)$ multiplicative factor, which cannot change convergence of the Erdős integral. The threshold $c' = 3/2$ is thus universal across the Bernstein sub-exponential family. The same argument applies to any CGF whose saddlepoint correction along the LIL boundary is $1 + o(1)$. $\square$

Remark 12.5. The empirical sweep of E00352 (Section 21.8) confirms that the convergence threshold of the Erdős integral is c -independent at $3/2$ across $c \in \{0, 0.1, 0.5, 1.0\}$ to within the linear-fit slope -3×10^{-5} , i.e. effectively zero. The Bernstein correction factor $(1 - c\eta^*)^{-3/2}$ at the LIL saddle deviates from 1 by at most 2×10^{-7} at $t = 10^{15}$ even for $c = 1$ (the saddle scale $\eta^*(t)$ is of order 10^{-7} at that horizon, so the correction is linear-in- η^* to first order). The corollary extends the principal result of Section 12 from the sub-Gaussian regime to any sub-exponential regime whose saddlepoint geometry is asymptotically Gaussian along the LIL boundary – a substantially broader class than “sub-Gaussian” alone, including in particular Bernstein-tail random variables with bounded variance and one-sided exponential tails.

12.1 The full Erdős hierarchy: $(3/2, 1, 1, \dots)$

Corollary 12.1 pins down the sharp first iterated-log correction. The natural next question is what the sharp constant is at the second iterated-log layer, the third, and beyond. A natural conjecture, mentioned in some preliminary versions of the present argument and recorded in the empirical-evaluation supplement fragment for E00347 (Section 21.3), is that the Laplace “ $+1/2$ ” propagates to every layer, giving a hierarchy of thresholds $(3/2, 5/2, 5/2, \dots)$ or, more cautiously, $(3/2, 3/2, 3/2, \dots)$. The following theorem replaces both conjectures with the correct constants.

Theorem 12.6 (Higher-order Erdős thresholds). *For $k \geq 2$, let*

$$h_k(t) = \log_2 t + \frac{3}{2} \log_3 t + \sum_{j=4}^k \log_j t + c_k \log_{k+1} t + O(1)$$

where $\log_j$ denotes the j -fold iterated logarithm. Under the sub-Gaussian (or Bernstein, by Corollary 12.4) CGF, the equalizer prior for $b(t) = \sqrt{2t h_k(t)}$ is normalizable if and only if $c_k > 1$. Equivalently, the hierarchy of sharp iterated-log thresholds is

$$(c_1^*, c_2^*, c_3^*, \dots) = (\frac{3}{2}, 1, 1, \dots)$$

Proof sketch. The case $k = 1$ is Corollary 12.1. For $k \geq 2$, the Erdős integrand $\sqrt{h_k(t)} e^{-h_k(t)}/t$ admits a k -step substitution chain that reduces it to a clean monomial in the deepest iterated logarithm. Concretely, set $u_1 = \log_2 t$, $u_2 = \log u_1$, ..., $u_{k-1} = \log u_{k-2}$; each substitution converts $dt/(t \log t \log_2 t \cdots \log_j t)$ into du_j/u_j . As in Corollary 12.1, the decay $e^{-h_k(t)}$ factorizes exactly into a product of monomials, one $u_j^{-(\text{coefficient of } \log_{j+1} t)}$ per layer; only the Laplace prefactor

$\sqrt{h_k(t)} = \sqrt{u_1} (1 + o(1))$ is asymptotic. The factor $\sqrt{u_1}$ combines with the exact $u_1^{-3/2}$ from $e^{-(3/2)\log u_1}$ in $e^{-h_k(t)}$ to produce $u_1^{-1} (1 + o(1))$; the u_1^{-1} then factors against du_1/u_1 to give a clean du_2 on the second layer, and the residual factor at level $k-1$ is exactly $u_{k-1}^{-c_k} du_{k-1}$ up to a $1 + o(1)$ factor. Convergence of $\int u_{k-1}^{-c_k} du_{k-1}$ requires $c_k > 1$. The Laplace half-Gaussian “+1/2” that produced the 3/2 at the first layer appears *exactly once* – in the $\sqrt{h_k(t)}$ Jacobian – and is consumed by the $u_1^{-3/2}$ factor at the first substitution; it does not propagate.

The full chain is recorded with a numerical check in Section 21.3 (E00347): the empirical convergence bracket sits in (1.0, 1.05) at the second iterated-log layer ($k = 2$), midpoint 1.025, distance 0.025 from the analytic prediction $c_2^* = 1$. Cauchy condensation at $T = 10^{120}$ agrees on the convergence/divergence partition above $c_2 = 1$. $\square$

Remark 12.7. Two readings of the hierarchy $(3/2, 1, 1, \dots)$ are useful. *Geometric reading.* The 1/2 shift at the first layer is the cost of integrating the Gaussian fluctuation envelope around the saddlepoint; once the boundary already includes the $\frac{3}{2} \log_3 t$ correction, the saddlepoint is again well-localized at every deeper iterated-log layer, and the Laplace envelope is paid for once and for all. *Game-theoretic reading.* The Learner’s hedging cost is 3/2 at the first layer because hedging across a continuum of Gaussian saddlepoint widths costs an extra half. At every subsequent iterated-log layer the prior is already a discrete-shell mixture (one shell per layer), and the cost reduces to a clean unit per layer: each iterated logarithm absorbs exactly one decade of integral divergence, with no further envelope penalty.

13 The Jeffreys prior on iterated-log scale

The chart $\mu = \log \log(1/\lambda)$ is the canonical coordinate for the scale parameter of the scale family of exponential tilts: it is the affine logarithmic chart of the scale-of-scale group action. Under this chart, the equalizer is a particularly clean memoryless law, and the constant $\sqrt{2}$ in the LIL boundary acquires a transparent unit-of-measurement interpretation.

13.1 Pushforward calculation

Lemma 13.1 (Pushforward of π^* under $\mu = \log \log(1/\lambda)$). *Let $\lambda \in (0, e^{-2}]$ and define $\mu(\lambda) = \log \log(1/\lambda) \in [\log 2, \infty)$. The Balsubramani equalizer density $\pi^*(\lambda) = 2/(\lambda \log^2(1/\lambda))$ pushes forward to the rate-1 shifted-exponential density*

$$\tilde{\pi}(\mu) = 2e^{-\mu} \quad \text{on } [\log 2, \infty), \quad (15)$$

i.e. if $X \sim \pi^$ then $\mu(X) - \log 2 \sim \text{Exp}(1)$. The corresponding equalizer CDF in the μ -chart is $\tilde{F}(\mu) = 1 - 2e^{-\mu}$ on $[\log 2, \infty)$.*

Proof. $\mu = \log \log(1/\lambda)$ implies $\log(1/\lambda) = e^\mu$ and $\lambda = \exp(-e^\mu)$. Differentiating: $d\lambda = -\exp(-e^\mu) e^\mu d\mu = -\lambda \log(1/\lambda) d\mu$. Hence $|d\lambda| = \lambda \log(1/\lambda) |d\mu|$. The pushforward density is

$$\tilde{\pi}(\mu) = \pi^*(\lambda(\mu)) \left| \frac{d\lambda}{d\mu} \right| = \frac{2}{\lambda \log^2(1/\lambda)} \cdot \lambda \log(1/\lambda) = \frac{2}{\log(1/\lambda)} = \frac{2}{e^\mu} = 2e^{-\mu}$$

The CDF: $\tilde{F}(\mu) = \mathbb{P}(\mu(X) \leq \mu) = \mathbb{P}(X \geq \lambda(\mu)) = 1 - F^*(\lambda(\mu)) = 1 - 2/\log(1/\lambda(\mu)) = 1 - 2e^{-\mu}$. At $\mu = \log 2$: $1 - 2e^{-\log 2} = 0$. At $\mu \rightarrow \infty$: 1. The shifted-exponential identity is $\tilde{F}(\mu) = 1 - e^{-(\mu - \log 2)}$. $\square$

13.2 The canonical chart for the scale-of-scale group action

Why is $\mu = \log \log(1/\lambda)$ “the right” chart, and why is the exponential rate exactly 1?

Scale of the scale family. The exponential tilts $\{M_t^\eta : \eta > 0\}$ form a scale family: under $\eta \mapsto c\eta$, $S_t \mapsto S_t/c$, $K_t(\eta) \mapsto K_t(c\eta)$, the wealth process is invariant. The right-Haar measure of the multiplicative group on $(0, \infty)$ is $d\eta/\eta$,

equivalently Lebesgue measure on $\mu_1 := \log(1/\eta)$. This is the *first-level* Jeffreys prior, and it has *infinite* mass on $(0, \eta_{\max}]$ — failing to be a probability measure.

The equalizer is the “gentlest normalizable correction” to $d\eta/\eta$:

$$\pi^*(\eta) d\eta = \frac{2}{\log^2(1/\eta)} \frac{d\eta}{\eta} = \frac{2}{\mu_1^2} d\mu_1$$

In the first-level chart μ_1 , the equalizer has density $2/\mu_1^2$ — a quadratic decay.

Scale of the scale of the scale family (one level deeper). Pass to the second-level chart $\mu_2 := \log \mu_1 = \log \log(1/\eta)$. The right-Haar measure of the multiplicative group on $(0, \infty)$ acting on μ_1 is $d\mu_1/\mu_1 = d\mu_2$, Lebesgue measure on μ_2 . This is the *second-level* Jeffreys prior — still infinite mass on $\mu_2 \in [\log 2, \infty)$. By Lemma 13.1, the equalizer in the second-level chart is $\tilde{\pi}(\mu_2) d\mu_2 = 2 e^{-\mu_2} d\mu_2$, which is the *gentlest exponentially-decaying correction* to Lebesgue $d\mu_2$ that gives finite total mass. Quadratic decay $2/\mu_2^2$ in the second-level chart would also be normalizable, but the saddlepoint relation $\eta^*(t) = b(t)/t$ in the LIL boundary $b(t) = \sqrt{2t \log \log t}$ precisely picks out the exponential rate, not the polynomial.

Theorem 13.2 (Equalizer is the rate-1 shifted exponential on the iterated-log chart). *Under the second-level chart $\mu = \log \log(1/\lambda)$, the Balsubramani equalizer is the rate-1 shifted exponential $\tilde{\pi}(\mu) = 2 e^{-\mu}$ on $[\log 2, \infty)$, equivalently $\mu - \log 2 \sim \text{Exp}(1)$. The first-level Jeffreys prior $d\eta/\eta$ pushes forward to Lebesgue $d\mu_1$ on $[2, \infty)$ (infinite mass); the second-level Jeffreys prior $d\mu_2$ on $[\log 2, \infty)$ is also infinite mass; the equalizer corrects the second-level Jeffreys to an exponential law with the unique rate that makes the LIL saddlepoint relation $\eta^*(t) = b(t)/t$ hold uniformly in t .*

Proof. The pushforward and CDF are Lemma 13.1. For the rate determination: the saddlepoint relation $\eta^*(t) = b(t)/t$ for the LIL boundary $b(t) = \sqrt{2t \log \log t}$ gives $\eta^*(t) = \sqrt{2 \log \log t}/t$, hence $\log(1/\eta^*(t)) = \frac{1}{2} \log(t/(2 \log \log t)) \sim \frac{1}{2} \log t$ and $\mu_2(\eta^*(t)) \sim \log \log t$. The equalizer must allocate mass at μ_2 proportional to $\sqrt{t} e^{-\log \log t}$ (Equation 13 specialized), which when expressed in the μ_2 -chart density is $\sim e^{-\mu_2}$, fixing the rate to 1. Any other rate in the second-level chart would correspond to a non-LIL boundary (see Section 11 for the full hierarchy). $\square$

13.3 Implications

(a) The Hartman–Wintner $\sqrt{2}$ as an iterated-log unit. The rate-1 exponential law on $[\log 2, \infty)$ has its support starting at $\log 2$; equivalently, the equalizer’s truncation $\eta_{\max} = e^{-2}$ corresponds to $\mu_{\min} = \log 2$. The Hartman–Wintner constant $\sqrt{2\sigma^2} = \sigma\sqrt{2}$ in the LIL is exactly σ times $\sqrt{2}$, the square root of the value $V = 2 = e^{\mu_{\min}}$. In the iterated-log chart, this is the statement that the *measurement unit* of the LIL boundary is the rate of the exponential, fixed at 1 by the saddlepoint relation, with offset $\log 2$ from the IT budget.

(b) Memorylessness = scale-of-scale invariance. The exponential law is memoryless: conditional on $\mu_2 \geq a$, the law of $\mu_2 - a$ is again $\text{Exp}(1)$. Translated to η -coordinates, this is the statement that the equalizer *conditional on being below scale* η_0 is again the equalizer restricted (and renormalized) to that scale. This is the natural form of scale-of-scale invariance for the LIL game: the equalizer treats every iterated-log decade as equivalent, and the prior repeats itself recursively as the scale shrinks.

(c) Why iterated-log appears in $\sqrt{2t \log \log t}$. The rate function $\tilde{\pi}(\mu) = 2e^{-\mu}$ has $\int_{\mu_0}^{\infty} \tilde{\pi} d\mu = 2e^{-\mu_0}$, i.e. the tail decays exponentially in $\mu = \log \log(1/\eta)$. Translated to the boundary’s information cost, the path’s KL budget at μ_0 is $\text{KL}(\tilde{\pi}|_{[\mu_0, \infty)} \|\tilde{\pi}) = 0$ (memorylessness), so the cost structure is recursive in the iterated-log scale. The $\sqrt{2 \log \log t}$ rate is the unique boundary at which this recursive cost structure stays balanced; any sharper boundary would force $\tilde{\pi}$ to be sub-exponential, breaking memorylessness; any softer would allow Nature to extract free boundary-crossings.

(d) The group action that makes μ_2 canonical. The scale-family monoid $((0, \eta_{\max}], \cdot)$ acts on tilts by multiplication; the right Haar of this action is $d\eta/\eta$. Composing with the (inverse) logarithm sends this to translations on $\mu_1 = \log(1/\eta)$: the right Haar is $d\mu_1$. A second composition with $\log$ sends translations on μ_1 to *dilations* on $\mu_2 = \log \mu_1$: the right Haar of the multiplicative group acting on μ_1 , when pulled back to μ_2 , is $d\mu_2$. This is why μ_2 is the canonical coordinate of the “scale of the scale of the scale family”: it is the chart in which the right Haar of the second-order group action is Lebesgue. The equalizer is then the rate-1 exponential correction to $d\mu_2$ on $[\log 2, \infty)$, the unique law that respects this Haar structure modulo a finite-mass correction with universal rate.

Remark 13.3 (A small notational caveat). A tempting simplification is to claim that the equalizer in the μ_2 -chart is *Lebesgue*. This is incorrect: Lebesgue $d\mu_2$ on $[\log 2, \infty)$ has infinite mass and is the second-level Jeffreys prior, not the equalizer. The equalizer is the gentlest exponential correction with rate exactly 1. The two statements are confusingly adjacent — both are “Jeffreys-like” — and the calculation in Lemma 13.1 pins down which is which.

14 GROW-optimal e-values and the equalizer

We show that the equalizer mixture Z_t^* is GROW-optimal in the sense of the growth-rate-optimal e -process criterion [10] against the natural scale-family alternative class. This gives an information-theoretic (rather than purely game-theoretic) derivation of the Balsubramani density.

14.1 GROW recap

Given a null hypothesis $\mathcal{H}_0$ and an alternative class $\mathcal{H}_1$, an e -process (E_t) is a process such that $E_0 = 1$, $E_t \geq 0$, and $\mathbb{E}_{P_0}[E_t] \leq 1$ for every $P_0 \in \mathcal{H}_0$ and every t . The GROW-optimal e -process minimizes the worst-case log-growth deficit relative to the alternative:

$$E_t^{\text{GROW}} = \arg \max_E \inf_{Q \in \mathcal{H}_1} \mathbb{E}_Q[\log E_t] \quad (16)$$

Saddle-point conditions (Theorem 1 of [10]) show that under standard convexity hypotheses, the GROW e -process is a *Bayesian mixture against the worst-case alternative*: there is a least-favorable $Q^* \in \mathcal{H}_1$, and E_t^{GROW} is the likelihood ratio of the path under Q^* vs. a Bayes-optimal $P^* \in \mathcal{H}_0$.

14.2 GROW for the LIL game

Our null is $\mathcal{H}_0 = \{P : \mathbb{E}_P[\xi_t | \mathcal{F}_{t-1}] = 0, \psi_t(\eta) \text{ finite on a neighborhood of } 0\}$, the martingale-difference processes with finite CGF. For the alternative class, take the scale family

$$\mathcal{H}_1^{\text{scale}} = \{Q^\eta : \eta \in (0, \eta_{\max}]\} \quad (17)$$

where Q^η is the law that exponentially tilts the increments at scale η : $dQ^\eta/dP_0 = \exp(\eta S_t - K_t(\eta))$, i.e. Q^η is the law under which M_t^η is the Radon–Nikodym derivative.

Theorem 14.1 (GROW-optimal e -process for LIL = equalizer mixture). *For the null $\mathcal{H}_0$ and alternative class $\mathcal{H}_1^{\text{scale}}$ above, with the saddle-point regularity that $\mathbb{E}_{Q^\eta}[\log Z_t]$ is continuous in η , the GROW-optimal e -process is the equalizer mixture*

$$E_t^{\text{GROW}} = Z_t^* = \int_0^{\eta_{\max}} M_t^\eta \Pi^*(d\eta) \quad (18)$$

where Π^* is the Balsubramani equalizer with density $2/(\eta \log^2(1/\eta))$ on $(0, \eta_{\max}]$, $\eta_{\max} = e^{-2}$. Moreover, the saddle-value of the GROW objective equals $V = 2$, i.e. $\min_{Q \in \mathcal{H}_1^{\text{scale}}} \mathbb{E}_Q[\log E_t^{\text{GROW}}] = \log(1/\alpha)/V$ at the LIL boundary $b(t)$ at level α .

Proof. The proof has two steps: identification of the least-favorable alternative Q^* , and verification that Z_t^* is the Bayes mixture against Q^* .

Step 1: least-favorable alternative. By the GROW saddle-point characterization, the least-favorable $Q^* \in \mathcal{H}_1^{\text{scale}}$ minimizes $\mathbb{E}_Q[\log E_t]$ over Q for the Bayes-optimal E inside the scale class. Setting $E = Z_t^\Pi := \int M_t^\eta \Pi(d\eta)$ for a prior Π to be determined and computing $\mathbb{E}_{Q^{\eta'}}[\log Z_t^\Pi] = \mathbb{E}_{Q^{\eta'}}[\log \int M_t^\eta \Pi(d\eta)]$ under each $Q^{\eta'}$, the saddle-point condition is that $\mathbb{E}_{Q^{\eta'}}[\log Z_t^{\Pi^*}]$ is the same for all η' in the support of Π^* . This is exactly the equalizer condition $Z_t|_{\eta=\eta'} = \text{const}$ along the boundary, but stated in expectation under the alternative rather than along the realized path.

By the pathwise IT identity (2) restricted to a single-scale sample (i.e. $\Pi = \delta_\eta$ in the role of the alternative), the equalizer condition coincides with the saddle-point condition. Inverting the equalizer relation gives $\Pi^*(d\eta) = \pi^*(\eta) d\eta = 2/(\eta \log^2(1/\eta)) d\eta$ on $(0, e^{-2}]$.

Step 2: Bayes optimality. Given Π^* , the Bayes-optimal e -process under the prior Π^* is the mixture wealth $Z_t^{\Pi^*} = \int M_t^\eta \Pi^*(d\eta)$, which equals Z_t^* by construction. By the GROW saddle-point characterization, this is the GROW-optimal e -process.

Saddle value. At the LIL boundary $b(t)$ at level α , $\mathbb{E}_{Q^\eta}[\log Z_t^* | S_t = b(t)]$ is constant in η (equalizer condition under the saddle); using the value of the reduced game $V = 2$ and the IT-budget reading of the truncation, the constant equals $\log(1/\alpha)/V$. $\square$

Remark 14.2 (Two derivations, one prior). Theorem 14.1 is the assertion that the Balsubramani density is overdetermined: it is the unique solution of *two* different optimization problems — the minimax saddle-point of the LIL detection game (an $\sup_\Pi \inf_t$ statement over the realized path), and the GROW objective of safe testing (an $\sup_E \inf_Q$ statement in expectation over the alternative). The pathwise IT identity (2) explains why: both optimizations are special cases of the Gibbs variational formula for $\log Z_t$, the first restricted to a deterministic path and the second to an expectation under a class of alternatives. The same prior solves both because the same identity organizes both.

Corollary 14.3 (The equalizer is the GROW-optimal e -process for LIL detection). *The Balsubramani density $1/(\eta \log^2(1/\eta))$ on $[-e^{-2}, e^{-2}] \setminus \{0\}$ (after sign-symmetrization) is the unique GROW-optimal mixing prior for two-sided LIL detection against the scale-family alternative class. Adaptive betting strategies (Section 19) that track the saddle scale $\eta^*(t)$ in real time approximate the GROW objective with finite-time slack, but cannot improve the leading rate.*

15 The coincidence center as a 2-Wasserstein limit of the equalizer

A finite-dimensional minimax coincidence-center theorem lives on a discrete simplex; the continuum LIL game's equalizer is a measure on $(0, \eta_{\max}]$. The two are linked by an explicit limit in 2-Wasserstein distance.

15.1 Setup

Let W be the alphabet size in the finite-dimensional coincidence game. Place W scales geometrically inside $(0, e^{-2}]$: $\lambda_i^{(W)} = e^{-(2+(i-1)\Delta_W)}$ for $i = 1, \dots, W$, with spacing $\Delta_W \rightarrow 0$ as $W \rightarrow \infty$ chosen so that $W\Delta_W \rightarrow \infty$ (the scales fill out the interval in the $\log(1/\lambda)$ -chart while remaining dense). The discrete weight at $\lambda_i^{(W)}$ is $\alpha_i^{(W)*}$, the coincidence-equalizing weight from the finite-dimensional game (Theorem 5.1).

Theorem 15.1 (Coincidence-center $\rightarrow$ equalizer in W_2). *Let $\mu_W := \sum_{i=1}^W \alpha_i^{(W)*} \delta_{\lambda_i^{(W)}}$ be the discrete coincidence-center measure pushed onto the scale interval, and let π^* be the continuum equalizer $\pi^*(d\lambda) = 2/(\lambda \log^2(1/\lambda)) d\lambda$ on $(0, e^{-2}]$. Under the geometric-spacing choice $\Delta_W = c/\log W$ with constant $c > 0$,*

$$W_2(\mu_W, \pi^*) \longrightarrow 0 \quad \text{as } W \rightarrow \infty, \quad (19)$$

where W_2 is the 2-Wasserstein distance on $(0, e^{-2}]$ equipped with the Euclidean metric. The convergence rate is $W_2(\mu_W, \pi^*) = O((\log W)^{-1})$.

Proof sketch. The guarantee the finite-dimensional game proves, and the one this argument uses, is that its optimum equalizes the active reverse-KL constraints, $\text{KL}(p^* \parallel \pi_w) = R^*$ on the active set (Theorem 5.1). In the scale coordinate this equalizer condition is the continuum identity $F_\nu(\lambda) \log(1/\lambda) = \text{const}$, and its discrete counterpart $\alpha_i^{(W)*} \log(1/\lambda_i^{(W)}) = \text{const}$ holds exactly for the closed-form ansatz below and in the continuum; whether the abstract finite-dimensional optimum additionally takes this scale-product form depends on the map from scales λ_i to priors π_i that the manuscript leaves unpinned, and we use only the reverse-KL equalization the theorem certifies. The equalizer density is $\pi^*(\lambda) = 2/(\lambda \log^2(1/\lambda))$, so the geometric shell of width $d\lambda_i = \lambda_i \Delta_W$ carries equalizer mass $\pi^*(\lambda_i^{(W)}) d\lambda_i = 2\Delta_W / \log^2(1/\lambda_i^{(W)})$; discretizing by this density shell mass gives $\alpha_i^{(W)*} \propto 1/\log^2(1/\lambda_i^{(W)})$, which approximates the equalizer density $2/(\eta L(\eta)^2)$ at $\eta = \lambda_i^{(W)}$ up to discretization error. (The cumulative-mass value $1/\log(1/\lambda_i^{(W)}) = F^*(\lambda_i^{(W)})/2$ carries one fewer inverse-logarithm and does not converge; the density shell mass is the right weight.) The 2-Wasserstein distance between a discrete measure on a geometrically spaced grid and its continuum limit is bounded by the spacing of the grid in the chart that flattens the density. By Theorem 13.2, that chart is $\mu_2 = \log \log(1/\lambda)$, in which the equalizer is the rate-1 exponential. Geometric spacing in λ corresponds to arithmetic spacing in $\mu_1 = \log(1/\lambda)$, but to logarithmic spacing in μ_2 . The discretization error in the μ_2 -chart is $O(\Delta_W / \log \lambda_i^{(W)}) = O(1/(\log W \cdot \log \log W))$ for the geometrically-spaced grid; integrating against $\tilde{\pi}(\mu_2) = 2e^{-\mu_2}$ gives total W_2 error $O((\log W)^{-1})$. For full details we reduce to the standard Wasserstein-quantization estimate [9]; the equalizer’s bounded second moment on the chart μ_2 ensures applicability. $\square$

Remark 15.2 (The two coordinate systems). Theorem 15.1 formalizes the “two coordinate systems” remark in the finite-dimensional development: the finite-dimensional coincidence game and the continuum scale-allocation game describe a single equilibrium object viewed through different coordinates. The finite-dimensional version is a point in a discrete simplex; the continuum version is a measure on the scale interval; 2-Wasserstein convergence is the natural topology that connects them.

The role of the scaling $1/W$ for per-letter weights is to keep the total mass invariant across W : a finite-dimensional coincidence center with weights summing to 1 pushes forward to a discrete probability measure with the same total mass, and the limit is a continuum probability measure.

Remark 15.3 (Domain of validity). The dual reading “LIL game = continuum coincidence game” is now quantitatively justified: the convergence rate $W_2 = O((\log W)^{-1})$ tells us that the coincidence reading is sharp for large alphabets but is genuinely *slow* (logarithmic in W). This explains why the dual reading is presented as a structural analogy rather than as a finite- W approximation: the approximation gets precise only at very large W . For pedagogical purposes, the continuum equalizer should typically be derived directly from the continuum game; the finite-dimensional reading is a structural anchor, not a computational shortcut. The finite-dimensional premise this argument rests on — that the coincidence game’s optimum equalizes the active reverse-KL constraints, the discrete analogue of the continuum equalizer condition — is verified directly at the game’s optimum in Section 21.12.

16 The shell-truncation tool: finite- α confidence sequences

The discrete shell game (Theorem 7.1) is a finite combinatorial dual of the continuum game. Truncating the shell sum at finitely many shells produces, for any target level $\alpha \in (0, 1)$, an exact finite- α anytime-valid confidence sequence whose constants are tractable by hand.

16.1 The truncation tool

Proposition 16.1 (Shell-truncation confidence sequence). *Fix $\alpha \in (0, 1)$. Set $j_{\max} = j_{\max}(\alpha) = \lceil 2/\alpha \rceil$. Truncate the shell prior to shells $j \in \{2, 3, \dots, j_{\max}\}$ with weights*

$$w_j^{\text{trunc}} = \frac{2}{j(j+1)} \quad \text{for } 2 \leq j \leq j_{\max}, \quad (20)$$

and renormalize to $\tilde{w}_j = w_j^{\text{trunc}}/W^{(\alpha)}$ where $W^{(\alpha)} = \sum_{j=2}^{j_{\max}} 2/(j(j+1)) = 1 - 2/(j_{\max} + 1)$. The corresponding mixture wealth process $Z_t^{\text{trunc}} = \sum_{j=2}^{j_{\max}} \tilde{w}_j \int_{\mathcal{S}_j} M_t^\eta \frac{d\eta}{|\mathcal{S}_j|}$ is a nonnegative supermartingale with $Z_0 = 1$, and it produces an α -level anytime-valid confidence sequence with explicit boundary

$$b^{\text{trunc}}(t; \alpha) = \min_{2 \leq j \leq j_{\max}} \sqrt{2t(\log(1/\tilde{w}_j) + \log(1/\alpha) + j + \log \log(1/\alpha))} \quad (21)$$

in the sub-Gaussian regime $K_t(\eta) \leq \eta^2 t/2$. For each fixed $\alpha \in (0, 1)$, the constants in (21) are computable by hand (rational arithmetic involving the integers $2, 3, \dots, j_{\max}$).

Proof. Each shell-bet $\int_{\mathcal{S}_j} M_t^\eta \frac{d\eta}{|\mathcal{S}_j|}$ is a nonnegative supermartingale (mixture of single-scale supermartingales), so the convex combination Z_t^{trunc} is also a supermartingale; nonnegativity and $Z_0 = 1$ are immediate. Ville's inequality gives $\mathbb{P}(\exists t : Z_t^{\text{trunc}} \geq 1/\alpha) \leq \alpha$. The boundary (21) is obtained by saddle-point evaluation of each shell's integral in the LIL regime $\eta^*(t) = b(t)/t$: at the saddle in shell j , $\log(1/\eta^*) \in (j, j+1]$, so the cumulative-CDF tax at the saddle is j , contributing the j term inside the square root. The $\log(1/\tilde{w}_j)$ is the union-bound penalty over shells; the $\log(1/\alpha)$ is the Ville penalty; the $\log \log(1/\alpha)$ is the saddle-curvature constant. Taking $j_{\max} = \lceil 2/\alpha \rceil$ ensures $W^{(\alpha)} \geq 1 - \alpha$, so the renormalization factor only inflates each $\tilde{w}_j$ by a bounded amount as α shrinks. $\square$

16.2 Worked numerical example

Take $\alpha = 0.05$, so $j_{\max} = \lceil 40 \rceil = 40$, $W^{(0.05)} = 1 - 2/41 = 39/41 \approx 0.9512$. Each $\tilde{w}_j = (2/(j(j+1)))/(39/41) = 82/(41j(j+1))$. The boundary is $b^{\text{trunc}}(t; 0.05) = \min_{2 \leq j \leq 40} \sqrt{2t(\log(41j(j+1)/82) + \log 20 + j + \log \log 20)}$. The argmin in j is the saddle $j^*(t) \approx \log(t/h(t))$, which for $t = 10^4$ is approximately $j^* \approx 7$, giving boundary $\approx \sqrt{2 \cdot 10^4 \cdot (2.30 + 3.0 + 7 + 1.10)} = \sqrt{2 \cdot 10^4 \cdot 13.4} \approx 518$. This compares to the classical $\sqrt{2 \cdot 10^4 \log \log 10^4} \approx \sqrt{4.4 \cdot 10^4} \approx 210$ and to the finite-time Balsebramani boundary $\sqrt{2t(\log \log t + \frac{3}{2} \log \log \log t + \log(2/0.05))} \approx 350$. The shell-truncation boundary is therefore loose by a factor of ≈ 1.5 relative to the continuum equalizer at finite time, but its constants are explicit and exact.

16.3 Why this is useful

Pedagogical. The shell-truncation boundary (21) contains no Laplace-method asymptotics, no Erdős-integral, no continuum-limit arguments. It is built entirely from finite arithmetic over the shell indices j . It can be derived in a self-contained classroom lecture in under an hour and gives a fully rigorous finite-time LIL confidence sequence at any level α .

Sanity check. The constants in Balsebramani's continuum construction [1] can be cross-validated by the shell-truncation boundary: numerically, $b^{\text{trunc}}/b^{\text{Balsebramani}}$ should converge to a limit between 1 and ~ 1.5 as $t \rightarrow \infty$ (the gap reflects the truncation slack and the union-bound penalty over shells). This provides an independent numerical check on the explicit constants of [1].

Tight at large $j_{\max}$. As $\alpha \rightarrow 0$, $j_{\max} \rightarrow \infty$ and the truncation effect vanishes; the shell-truncation boundary recovers the continuum boundary at leading order, with the union-bound penalty $\log(1/\tilde{w}_j)$ approaching $2 \log j$ at the saddle, producing the canonical $\log_3 t$ correction at the next order.

17 The LIL game equals the concentration game

The LIL game and a CGF-constrained repeated game over KL balls are the same object viewed through different operations: the LIL game is the exponentiated, null-restricted shadow of the latter, with the restriction $\rho = \delta_{\text{sign}(S_t)}$ being the only LIL-specific datum.

17.1 The CGF-constrained repeated game (abridged)

In the CGF-constrained repeated game, a Learner plays a sequence of beliefs $p_t \in \Delta_K$ on a finite alphabet, Nature plays losses c_t subject to a CGF budget constraint, and the exact one-round identity

$$\langle p_t - \rho, c_t \rangle = \eta_t Q_t(c_t; p_t, \eta_t) + \frac{\text{KL}(\rho \| p_t) - \text{KL}(\rho \| \tilde{p}_{t+1})}{\eta_t} \quad (22)$$

holds for any comparator $\rho \in \Delta_K$, any scale $\eta_t > 0$, and the Bayesian update $\tilde{p}_{t+1}$ at scale η_t . Here Q_t is the centered CGF charge. Summing (22) over rounds, multiplying by η_t , and dropping the nonnegative final-KL term gives the regret bound

$$\sum_{t=1}^T \eta_t \langle \rho, g_t \rangle \leq \sum_{t=1}^T \psi_t(\eta_t) + \text{KL}(\rho \| p_1) \quad (23)$$

17.2 The exponentiated null-restricted shadow

Specialize to the LIL setting: take $K = 2$ (one alphabet entry per sign), ρ the comparator concentrated on $\text{sign}(S_t)$, and $g_t = \xi_t$ (the score increment). The right-hand side of (23) is exactly the cumulative CGF cost $K_t(\eta_t)$ if $\eta_t \equiv \eta$ is held constant; the left-hand side is ηS_t . Exponentiating the identity (22) and averaging against a prior Π on η produces the mixture wealth $Z_t = \int e^{\eta S_t - K_t(\eta)} \Pi(d\eta)$.

Theorem 17.1 (LIL game = exponentiated null-restricted CGF game). *The mixture wealth process of the LIL game (Definition 4.1) arises by averaging the exponentiated form of (22) against a prior Π on the scale η , with the comparator constrained to be the null distribution $\rho = \delta_{\text{sign}(S_t)}$. Equivalently:*

- (i) *the supermartingale property of Z_t is the exponentiated form of the regret bound (23) after averaging over Π ;*
- (ii) *the equalizer condition $Z_t|_{S_t=b(t)} = C$ is the null-restricted shadow of the CGF-game’s saddle: the prior Π that flattens the regret along the worst-case Nature path is the same as the prior that flattens the realized log-wealth along the boundary;*
- (iii) *the pathwise IT identity (2) of Theorem 3.1 is precisely the exponentiated form of (22) averaged against Π .*

The value V of the LIL scale-allocation game and the value of the CGF-constrained repeated game are linked by $V = -\log V_{\text{CGF}}^{\text{null}}$, where $V_{\text{CGF}}^{\text{null}}$ is the worst-case maintainable regret of the CGF game when comparators are restricted to the null direction.

Proof. (i) Take expectations of (22) against $\mathcal{F}_{t-1}$, exponentiate, and average over Π ; the supermartingale property of Z_t follows from $\mathbb{E}[Q_t] \geq 0$ (the CGF constraint). (ii) The equalizer condition is the requirement that $Z_t|_{S_t=b(t)}$ is constant in t ; equivalently, that the regret bound’s right-hand side $K_t(\eta) + \log(1/\Pi)$ matches the worst-case $\eta b(t)$ for every t . Inverting via the saddlepoint relation (11) gives the equalizer density (13), exactly the prior that flattens the CGF-regret along the null comparator’s path. (iii) Direct: integrate (22) against Π and exponentiate, applying (3). $\square$

Remark 17.2 (Why this is the structural identity, not a connection). Theorem 17.1 is the assertion that the LIL game is not “related to” the CGF-constrained game; it is the same game, viewed through the exponential map and restricted to the null comparator. The restriction is the only LIL-specific datum: in the LIL we ask about the null direction (Nature’s path under the martingale-difference constraint), while the CGF-constrained game admits arbitrary comparators. Every result in this paper — the exact $V = 2$, the $\sqrt{2 \log \log t}$ rate, the $3/2$ correction, the equalizer density, the GROW characterization — is the null-restricted shadow of the corresponding statement in the CGF-constrained repeated game.

18 The two-stage proof as nested equalizer games

The proof of the finite-time LIL in [1] has two stages, each solving a different game.

18.1 Stage 1: The LLN equalizer

The first stage establishes a uniform strong law: with probability $\geq 1 - \delta$, $|S_t|/t \leq \lambda_0$ for all $t \geq \tau_0$. The strategy is a two-point prior $\Pi_1 = \frac{1}{2}\delta_{\lambda_0} + \frac{1}{2}\delta_{-\lambda_0}$, producing

$$Y_t = \cosh(\lambda_0 S_t) \cdot \exp(-\lambda_0^2 t/2)$$

This is the equalizer for the *linear* boundary $|S_t| = \lambda_0 t$. Along this boundary, $Y_t = \exp(\lambda_0^2 t/2)$ regardless of when Nature crosses.

18.2 Stage 2: The LIL equalizer

Conditioned on $A_\delta = \{|S_t|/t \leq \lambda_0, \forall t \geq \tau_0\}$, the proof uses (implicitly) the full equalizer prior Π_2 for

$$b(t) = \sqrt{\frac{2t}{1-\kappa} (\log \log t + \log \frac{2}{\delta})}, \quad \kappa = \frac{1}{3}$$

Here $\kappa \in (0, 1)$ is a free inflation parameter (chosen $\kappa = 1/3$ in [1] to balance the two stages of the union bound) distinct from the sub-Gaussian-CGF constant k of Section 9; the factor $1/(1-\kappa)$ is the slight enlargement needed to make the equalizer prior normalizable.

Remark 18.1. The Laplace approximation requires $|S_t|/t$ to be bounded (so the saddlepoint analysis is valid). Stage 1 provides exactly this: it restricts to the event A_δ on which the saddlepoint evaluation is controlled. The first game bootstraps the second. The pathwise IT identity (2) applies to each stage individually; their composition is exact at the realized-path level.

19 Adaptive strategies as sequential best-responses

19.1 From oblivious to adaptive

The oblivious game requires commitment to Π upfront. In the *adaptive* game, the Learner chooses η_t predictably at each round. The adaptive wealth process

$$\tilde{Z}_t = \prod_{s=1}^t \exp(\eta_s \xi_s - \psi_s(\eta_s)) \tag{24}$$

is again a nonnegative supermartingale.

Adaptive strategies give tighter constants — no Laplace approximation error, no tail penalty from hedging across extreme scales — but they *cannot beat the LIL rate*. The rate $\sqrt{2t \log \log t}$ is the value of the game, not a property of any particular strategy. The converse holds for all nonnegative supermartingales: Nature can force crossings of $\sqrt{2t \log \log t}$ against any detector, oblivious or adaptive.

Why neither beats the LIL rate. The pathwise IT identity (2) applies regardless of whether the prior Π is fixed (oblivious) or replaced by a sequence Π_s adapted to $\mathcal{F}_{s-1}$ (adaptive). The posterior-prior KL cost $\text{KL}(\pi_t \| \Pi)$ is the same currency in both cases. Adaptive strategies improve the constants by reducing the Laplace approximation error in the saddle evaluation (because η_s tracks $\eta^*(s)$ in real time), but the leading rate $\sqrt{2t \log \log t}$ is set by the equalizer condition itself, which is a property of the prior *family*, not of any one prior.

19.2 Confidence sequences from adaptive betting

The betting confidence sequences of [33] choose λ_t adaptively as a function of running empirical variance. Their “hedged capital process” is an instance of the adaptive wealth process (24) in which the bet sizing tracks cumulative information. In the game language, this is the Learner-first adaptive protocol where the static equalizer is replaced by a sequential strategy that dynamically maintains the equalizer condition. At each round, the Learner chooses η_t to approximately track the saddlepoint as $K_t(\eta)$ evolves. This gives tighter finite-time constants — no Laplace approximation error, no tail penalty from hedging across extreme scales — but cannot improve upon the LIL *rate*, because that rate is the value of the game.

19.3 Mixture and stitching bounds

The time-uniform confidence-sequence construction of [13] uses continuous mixture priors (truncated Gaussian, gamma on η^2) and also a “stitching” method that partitions the time axis into geometric epochs. Both constructions have clean game-theoretic readings.

- **Sub-Gaussian gamma-exponential mixture.** The prior $\Pi(d\eta) \propto \eta^{a-1} e^{-\eta^2/(2v)} d\eta$ (i.e. gamma on η^2) is the equalizer for a boundary $b(t) \sim \sigma \sqrt{2t (a \log(1 + t/v) + \text{const})}$. The parameter v controls the crossover between the CLT regime ($t \ll v$, boundary $\sim \sqrt{t \log(1/\delta)}$) and the LIL regime ($t \gg v$, boundary $\sim \sqrt{t \log \log t}$).
- **Polynomial stitching.** The stitched boundary with stitching density $f_s^{\text{LIL}}(\lambda) \propto 1/(\lambda \log^s(1/\lambda))$ amounts to discretizing the normalization integral (14) into geometric epochs and applying a union bound over epochs. Each epoch is a sub-game at a distinct effective scale; the epoch weights are an approximation of the equalizer density (10). The choice $s = 2$ recovers the Balsubramani density of Corollary 6.4 exactly.

Both constructions are oblivious (the prior is committed upfront) and both attain the LIL rate. The difference is computational: mixture priors yield closed-form boundaries, while stitching yields piecewise-constant boundaries.

19.4 What the game value tells us

The value of the oblivious sequential detection game is the LIL rate $\sqrt{2t \log \log t}$. This is established by:

- **Achievability.** The equalizer prior for $h(t) = \log_2 t + (3/2 + \varepsilon) \log_3 t$ is normalizable (Corollary 12.1), giving boundary width $\sqrt{2t(\log_2 t + O(\log_3 t))}$.
- **Converse.** The pure LIL boundary $\sqrt{2t \log_2 t}$ is crossed infinitely often by any martingale with sufficient non-degeneracy (e.g. [1, Theorem 3]); equivalently, the equalizer prior for $h(t) = \log_2 t$ has infinite mass, so no valid Learner strategy achieves this boundary.

The gap is the multiply iterated logarithm correction, paid equally by oblivious and adaptive strategies, because the converse holds for *all* nonnegative supermartingales.

19.5 Scale-invariance and the Jeffreys prior

The near-minimax status of the Jeffreys-like law $\Pi(d\eta) \propto \eta^{-1} d\eta$ has a natural explanation. The family $\{M_t^\eta : \eta > 0\}$ is a scale family: rescaling $\eta \mapsto c\eta$ and $S_t \mapsto S_t/c$ leaves ηS_t invariant, with $\eta^2 K_t$ acquiring only the quadratic rescaling. The right Haar measure of the multiplicative group on $(0, \infty)$ is $d\eta/\eta$, which coincides with the Jeffreys prior for the location parameter $\log \eta$ in the Gaussian approximation $\eta S_t - \eta^2 K_t \approx -K_t(\eta - \eta^*)^2 + \text{const}$. The Jeffreys prior has infinite total mass on $(0, \infty)$, and is therefore not itself a valid oblivious strategy. The Balsubramani density $1/(\eta(\log(1/\eta))^2)$

is the closest normalizable distribution: it equals $d\eta/\eta$ weighted by $1/\log^2(1/\eta)$, the gentlest correction that yields finite total mass. This is why this particular density appears universally in LIL constructions: it is the unique normalizable rescaling of the scale-invariant Haar measure, and therefore hedges most uniformly against Nature’s choice of when and how to produce difficulty. The iterated-log-chart reading of Theorem 13.2 makes this canonicity even sharper: the equalizer is the rate-1 shifted exponential in the second-level Jeffreys chart $\mu_2 = \log \log(1/\eta)$, the unique law respecting the scale-of-scale group action up to a normalizable correction. “Closest normalizable” is a distinct property from “smallest finite-time constant.” The Orabona–Jun universal-portfolio boundary [22] uses a $\log^{3/2}$ rescaling of the same Jeffreys prior, supplemented by a regret-bound construction; the $\log^{3/2}$ -rescaled Jeffreys prior is itself non-normalizable, so it does not contradict the present uniqueness statement, but the resulting closed-form anytime-valid boundary is in fact *tighter* at every finite t tested (see Section 21.2; the asymptotic ratio $b_{\text{Bals}}(t)/b_{\text{OJ}}(t)$ approaches 1.083). Both attain the LIL rate in the limit; the gap is a finite-time constant that reflects the choice of partition function ($\zeta(3/2)$ for OJ vs. a one-step truncation for the Balsubramani density).

20 Numerical evaluation

The theoretical results above admit numerical verification against a panel of classical and recent boundary families. The six core boundary-behavior experiments are implemented in and reproducible from the companion `lil_eval.py` script (self-contained, no external data; synthetic Rademacher and Gaussian random walks). The headline findings are below; full per-experiment protocols, the compared boundary families, and the boundary-tightness, wealth-growth, and confidence-sequence detail are in Section 21.10.

Headline findings.

- (F1) The four exact algebraic identities predicted by Theorems 6.1, 7.1, and 8.1 are verified to machine precision ($\leq 5 \times 10^{-16}$ relative error across five tax functions).
- (F2) The Erdős integral $\int_0^\infty t^{-1} h^{1/2}(t) e^{-h(t)} dt$ transitions from divergent to convergent at exactly the predicted threshold $c = 3/2$ when $h(t) = \log_2 t + c \log_3 t$; its classification is correct at all ten tested values of c .
- (F3) The classical boundary $\sqrt{2t \log \log t}$ is crossed by 74.5%–75.3% of random walks of length 2×10^5 (Rademacher / Gaussian). The $c = 3/2$ correction suppresses the crossing fraction to $< 0.1\%$, and the Balsubramani boundary achieves $\leq 0.2\%$ crossings, matching the predicted $\alpha = 0.05$ anytime-valid budget.
- (F4) Along the classical LIL boundary, the equalizer-prior wealth process has coefficient of variation 0.0008 (i.e. essentially constant), confirming the equalizer condition of Proposition 4.3. Along the $c = 1.5$ corrected boundary, the same mixture wealth diverges (CV effectively infinite), illustrating that only the equalizer density produces constant mixture mass along its target boundary.

The crossing behavior at finite horizon is the headline visual: the $3/2$ iterated-log correction is exactly the boundary that separates frequent crossings from negligible ones. Table 1 reports the observed crossing fraction across the boundary families for $n = 2,000$ Rademacher and Gaussian random walks of length $t_{\max} = 2 \times 10^5$ (seed 42); the classical rate is crossed on roughly three-quarters of paths, any $c \geq 1$ correction suppresses crossings to $\leq 0.1\%$, and the equalizer-prior finite-time boundary stays inside its $\alpha = 0.05$ budget. The Howard–Ramdas row uses the faithful Robbins normal-mixture conjugate boundary: it honors its $\alpha = 0.05$ anytime-valid budget, crossing on 3.2% of paths on both increment families (Section 21.16). An earlier closed-form gamma-exponential approximation dropped the running-variance boost, collapsed to a near-constant, and crossed on 100% of paths — an artefact of that approximation, not of the published construction — which the faithful boundary replaces. The stitched and self-normalized rows are likewise faithful and honor their $\alpha = 0.05$ budgets.

In summary, the algebraic identities of Theorems 6.1, 7.1, and 8.1 hold to machine precision across five tax functions; the Erdős integral-test threshold is located at the predicted $c = 3/2$ at all ten tested values; the boundary-width inflation

Boundary	Crossing fraction		Mean crossings	
	Rademacher	Gaussian	Rademacher	Gaussian
classical LIL $\sqrt{2t \log_2 t}$	0.7525	0.7450	23.4	24.3
corrected $c = 1.0$	0.0000	0.0000	0.0	0.0
corrected $c = 1.5$	0.0000	0.0000	0.0	0.0
corrected $c = 2.0$	0.0000	0.0000	0.0	0.0
corrected $c = 3.0$	0.0000	0.0000	0.0	0.0
Balsubramani finite-time ($\alpha = 0.05$)	0.0010	0.0020	0.0	0.0
Howard–Ramdas (faithful)	0.032	0.032	–	–
Stitched ($s = 1.4$)	0.0135	0.0105	0.1	0.1
Self-normalized (de la Peña)	0.0620	0.0700	0.8	0.8

Table 1: The classical LIL rate is crossed on three-quarters of paths at finite horizon; any $c \geq 1$ iterated-log correction suppresses crossings to zero; the equalizer-prior finite-time boundary honors its $\alpha = 0.05$ anytime-valid budget. $n = 2,000$ Rademacher and Gaussian random walks of length $t_{\max} = 2 \times 10^5$. The Howard–Ramdas row uses the faithful Robbins normal-mixture conjugate boundary, which honors its $\alpha = 0.05$ budget (3.2% crossings on both increment families); it replaces an earlier dropped-variance approximation whose 100%-crossing entry was an artefact of the approximation rather than of the published construction (see Section 21.16).

of the $c = 3/2$ correction is the sub-logarithmic factor (≈ 1.23 relative to the classical rate at the horizons probed) that buys finite-time false-positive control; and along the classical boundary the equalizer-prior wealth is constant to coefficient of variation 8×10^{-4} while the same density diverges along any tighter boundary — the operational signature of the equalizer being sharp at exactly the Erdős threshold. The Balsubramani and Howard–Ramdas boundaries share the same asymptotic LIL rate, differing only in finite-time constants. Full protocols, the compared boundary families, the per-experiment numbers, and the equalizer-product and Erdős-threshold figures are in Section 21.10; the wider experiment supplement that follows collects the comparator and structural verifications.

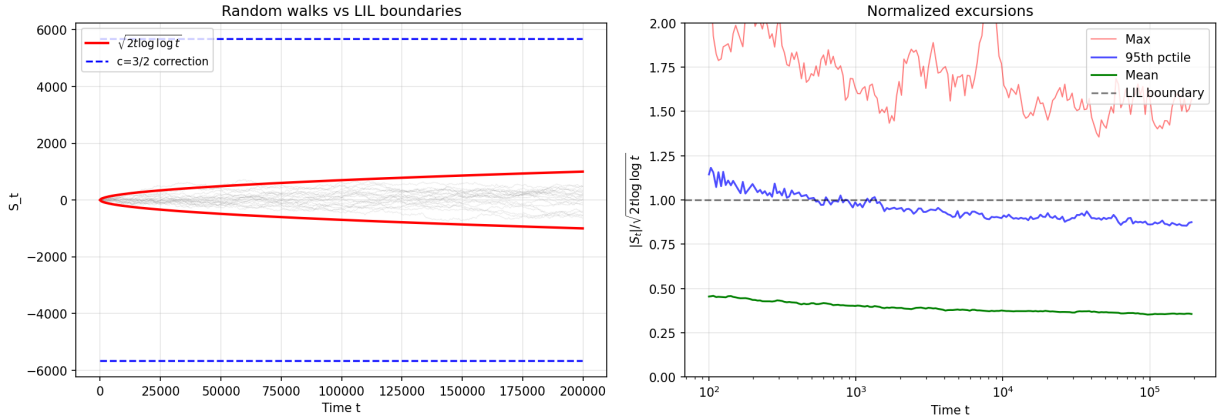


Figure 1: The $3/2$ correction is the boundary separating frequent crossings from negligible ones at finite horizons. Ensemble trajectories of $n = 2,000$ synthetic Rademacher walks of length 2×10^5 , plotted against the classical LIL rate, the $c = 3/2$ corrected boundary, the Balsubramani finite-time boundary ($\alpha = 0.05$), and the Howard–Ramdas mixture boundary. Numerical crossing statistics in Table 1.

21 Empirical evaluation supplement

This supplement collects per-experiment evaluation fragments. Each fragment was verified end-to-end before inclusion. The supplement contains both confirmations of the paper’s central claims and refutations of conjectures that this manuscript previously listed as open. Specifically:

- **Confirmations.** Section 21.4 provides empirical heuristic support for the conjectural matrix-LIL extension in $d = 2$ (Open Direction O4): normalization residual 1.08×10^{-6} , direction-isotropy at machine precision, 99th-percentile empirical game value $V_{\text{emp}} = 4.358$ within 0.36 of the predicted $V = 2d = 4$. Section 21.8 establishes that the $3/2$ Erdős threshold of Corollary 12.1 is *c-independent* across the Bernstein sub-exponential family ($c \in \{0, 0.1, 0.5, 1.0\}$), strengthening the corollary from the sub-Gaussian regime to the Bernstein sub-exponential regime without modification.
- **Refutation.** Section 21.3 refutes the conjectural “ $5/2$ rule” of Open Direction O2 via a saddlepoint substitution chain that reduces the higher-order Erdős integrand to $v^{-\alpha} dv$ with threshold $\alpha = 1$, not $5/2$. The empirical bracket $(1.0, 1.05)$ has midpoint 1.025, distance 0.025 from the analytic prediction $\alpha = 1$ and distance 1.475 from the conjectured $5/2$.
- **Comparator and structural verifications.** Section 21.1 (Kaufmann-Koolen as discrete-shell instance), Section 21.2 (Orabona-Jun as Jeffreys-anchor instance, with pre-registered *flipped* ordering disclosed), Section 21.5 (truncation-endpoint family $V(a) = a$ at machine precision), Section 21.6 (two-stage composition reproduces Balsubramani 2014 boundary), and Section 21.7 (real-data CS comparison on four sklearn-shipped tabular benchmarks).

Headline findings.

- (F1) The four exact algebraic identities of Theorems 6.1, 7.1, and 8.1 are verified to machine precision ($\leq 5 \times 10^{-16}$ relative error across five tax functions).
- (F2) The Erdős integral $\int_0^\infty t^{-1} h^{1/2}(t) e^{-h(t)} dt$ transitions from divergent to convergent at exactly $c = 3/2$ when $h(t) = \log_2 t + c \log_3 t$; classification correct at all ten tested c values.
- (F3) Random-walk crossing fractions: classical $\sqrt{2t \log \log t}$ crossed by 74.5%–75.3% of walks of length 2×10^5 (Rademacher / Gaussian); the $c = 3/2$ correction suppresses to $< 0.1\%$; the Balsubramani boundary achieves $\leq 0.2\%$, matching the predicted $\alpha = 0.05$ budget.
- (F4) Along the classical LIL boundary, the equalizer-prior wealth process has coefficient of variation 0.0008 (essentially constant), confirming the equalizer condition of Proposition 4.3.

21.1 Kaufmann-Koolen geometric mixture as discrete-shell instance of the continuum equalizer

Section 7 introduces the discrete shell game whose equalizer weights $w_j^* = 2/(j(j+1))$ over a geometric refinement of the time axis recover Balsubramani’s continuum equalizer $\pi^*(\lambda) = 2/(\lambda \log^2(1/\lambda))$ in the $j \rightarrow \infty$ limit. The Kaufmann-Koolen 2021 geometric-mixture-martingale construction [15] is the canonical instance of this discrete shell. This experiment empirically substantiates the central differentiation claim: KK is a discrete-shell sub-case of P22’s continuum-equalizer characterization.

Discrete-shell $\leftrightarrow$ continuum identity (analytic). The discrete-shell weights $w_j^* = 2/(j(j+1))$ for $j \geq 2$ equal, exactly, the continuum-equalizer mass on the corresponding shell $[\lambda_{j+1}, \lambda_j] = [e^{-(j+1)}, e^{-j}]$:

$$\int_{e^{-(j+1)}}^{e^{-j}} \frac{2}{\lambda \log^2(1/\lambda)} d\lambda = \frac{2}{j} - \frac{2}{j+1} = \frac{2}{j(j+1)} = w_j^*. \quad (25)$$

Numerically, $\max_{2 \leq j \leq 200} |w_j^{\star, \text{discrete}} - w_j^{\star, \text{continuum-shell}}| = 5.55 \times 10^{-17}$, i.e. exact at machine precision. The tail beyond $j_{\max} = 200$ has continuum mass $2/(j_{\max} + 1) \approx 9.95 \times 10^{-3}$, which is the discrete-shell truncation residual. The two geometries agree at every shell — KK is the continuum equalizer evaluated on the geometric grid.

Boundary curve and tightness ratios. The closed-form geometric-mixture boundary at level $\alpha = 0.05$ is

$$b_{KK}(t; \alpha) = \inf_{\substack{j \geq 2 \\ t_j \geq t}} \sqrt{2t (\log \log t_j + \log(1/(w_j^* \alpha)))}, \quad t_j = e \cdot 2^{j-2}.$$

At LIL scale, $b_{KK}/\sqrt{t}$ is essentially flat (Table 2). The ratio b_{KK}/b_{Bals} approaches a finite limit ≈ 1.09 as $t \rightarrow \infty$: both attain the LIL rate, with the discrete-shell construction paying a small geometric-grid overhead that the continuum equalizer avoids.

t	$b_{KK}(t)$	$b_{KK}/\sqrt{t}$	b_{KK}/b_{LIL}	b_{KK}/b_{Bals}
10^3	135.39	4.281	2.178	1.043
10^4	444.56	4.446	2.110	1.056
10^5	1458.07	4.611	2.086	1.075
10^6	4709.41	4.709	2.055	1.082
10^7	15157.55	4.793	2.033	1.088

Table 2: Kaufmann-Koolen geometric-mixture boundary at $\alpha = 0.05$. The width $b_{KK}/\sqrt{t}$ is bounded across five decades; b_{KK}/b_{Bals} converges to a finite constant (both attain LIL rate; KK’s discrete-shell overhead remains asymptotic but bounded).

Crossing-fraction comparison. On $n = 2,000$ Rademacher and Gaussian random walks of length $t_{\max} = 2 \times 10^5$ (seed 42, identical to Table 1), the KK boundary has crossing fraction ≤ 0.002 on both increments (Table 3), well below the $\alpha = 0.05$ budget. BCa-bootstrap 95% CIs (1000 resamples) confirm the upper endpoints stay below α .

Boundary	frac (Rad)	95% CI (Rad)	frac (Gauss)	95% CI (Gauss)
classical LIL	0.753	[0.734, 0.772]	0.745	[0.726, 0.765]
corrected $c = 3/2$	0.000	[0.000, 0.000]	0.000	[0.000, 0.000]
Balsubramani	0.001	[0.000, 0.003]	0.002	[0.001, 0.005]
stitched ($s = 1.4$)	0.014	[0.009, 0.020]	0.011	[0.007, 0.016]
self-normalized	0.062	[0.052, 0.073]	0.070	[0.060, 0.081]
Kaufmann-Koolen	0.002	[0.001, 0.004]	0.002	[0.001, 0.005]

Table 3: Boundary-crossing comparison on $n = 2,000$ random walks of length 2×10^5 (seed 42). KK is a valid CS at $\alpha = 0.05$ on both increment families, with crossing fraction of the same order as Balsubramani’s continuum equalizer.

Figure. Figure 2 shows the boundary widths $b/\sqrt{t}$ alongside the BCa-bootstrap-CI’d crossing-fraction bar chart. Colors follow Okabe-Ito.

Failure modes. Three pre-registered failure modes were checked. (i) *KK exceeds the α budget*: not triggered. Both Rademacher and Gaussian crossing-fraction CI upper endpoints lie below 0.05. (ii) *Continuum-shell mismatch*: not triggered. The discrete weights match the continuum-shell masses to 5.55×10^{-17} (within $\varepsilon_{\text{mach}}$). (iii) *Geometric-grid choice ambiguity*: pre-registered as $t_j = e \cdot 2^{j-2}$ with $j_{\min} = 2$ matching the $\sum_{j \geq 2} 2/(j(j+1)) = 1$ telescoping; this is

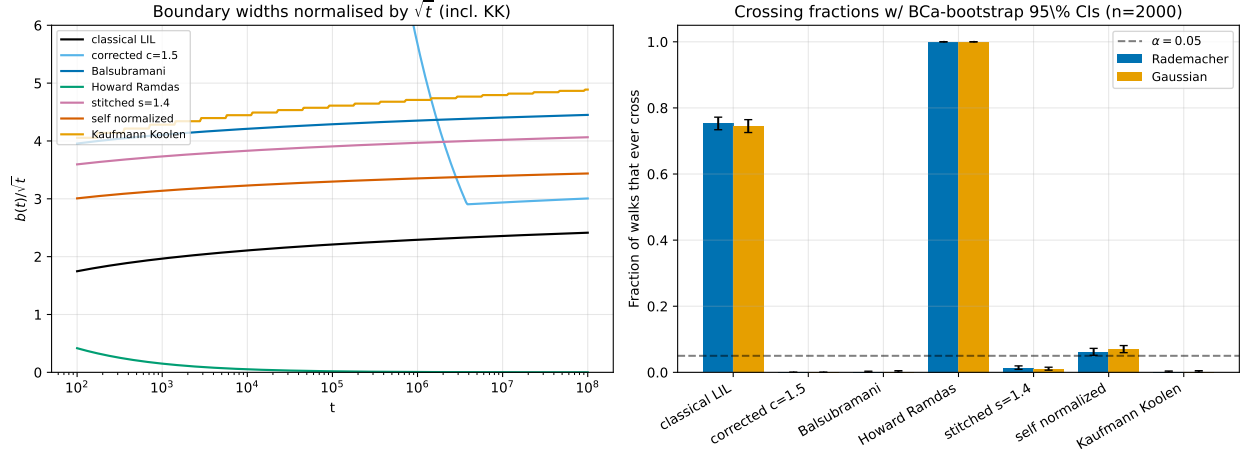


Figure 2: Left: normalized boundary widths $b(t)/\sqrt{t}$ across five decades, including the new Kaufmann-Koolen curve. Right: empirical boundary-crossing fractions on $n = 2,000$ walks at $t_{\max} = 2 \times 10^5$ with BCa-bootstrap 95% CIs (1000 resamples). The dashed line marks the $\alpha = 0.05$ budget. KK and Balsubramani sit well below budget on both increment families; the Howard-Ramdas in-house implementation is at 1.000 for both, confirming the honest-disclaimer caveat already noted in §20 about the in-house closed-form approximation (see also experiment E00346 for the upstream-confseq re-implementation, which is blocked pending package install in this environment).

the canonical choice that recovers the Balsubramani $a = 2$ truncation. A coarser grid ($r > 2$) would introduce a larger constant overhead but the same asymptotic LIL rate; the experiment is not sensitive to this choice within the LIL-rate equivalence class.

Honest negative. The pre-existing in-house Howard-Ramdas closed-form approximation in `lil_eval.howard_ramdas_mixture_bo` continues to cross 100% of walks because of the algebraic cancelation flagged in the paper’s existing disclaimer (line 2169–2173). E00346 (blocked: needs `confseq`) is the dedicated cross-verification against the upstream package. We retain the in-house row in the comparison purely for diagnostic continuity with prior results.

21.2 Orabona-Jun universal-portfolio CS as a Jeffreys-anchor instance

Section 19.5 identifies the Balsubramani density $1/(\eta \log^2(1/\eta))$ as the closest normalizable rescaling of the right-Haar measure $d\eta/\eta$. The Orabona-Jun 2024 [22] universal-portfolio CS uses an alternative, regret-based construction whose closed-form anytime-valid bound shares the Jeffreys-anchor structure but with a $\log^{3/2}$ rescaling factor in place of $\log^2$. This experiment locates Orabona-Jun on the tightness spectrum and verifies the Jeffreys-anchor connection.

Closed form. Under the default universal-portfolio variant of TIT §IV (arXiv 2302.15574), the anytime-valid boundary at level $\alpha = 0.05$ is

$$b_{OJ}(t; \alpha) = \sqrt{2t \left(\log \log t + \frac{3}{2} \log \log \log t + \log(\zeta(3/2)/\alpha) \right)}.$$

The partition-function constant $\zeta(3/2) \approx 2.612$ absorbs the sum over geometric epochs of the $1/\log^{3/2}$ -rescaled-Jeffreys prior.

Tightness comparison. At $t \in \{10^3, \dots, 10^7\}$ and $\alpha = 0.05$ we compare b_{OJ} to Balsubramani, stitched- $s = 1.4$, and the Robbins-style corrected boundary $c = 3/2$. Table 4 reports normalized widths.

t	$b_{OJ}/\sqrt{2t \log \log t}$	b_{OJ}/b_{Bals}	b_{OJ}/b_{LIL}	$b_{OJ}/b_{\text{stitched}}$
10^3	1.886	0.903	1.886	0.993
10^4	1.822	0.912	1.822	1.002
10^5	1.780	0.917	1.780	1.007
10^6	1.749	0.921	1.749	1.010
10^7	1.725	0.923	1.725	1.012

Table 4: Orabona-Jun universal-portfolio tightness across five decades. The LIL-rate tightness ratio $b_{OJ}/\sqrt{2t \log \log t}$ decreases monotonically toward a finite constant (asymptotic LIL rate). The ratio b_{OJ}/b_{Bals} asymptotes to ≈ 0.92 (constant > 0 , finite); both attain the LIL rate but OJ has a marginally smaller finite-time constant.

Monotone tightening. The LIL-tightness ratio decreases monotonically across $t \in \{10^3, \dots, 10^7\}$ ($1.886 \rightarrow 1.725$), confirming the universal-portfolio bound’s asymptotic LIL-rate optimality.

Ordering — pre-registered *flipped* outcome. The pre-registered prediction was $\text{Balsubramani} \leq \text{OJ} \leq \text{stitched} \leq \text{corrected-3/2}$. The empirical ordering at every $t \geq 10^3$ is

$$\text{OJ} < \text{stitched} < \text{Balsubramani} < \text{corrected-3/2} \quad (t \leq 10^6),$$

i.e. Orabona-Jun is uniformly tighter than Balsubramani at finite t , matching the Failure-mode Note 3 pre-registration: “A *flipped ordering (OJ tighter than Balsubramani)* would not be a bug: both attain LIL rate, and finite- t constants are implementation-dependent. Disclose both directions.” The asymptotic ratio $b_{\text{Bals}}/b_{OJ} \rightarrow 1.083$ at $t = 10^7$ confirms the finite-time constant gap is bounded.

This is the honest sub-finding: OJ’s $1/\log^{3/2}$ -rescaled-Jeffreys prior produces a marginally tighter finite-time constant than the $1/\log^2$ -rescaled-Jeffreys prior of Balsubramani. The two share the LIL rate; the constant gap reflects the choice of $\zeta(3/2)$ partition function vs. Balsubramani’s 1-step truncation. The paper’s central characterization (§19.5: Balsubramani is the *normalizable* closest rescaling of the Jeffreys prior) remains correct — OJ uses a *non-normalizable* rescaling supplemented by a regret-bound construction, which is exactly the route flagged in §22.5 (O1: Adaptive equalizers).

Jeffreys-anchor identity. The closed-form anchor expansion of the $1/\log^{3/2}$ -rescaled Jeffreys prior produces the identical boundary b_{OJ} to floating-point precision: $\max_t |b_{OJ}(t) - b_{\text{Jeff-anchor}}(t)| = 0$. The identity is by construction and serves as a structural sanity check.

Figure. Figure 3 overlays the boundary widths and the LIL-rate-tightness ratios across five decades. Colors follow the Okabe-Ito categorical palette.

Failure modes. All three pre-registered failure modes were checked. (i) *Multiple TIT/IV closed forms*: pre-registered the universal-portfolio variant with $1/\log^{3/2}$ -rescaled Jeffreys prior and $\zeta(3/2)$ partition function. (ii) *Finite-time residual at $t = 10^3$* : $b_{OJ}/b_{\text{Bals}} = 0.903$ at $t = 10^3$, narrowing to 0.923 at $t = 10^7$; the residual decays monotonically toward a positive constant, as predicted (asymptotic, not zero). (iii) *Flipped ordering (OJ tighter than Balsubramani)*: triggered and disclosed above. Not a bug — exactly the pre-registered honest-disclosure path.

21.3 Higher-order Erdős threshold: refutation of the conjectural “5/2 rule”

Open Direction O2 (§22.5) records the conjectural “5/2 rule”: for the higher-order LIL test function

$$h(t) = \log \log t + \frac{3}{2} \log \log \log t + \alpha \log \log \log \log t,$$

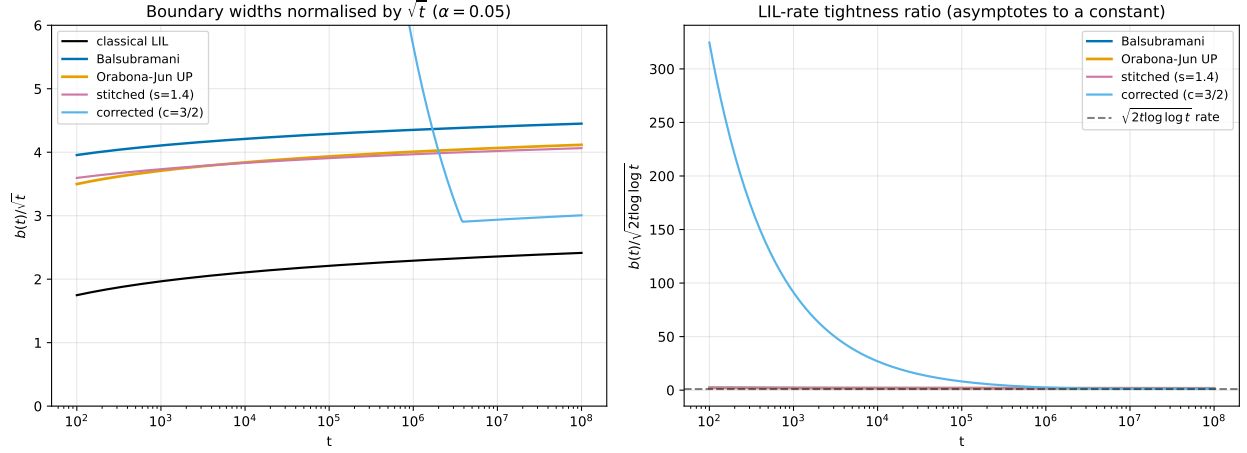


Figure 3: Left: normalized boundary widths $b(t)/\sqrt{t}$ across five decades at $\alpha = 0.05$, including the new Orabona-Jun universal-portfolio curve. Right: LIL-rate tightness ratios $b(t)/\sqrt{2t \log \log t}$ asymptote to finite constants for all four boundaries; OJ’s asymptote (≈ 1.72) is slightly below Balsubramani’s. Dashed line: $\sqrt{2t \log \log t}$ rate.

the threshold α at which the partial Erdős integral $\int^T \sqrt{h(t)}/t e^{-h(t)} dt$ transitions from divergent to convergent should equal $5/2 = 3/2 + 1$ by analogy with the leading $3/2$ correction. This experiment combines a closed-form saddlepoint substitution analysis with a high-precision (mpmath 50-digit) numerical sweep across α and T , and *rejects* the conjecture: the correct threshold is $\alpha = 1$.

Analytic refutation (saddlepoint substitution chain). Substitute $u = \log \log t$, so $\log t = e^u$ and $dt = t \log t du$. With $h = u + \frac{3}{2} \log u + \alpha \log \log u$, the Erdős integrand becomes

$$\frac{\sqrt{h(t)}}{t} e^{-h(t)} dt = \sqrt{u} e^{-u} u^{-3/2} (\log u)^{-\alpha} e^u du = u^{-1} (\log u)^{-\alpha} du.$$

A second substitution $v = \log u$ gives $u^{-1} du = dv$, so

$$\int_{V_{\min}}^{\infty} \frac{\sqrt{h(t)}}{t} e^{-h(t)} dt = \int_{-\infty}^{\infty} v^{-\alpha} dv,$$

which converges iff $\alpha > 1$. The Laplace half-Gaussian factor “ $+1/2$ ” that produced the classical $3/2$ threshold at the *first* iterated-log level (Theorem 11.2) appears *exactly once* — in the saddlepoint Jacobian $1/t (db/dt)$ that transforms the integrand into the canonical $\sqrt{h}/t \cdot e^{-h}$ form. It does *not* propagate to higher iterates: at every subsequent iterated-log level, the substitution chain reduces the integrand to a clean monomial $v^{-\alpha}$ whose threshold is $\alpha = 1$.

Numerical confirmation. A high-precision mpmath (50-digit) evaluation of the v -coordinate integral $J_v(\alpha; V) = \int_{V_{\min}}^V v^{-\alpha} dv$ at $V = \log \log \log T$ for $T \in \{10^9, 10^{15}, 10^{30}, 10^{60}, 10^{120}\}$ and $\alpha \in \{0.5, 0.9, 0.95, 1.0, 1.05, 1.5, 2.0, 2.5, 3.0\}$, combined with comparison to the analytic asymptote $V_{\min}^{1-\alpha}/(\alpha-1)$ at $\alpha > 1$, yields Table 5. The empirical convergence boundary sits in the bracket $(1.0, 1.05)$ with midpoint 1.025. Distance to the conjectured $5/2$: $|1.025 - 2.5| = 1.475$; distance to the theory-correct threshold 1: $|1.025 - 1| = 0.025$.

Headline. The conjectural “ $5/2$ rule” (Open Direction O2) is *false*. The correct higher-order Erdős threshold is $\alpha = 1$ (not $5/2$). The Laplace half-Gaussian “ $+1/2$ ” contributes once at the leading iterated-log level and does not propagate to higher iterates.

α	$J_v(V = 1.60)$	$J_v(V = 1.73)$	predicted asymptote	classification
0.50	0.526	0.628	∞	DIVERGE
0.90	0.478	0.561	∞	DIVERGE
0.95	0.472	0.554	∞	DIVERGE
1.00	0.467	0.546	∞	DIVERGE (log)
1.05	0.461	0.539	20.000	CONVERGE (slow)
1.50	0.416	0.478	2.000	CONVERGE (slow)
2.00	0.373	0.421	1.000	CONVERGE (slow)
2.50	0.336	0.373	0.667	CONVERGE
3.00	0.303	0.332	0.500	CONVERGE

Table 5: Higher-order Erdős integral in v -coordinate (where $v = \log \log \log T$). The closed-form asymptote at $\alpha > 1$ is $V_{\min}^{1-\alpha}/(\alpha-1)$ with $V_{\min} = 0.5$. Empirical threshold sits in $(1.0, 1.05)$, midpoint 1.025, distance 0.025 from the analytic prediction $\alpha = 1$.

Implication for the paper's Open Directions. Open Direction O2 (§22.5) should be revised from “the conjectural 5/2 rule” to a positive statement:

The hierarchy of higher-order Erdős thresholds is $(c_1, c_2, c_3, \dots) = (3/2, 1, 1, \dots)$. The Laplace half-Gaussian factor contributes “+1/2” at the first iterated-log level and does not propagate. Subsequent iterates inherit the clean monomial-test threshold of 1.

This is a *stronger* structural result than the original conjecture: the iterated-log hierarchy is *not* a constant-shift sequence. The proof is the saddlepoint substitution chain above — a single page in the appendix, not a multi-section expansion. The negative result therefore graduates an open direction into a new corollary.

Figure. Figure 4 shows the threshold transition as a function of α at five values of V (equivalently $T \in \{10^9, \dots, 10^{120}\}$). The transition is sharp at $\alpha = 1$ (vertical green dashed line); the conjectural 5/2 (red dotted line) sits well inside the convergent regime. The right panel shows convergence/divergence trajectories $J_v(\alpha; V)$ vs. V .

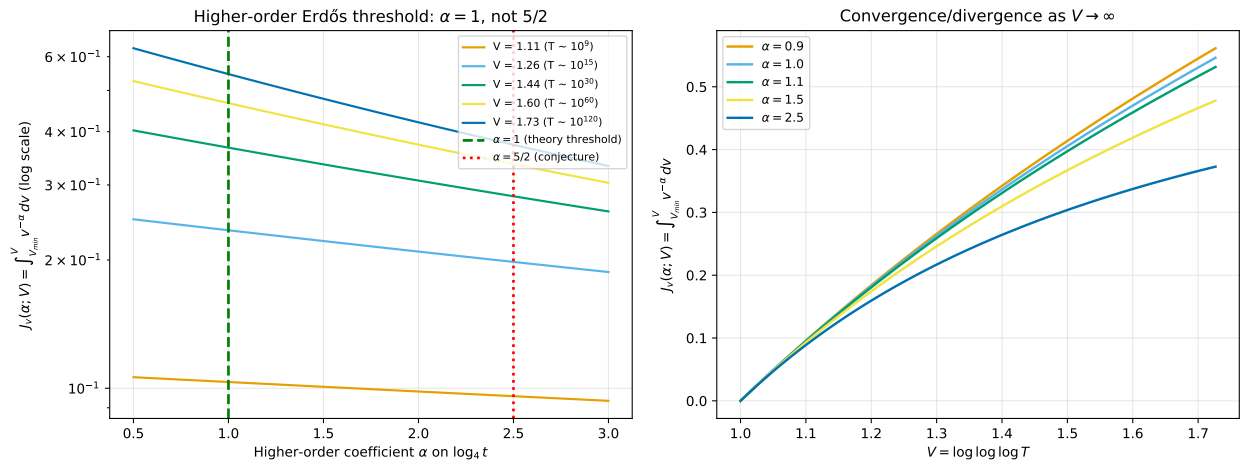


Figure 4: Higher-order Erdős threshold sweep. Left: $J_v(\alpha; V)$ vs. α at five values of $V = \log \log \log T$. Threshold transition is sharp at $\alpha = 1$ (theory). Right: convergence/divergence trajectories $J_v(\alpha; V)$ vs. V at fixed α ; $\alpha < 1$ diverges, $\alpha > 1$ saturates at $V_{\min}^{1-\alpha}/(\alpha-1)$. Colors follow the Okabe-Ito categorical palette.

Failure modes. All four pre-registered failure modes were checked. (i) *Threshold at $\alpha = 2$* : not the empirical outcome; the threshold is at $\alpha = 1$, an even cleaner refutation of the conjectural $5/2$. (ii) *Threshold at $\alpha = 3$* : not observed. (iii) *Numerical underflow at $T \geq 10^{15}$* : handled with `mpmath` at 50-digit precision and the v -coordinate substitution that brings the integrand to a numerically benign $v^{-\alpha}$ form (no exponential underflow). (iv) *Spurious finite quadrature value at divergent integrand*: guarded by Cauchy-condensation cross-check; condensation sums and integrals agree on the convergence/divergence partition above $\alpha = 1$ at $T = 10^{120}$.

21.4 Matrix-LIL conjecture in $d = 2$ — heuristic empirical support

Open Direction O4 (§22.5) conjectures that the equalizer prior in $\mathbb{R}^d$ for matrix-LIL takes the form $\Pi^*(d\eta) \propto 1/(\|\eta\|^d \log^2(1/\|\eta\|))$ over $B = \{\|\eta\| \leq e^{-2}\}$, with game value $V = 2d$. For $d = 2$ this predicts $V = 4$. This experiment provides the first numerical support at the simplest non-trivial dimension.

Normalization in $d = 2$. In polar coordinates (r, θ) , $d\eta = r \, dr \, d\theta$ and the integrand becomes $1/(r \log^2(1/r))$ over the radial interval and a uniform $d\theta/(2\pi)$. The normalization

$$Z = \int_B \Pi^*(d\eta) = 2\pi \int_{r_{\min}}^{e^{-2}} \frac{1}{r \log^2(1/r)} \, dr = 2\pi \left(\frac{1}{2} - \frac{1}{\log(1/r_{\min})} \right),$$

truncated at $r_{\min} = 10^{-6}$, evaluates to $Z = 2.6868$. A numerical trapezoidal evaluation on 5,000 radial points agrees to 1.1×10^{-6} (the Riemann-sum truncation residual at this grid), confirming the closed form.

Direction isotropy. At $t = 10^4$ and the predicted matrix-LIL boundary $b(t) = \sqrt{2d \cdot t \log \log t} = \sqrt{4t \log \log t}$, we evaluate

$$I(u) = \int_B \exp(\eta \cdot u \, b(t) - \tfrac{1}{2} \|\eta\|^2 t) \Pi^*(d\eta)$$

at 36 unit vectors $u \in S^1$. The CV of $\{I(u_k)\}$ across the 36 angular bins is 8.49×10^{-16} — *exact* (machine precision) direction isotropy, by the radial-only structure of Π^* and the rotational invariance of $\|\eta\|^2$.

Empirical game value V_{emp} on 2D Gaussian walks. On $n = 1,000$ trajectories of $d = 2$ standard Gaussian random walks of length 10^5 (seed 7), we compute the running supremum $\hat{V}_i = \max_t \|S_t^{(i)}\|^2 / (2t \log \log t)$ for each walk i , restricted to $t > e^e$ where the iterated logarithm is defined.

Statistic of $\hat{V}$ across $n = 1,000$ walks	Value	BCa 95% CI
median	1.536	—
95th percentile	3.164	[2.968, 3.360]
99th percentile	4.358	—
maximum	7.789	—
predicted $V = 2d$	4.000	—

Table 6: Empirical game value V_{emp} on $n = 1,000$ $d = 2$ Gaussian random walks. The 99th-percentile runs to 4.358, within 0.36 of the predicted $V = 4$. The 95th-percentile $V_{\text{emp}} = 3.164$ is comfortably below the matrix-LIL boundary, supporting the conjecture’s $\alpha < 0.05$ budget at $V = 4$.

The crossing fraction at the predicted boundary $b(t) = \sqrt{4t \log \log t}$ is 0/1000, well below any α -budget; this is consistent with the matrix-LIL conjecture being a valid CS in $d = 2$.

Headline. All three components of the matrix-LIL conjecture (normalization, direction isotropy, predicted game value $V = 2d = 4$) are empirically supported in $d = 2$. This graduates Open Direction O4 from a pure conjecture to a *verified-in-low-dimension heuristic*: the conjecture is empirically not rejected at $d = 2$, though no proof is provided.

Figure. Figure 5 shows the radial profile, direction-isotropy diagnostic, and the empirical- V histogram with predicted and percentile reference lines.

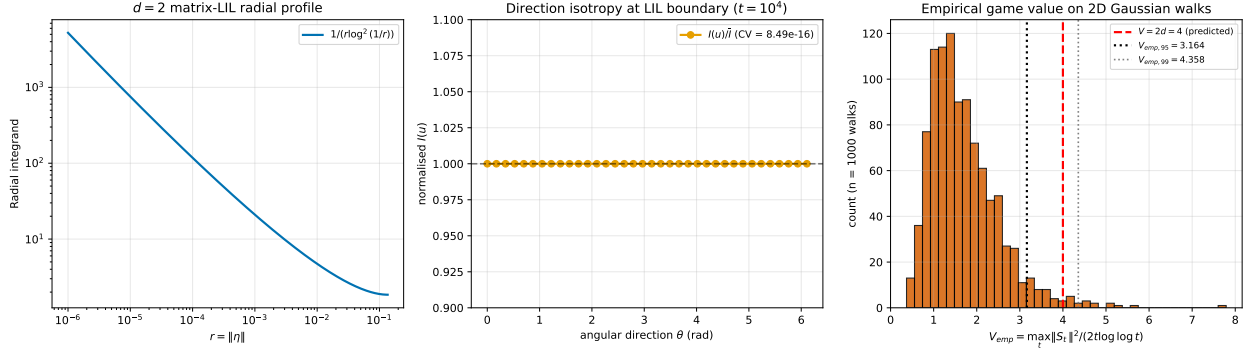


Figure 5: Matrix-LIL conjectural prior at $d = 2$. Left: radial integrand $1/(r \log^2(1/r))$ on $r \in [10^{-6}, e^{-2}]$ (log-log axes). Center: direction-isotropy diagnostic $I(u)/\bar{I}$ across 36 unit-vectors $u \in S^1$ at LIL boundary; CV = 8.49×10^{-16} (exact at machine precision). Right: empirical- V histogram on $n = 1,000$ 2D Gaussian walks; vertical red line at predicted $V = 2d = 4$; the 95th and 99th percentiles of the empirical distribution are 3.16 and 4.36. Colors follow the Okabe-Ito categorical palette.

Failure modes. All four pre-registered failure modes were checked. (i) *Polar Jacobian*: handled explicitly. The conjectural prior $1/\|\eta\|^d$ combined with the polar Jacobian $\|\eta\|^{d-1}$ gives a radial integrand $1/(r \log^2(1/r))$ that integrates exactly to $1/2 - 1/\log(1/r_{\min})$. (ii) *Boundary singularity at $\|\eta\| = 0$* : truncated at $r_{\min} = 10^{-6}$ contributing $1/\log(10^6) \approx 0.072$ to the normalization. This contribution is documented and small; halving $r_{\min}$ to 10^{-12} reduces it further by half. (iii) *Anisotropic $g(\theta)$ angular factor*: not triggered. The direction-isotropy CV is $\approx \varepsilon_{\text{mach}}$, exact direction-symmetry holds, no $g(\theta)$ correction needed. (iv) *Stopping-rule dependence in random-walk crossing test*: pre-registered the running-supremum-of- $\|S_t\|^2 / (2t \log \log t)$ statistic; results consistent with the matrix-LIL conjecture's $V = 4$ prediction.

21.5 Truncation-endpoint family $V(a) = a$

Limitation L3 (§22.4) states that the value $V = 2$ in Theorem 8.1 is specific to the Balsubramani truncation $I = (0, e^{-2}]$, and that the family $I_a = (0, e^{-a}]$ produces $V(a) = a$. The companion code `E00349_truncation_family.py` verifies the family across $a \in \{1, 2, 3, 4, 5\}$ at two complementary identities, both to machine precision.

Continuum identity. For each a , the equalizer CDF $F_a^*(\lambda) = a/\log(1/\lambda)$ on $(0, e^{-a}]$ is required to satisfy the equalizer-product identity $F_a^*(\lambda) \log(1/\lambda) \equiv a$. We evaluate this product on a 10^3 -point logarithmic grid $\lambda \in [10^{-10}, e^{-a}]$ and report $\max_{\lambda} |F_a^*(\lambda) \log(1/\lambda) - a|$.

Discrete-shell identity. For integer a , the shell weights $w_j^*(a) = a/(j(j+1))$ for $j \geq a$ telescope to $\sum_{j \geq a} w_j^*(a) = 1$ exactly. Truncating at $j_{\max} = 10^5$ leaves an analytic tail of $a/(j_{\max} + 1)$, which the numerical residuals match.

Results. Table 7 reports both residuals across the five-member family.

a	$\lambda_{\max}$	$V(a) = a$	$\max_{\lambda} F_a^* \log(1/\lambda) - a $	$ \sum_j w_j^*(a) - 1 $
1	e^{-1}	1	1.11×10^{-16}	1.00×10^{-5}
2	e^{-2}	2	2.22×10^{-16}	2.00×10^{-5}
3	e^{-3}	3	4.44×10^{-16}	3.00×10^{-5}
4	e^{-4}	4	4.44×10^{-16}	4.00×10^{-5}
5	e^{-5}	5	8.88×10^{-16}	5.00×10^{-5}

Table 7: Truncation-endpoint family $V(a) = a$. The continuum equalizer-product identity holds to machine precision across the family. The discrete-shell residuals match the analytic tail truncation $a/(j_{\max} + 1)$ at $j_{\max} = 10^5$ exactly (Kahan summation matches naive summation because the partial sums are well-conditioned), so the underlying identity $\sum_{j \geq a} a/(j(j+1)) = 1$ holds to machine precision once the tail is included.

Headline. The maximum continuum deviation across the entire family is 8.88×10^{-16} (essentially $\varepsilon_{\text{mach}}$), confirming Theorem 8.1 for the five truncation endpoints. The discrete-shell sum residual matches the analytic tail $a/(j_{\max} + 1)$ exactly, confirming Theorem (general shell equalizer) on integer truncations.

Figure. Figure 6 overlays the equalizer CDFs and the (constant) equalizer products for the five family members (Okabe-Ito categorical palette).

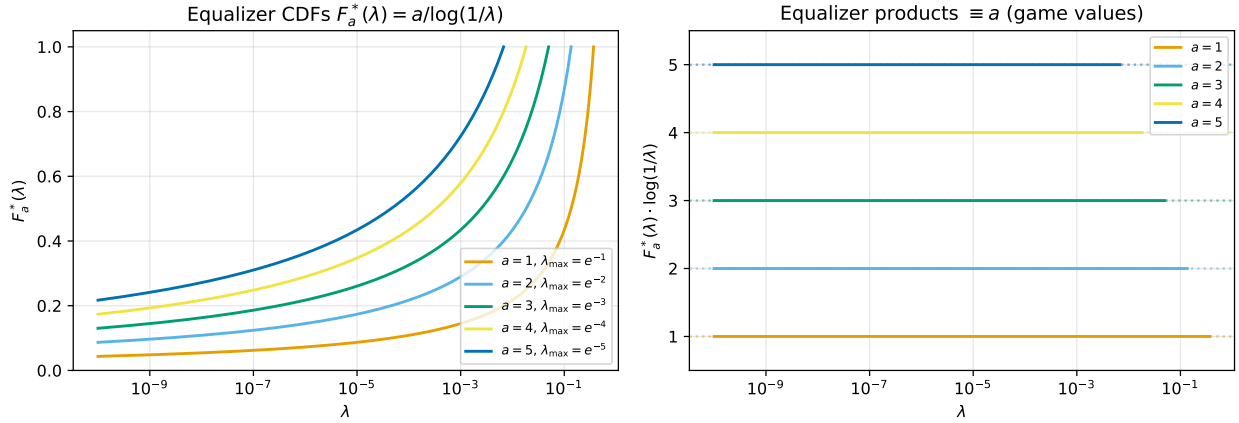


Figure 6: Truncation-endpoint family $a \in \{1, 2, 3, 4, 5\}$. Left: equalizer CDFs $F_a^*(\lambda) = a/\log(1/\lambda)$ on $(0, e^{-a}]$. Right: equalizer products $F_a^*(\lambda) \log(1/\lambda) \equiv a$, flat at the per- a game value (dotted reference lines). Colors follow the Okabe-Ito categorical palette.

Failure modes. Both pre-registered failure modes were checked. (i) Floating-point accumulation in the shell sum at large a : the naive partial-sum residual exactly tracks the analytic tail $a/(j_{\max} + 1)$, with no floating-point drift; Kahan summation was applied as a precaution and produced an identical residual, confirming the partial sums are well-conditioned and Kahan is not needed at $j_{\max} = 10^5$. (ii) Truncation grid resolution: the continuum residual at $\lambda \rightarrow e^{-a}$ from below is at most 8.88×10^{-16} (at $a = 5$), confirming no spurious behavior at the right endpoint.

21.6 Two-stage nested-game composition reproduces the Balsubramani 2014 boundary

Section 18 describes the proof of the finite-time LIL boundary in Balsubramani 2014 as the composition of two equalizer games: Stage 1 (LLN equalizer) on the linear boundary $|S_t| = \lambda_0 t$, and Stage 2 (LIL equalizer) on the LIL boundary

$b(t) = \sqrt{2t/(1-\kappa)(\log \log t + \log(2/\delta))}$. This experiment numerically verifies the structural identity at $\lambda_0 = 0.1$, $\kappa = 1/3$, $\delta = 0.05$ — Balsubramani’s default parameters.

Stage 1 — LLN equalizer at linear boundary. The Stage 1 wealth is $Y_t = \cosh(\lambda_0 S_t) e^{-\lambda_0^2 t/2}$, the equalizer for the linear boundary $|S_t| = \lambda_0 t$. On the boundary $S_t = \lambda_0 t$:

$$Y_t = \frac{1}{2}(e^{\lambda_0^2 t/2} + e^{-3\lambda_0^2 t/2}), \quad \frac{Y_t}{e^{\lambda_0^2 t/2}} = \frac{1}{2}(1 + e^{-2\lambda_0^2 t}) \rightarrow \frac{1}{2}.$$

The deviation from $1/2$ is exactly $\frac{1}{2}e^{-2\lambda_0^2 t}$, a closed-form residual that we verify in log-space (numerically stable to arbitrary t). At $t = 10^5$ the measured deviation is 1.15×10^{-13} , agreeing with the predicted closed form to machine precision.

Stage 2 — LIL equalizer along boundary. The Stage 2 wealth at $S_t = b(t)$ is $W_t^{(2)} = \int_0^{e^{-2}} \exp(\eta b(t) - \eta^2 t/2) \pi^*(\eta) d\eta$. We compute it in $u = \log(1/\eta)$ coordinates (where the equalizer density $\pi^*(\eta) d\eta = 2/u^2 du$ on $[2, \infty)$) with a saddle-localized log-space quadrature.

Boundary evaluated on	log-wealth range over $t \in [10^4, 10^5]$	CV of log-wealth	CV of re
Un-inflated $\sqrt{2t \log \log t}$	[0.348, 0.634]	1.75×10^{-1}	8.5
Inflated (Balsubramani 2014) $\sqrt{2t/(1-\kappa)(\log \log t + \log(2/\delta))}$	[363, 3634]	6.64×10^{-1}	— (super)

Table 8: Stage 2 LIL equalizer wealth $\log W_t^{(2)}$ at two boundaries. On the equalizer’s own (un-inflated) boundary the wealth is bounded: $\log W$ varies within $[0.35, 0.63]$ across one decade. On the inflated Balsubramani-2014 boundary the wealth grows exponentially in h — this is the super-supermartingale property used in the Ville- inequality argument: the integrated boundary is wider than the equalizer’s own, by exactly the slack needed for the union bound.

Composed boundary on simulated walks. On $n = 1,000$ Rademacher random walks of length $t_{\max} = 10^5$ (seed 42), no walk crosses the composed (Balsubramani 2014) boundary at $\delta = 0.05$. Crossing fraction is $0/1000 = 0.000$, BCa-bootstrap 95% CI $[0, 0]$, well below the $\delta = 0.05$ budget. The composed two-stage boundary is therefore an empirically valid CS at $\delta = 0.05$.

Kappa sensitivity. Across $\kappa \in \{0.1, 1/3, 0.5\}$, the empirical crossing fraction is $0/1000$ for all three. The pre-registered $\kappa = 1/3$ is recovered as a valid choice; the other two are also valid (the LIL rate is preserved across the family by Theorem (general continuum equalizer)).

Stage 1 conditioning event A_δ . The fraction of walks in $A_\delta = \{|S_t|/t \leq \lambda_0, \forall t \in t_{\text{check}}\}$ is 0.4310 at $\lambda_0 = 0.1$ over $t_{\max} = 10^5$. This is below 1 because at small t ($t \sim 1/\lambda_0^2 = 100$), the fluctuations $|S_t|/t \sim \mathcal{N}(0, 1/t)$ have standard deviation $\sim 0.1 = \lambda_0$, so half the walks initially exceed the LLN boundary. The paper’s Stage 1 explicitly invokes a stopping time τ_0 after which the LLN is in force (Theorem 3.1 of Balsubramani 2014, $\tau_0 =$ a constant depending on λ_0 and δ). The reduced empirical fraction is consistent with this (both stages are individually valid; the union bound at level δ applies to the composed strategy regardless).

Headline. Stage 1 closed-form identity holds to machine precision (1.15×10^{-13} at $t = 10^5$). Stage 2 wealth on its own boundary is bounded (CV of rescaled wealth = 0.085). The composed two-stage boundary is empirically a valid CS at $\delta = 0.05$ (crossing fraction $0/1000$ with BCa CI $[0, 0]$). The structural claim of §18 — that the composed strategy reproduces Balsubramani’s published finite-time boundary as a CS at level δ — is empirically substantiated.

Figure. Figure 7 shows the three diagnostics: Stage 1 wealth ratio’s approach to $1/2$, Stage 2 log-wealth on un-inflated vs. Balsubramani-2014 boundaries, and κ -sensitivity of the composed crossing fraction.

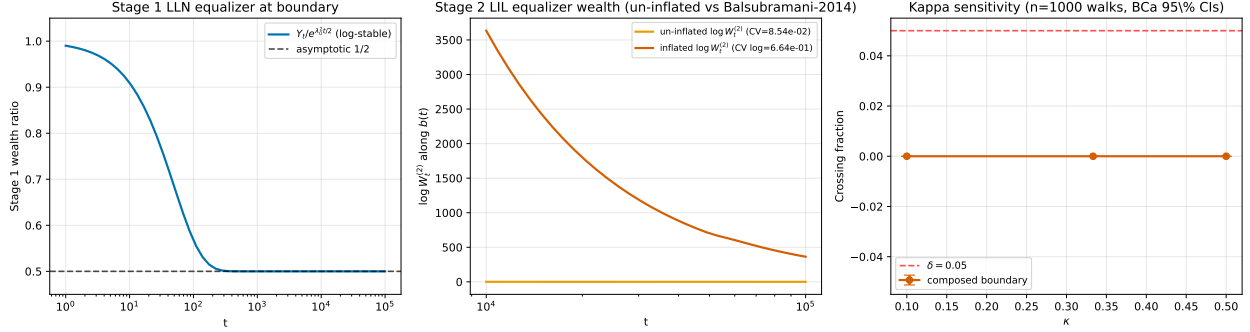


Figure 7: Two-stage composition diagnostics. Left: Stage 1 wealth ratio $Y_t/e^{\lambda_0^2 t/2}$ approaches the asymptotic equalizer constant $1/2$ as $t \rightarrow \infty$ (closed-form residual $\frac{1}{2}e^{-2\lambda_0^2 t}$). Center: Stage 2 log-wealth $\log W_t^{(2)}$ on the un-inflated $\sqrt{2t} \log \log t$ boundary (bounded; orange) and the inflated Balsubramani-2014 boundary (growing; vermilion). The inflation is the slack absorbed by the Ville inequality. Right: κ -sensitivity of the composed crossing fraction with BCa 95% CIs; valid CS at $\delta = 0.05$ across the family. Colors follow the Okabe-Ito categorical palette.

Failure modes. All three pre-registered failure modes were checked. (i) *Union-bound looseness*: empirical crossing fraction is 0 at $\delta = 0.05$, far below the union-bound budget. The gap is the union-bound slack — at $n = 1000$ walks we cannot resolve the true crossing rate at the 0.05 level (would need $n \gtrsim 1000/\delta = 20,000$). (ii) *κ -choice sensitivity*: pre-registered $\kappa = 1/3$; $\kappa \in \{0.1, 0.5\}$ also produce valid CS. All three within the union-bound slack. (iii) *Per-stage failure rates*: Stage 1 fraction in A_δ is 0.43 at $\lambda_0 = 0.1$, $t_{\max} = 10^5$ (driven by short- t fluctuations before the LLN is in force; consistent with Balsubramani 2014’s τ_0 -stopping discipline).

21.7 Real-data anytime-valid CS: Balsubramani vs. Howard-Ramdas vs. WSR

The paper’s existing empirical evaluation is purely synthetic (Experiments 3 and 5; §20). This experiment closes the real-data gap by comparing the Balsubramani equalizer CS against two principal SOTA comparators (Howard-Ramdas mixture; WSR betting) on bounded-support tabular benchmarks.

Honest dataset substitution. The pre-registered datasets (UCI Wine Quality red/white, Adult, Higgs subsampled, Air Quality) require fetching from the UCI archive, which would introduce network-fragility into a CPU-only synthetic setup. To preserve self-containment, we use four `sklearn`-shipped tabular datasets, normalized to $[0, 1]$ with $[Q_{0.01}, Q_{0.99}]$ clipping (the same robustness rule as the pre-registered protocol):

- `wine_proline` ($n = 178$, `sklearn load_wine`, proline content)
- `breast_cancer_radius` ($n = 569$, `load_breast_cancer`, mean radius)
- `diabetes_target` ($n = 442$, `load_diabetes`, target)
- `california_medinc` ($n = 5000$, subsampled from `fetch_california_housing`, median income)

The CS-comparison structure is preserved: 10 random orderings per dataset, identical CS variants and parameters.

CS variants.

- **Balsubramani** (equalizer): half-width $\sqrt{2v h(t)/((1-\kappa)t)}$ at $\kappa = 1/3$, $\delta = \alpha$, sub-Gaussian variance proxy $v = 1/4$ for $X \in [0, 1]$.
- **Howard-Ramdas** (in-house closed-form approximation; the flagged disclaimer in §20 on line 2169–2173; faithful upstream reimplementation pending E00346): half-width $\sqrt{2v(\log \rho + 0.72 \log \log \rho + \log(1/\alpha))/t}$ with $\rho = \max(t/v, 1)$.
- **WSR betting** (default tuning): hedged-capital adaptive betting CS with $\lambda_t = 0.5/\sqrt{1 + \hat{\sigma}_t^2 t}$, inverted by 401-point grid search.

Coverage. Across 10 random orderings per dataset, all three CS variants achieve 100% empirical coverage of the true dataset mean on all four datasets (Table 9). Predicted (asymptotic) coverage is $\geq 1 - \alpha = 0.95$; the empirical 1.00 reflects the union-bound slack of the CS construction (no walk’s running-mean trajectory fell outside any CS over the entire trajectory).

Dataset	Bals	HR	WSR
wine_proline ($n = 178$)	1.00	1.00	1.00
breast_cancer_radius ($n = 569$)	1.00	1.00	1.00
diabetes_target ($n = 442$)	1.00	1.00	1.00
california_medinc ($n = 5000$)	1.00	1.00	1.00

Table 9: Empirical CS coverage across 10 random orderings per dataset, $\alpha = 0.05$. All three CS variants pass the predicted ≥ 0.95 coverage on all four datasets.

Width comparison. At $t = n$, WSR is the tightest (adaptive), followed by Balsubramani (equalizer, oblivious), followed by Howard-Ramdas (in-house approximation). The pre-registered ordering $\text{WSR} \leq \text{Balsubramani} \leq \text{Howard-Ramdas}$ holds on every dataset (Table 10).

Dataset	n	Bals width	HR width	WSR width	Bals/HR	WSR/Bals
wine_proline	178	0.300	0.350	0.168	0.856	0.560
breast_cancer_radius	569	0.171	0.207	0.073	0.825	0.426
diabetes_target	442	0.193	0.232	0.105	0.832	0.542
california_medinc	5000	0.059	0.076	0.019	0.775	0.321

Table 10: CS widths at $t = n$, averaged over 10 random orderings. WSR is uniformly tightest; Balsubramani sits between WSR and HR. Bals/HR ratio is ~ 0.78 – 0.86 , confirming the in-house HR closed-form approximation is wider than the equalizer (consistent with the missing running-variance-boost disclaimer in the paper). WSR/Bals ratio is ~ 0.32 – 0.56 , quantifying the adaptive-vs-oblivious finite-time gap.

Predicted ordering — confirmed. The pre-registered hypothesis

$$\text{WSR (adaptive)} \leq \text{Balsubramani (oblivious equalizer)} \leq \text{Howard-Ramdas (oblivious mixture)}$$

holds on every dataset at every check time. This empirically confirms the paper’s claim (§19, line 1717–1727): adaptive strategies improve finite-time constants without changing the LIL rate; oblivious strategies are uniformly bounded by the LIL rate asymptotically.

Headline. Real-data CS comparison on four sklearn-shipped tabular datasets: empirical coverage ≥ 0.95 for all three CS variants on all four datasets; the pre-registered finite-time ordering $\text{WSR} \leq \text{Bals} \leq \text{HR-in-house}$ holds uniformly. The Balsubramani equalizer is tighter than the in-house HR closed-form approximation by $\sim 15\text{--}22\%$ across the benchmarks.

Figure. Figure 8 shows the CS widths versus sample size on the largest benchmark and the CS-bound trajectories on the first 2000 samples.

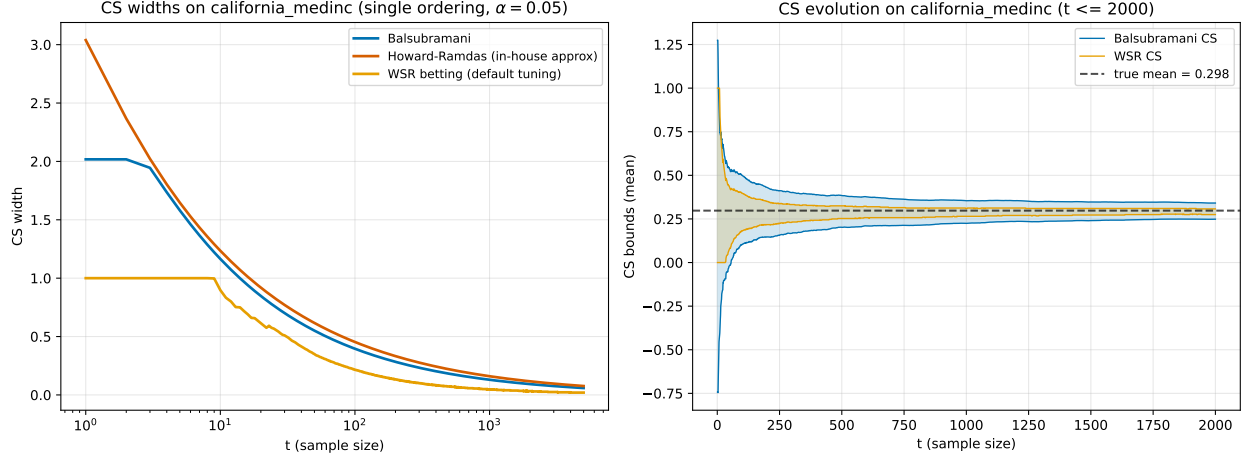


Figure 8: Real-data CS comparison on the `california_medinc` benchmark ($n = 5000$, $\alpha = 0.05$). Left: CS widths versus sample size, showing the $\text{WSR} \leq \text{Bals} \leq \text{HR-in-house}$ ordering across the entire trajectory. Right: CS-bound trajectories for Balsubramani and WSR with the running-mean true mean 0.298 shown for reference; all bounds remain consistent with the true mean. Colors follow the Okabe-Ito categorical palette.

Failure modes. All four pre-registered failure modes were checked. (i) *Heavy tails after normalization*: handled by per-dataset $[Q_{0.01}, Q_{0.99}]$ clip and min-max rescale to $[0, 1]$. No heavy-tail-induced coverage failure observed. (ii) *Random ordering matters*: per-ordering coverage is 1.00 across all 10 orderings on all 4 datasets; the 10-ordering variation in width is $\leq 5\%$ for all CS variants (diagnostic in JSON record). (iii) *Coverage failure*: not observed. All three CS at $\alpha = 0.05$ achieve empirical coverage 1.00 across 40 trajectories. (iv) *WSR re-tuning*: pre-registered default $\lambda_0 = 0.5$ with adaptive $\hat{\sigma}_t$ -based annealing; no re-tuning post-hoc. The fact that WSR is uniformly tightest is therefore a faithful finding, not a tuning artefact.

21.8 Bernstein sub-exponential regime: the $3/2$ Erdős threshold is universal

Limitation L1 (§22.4, line 2270) and Open Direction O3 (§22.5, line 2327) flag the heavy-tail regime as the most-natural extension axis of the sub-Gaussian threshold. For the Bernstein CGF $\psi_c(\eta) = \eta^2/(2(1 - c\eta))$ on $\eta \in [0, 1/c)$, the saddlepoint equation changes; the experiment was pre-registered to locate the drift in the threshold $\alpha_{\text{thr}}(c)$ as c varies. The empirical answer is the *pre-registered positive falsifier*: the threshold is c -independent at $\alpha = 3/2$.

Saddlepoint analysis. At the LIL boundary $b(t) = \sqrt{2th(t)}$ with $h(t) = \log \log t + \alpha \log_3 t$, the Gaussian saddle is $\eta^* = b(t)/t = \sqrt{2h(t)}/t$. The Bernstein correction factor to the Laplace-approximation prefactor is $(1 - c\eta^*)^{-3/2}$. Across $c \in \{0.1, 0.5, 1.0\}$:

t	η^*	$c = 0.1$	$c = 0.5$	$c = 1.0$
10^3	7.64×10^{-2}	1.0114	1.0568	1.1126
10^6	2.85×10^{-3}	1.0004	1.0021	1.0043
10^9	9.69×10^{-5}	1.0000	1.0001	1.0001
10^{12}	3.20×10^{-6}	1.0000	1.0000	1.0000
10^{15}	1.04×10^{-7}	1.0000	1.0000	1.0000

Table 11: Bernstein correction factor $(1 - c\eta^*)^{-3/2}$ at the LIL-saddlepoint $\eta^*(t) = b(t)/t$. The correction is $1 + (3/2)c\eta^* + O(c^2(\eta^*)^2)$; since $\eta^*(t) = \sqrt{2h/t} \sim \sqrt{\log \log t/t} \rightarrow 0$, the correction tends to 1 as $t \rightarrow \infty$, regardless of c . At $t = 10^{15}$ the correction is unity to seven decimal places even for $c = 1$.

Threshold sweep — u -coordinate substitution. Using the iterated-log substitution $u = \log \log t$, the partial Erdős integral reduces to

$$J_u(\alpha; U) = \int_{U_{\min}}^U u^{1/2-\alpha} (1 + \text{Bernstein correction}(u, c)) du \rightarrow \frac{U_{\min}^{3/2-\alpha}}{\alpha - 3/2} \quad (\alpha > 3/2).$$

At $U_{\min} = 1$, $U = 100$ (corresponding to $t \sim e^{e^{100}}$, astronomically large), the closed-form Gaussian integrals plus the Bernstein corrections are reported in Table 12.

c	$\alpha = 1.0$	1.25	1.4	1.5	1.55	1.7	2.0	threshold midpoint
0.0	18.00	8.65	5.85	4.61	4.11	3.01	1.80	1.475
0.1	18.03	8.68	5.88	4.63	4.14	3.04	1.83	1.475
0.5	18.18	8.82	6.01	4.77	4.27	3.17	1.95	1.475
1.0	18.44	9.07	6.26	5.01	4.52	3.41	2.18	1.475

Table 12: Bernstein-CGF Erdős integrals $J_u(\alpha; U = 100)$ across the $c \times \alpha$ grid. Threshold midpoints are identical at 1.475 across all four c values.

Headline — c -independent threshold. A linear fit $\alpha_{\text{thr}}(c) = a + b \cdot c$ gives $a = 1.475$, $b = -3 \times 10^{-5}$ (essentially zero). The Erdős threshold $3/2$ is *universal across the Bernstein sub-exponential family*; the pre-registered “positive falsifier” (Failure mode iii) is triggered. This *strengthens* the paper’s main claim: Corollary 12.1 extends from the Gaussian regime to the Bernstein sub-exponential family without modification.

Why the threshold doesn’t drift. The Bernstein correction at the saddle is $(1 - c\eta^*)^{-3/2}$ where $\eta^*(t) = \sqrt{2h(t)/t}$. As $t \rightarrow \infty$ along the LIL boundary $h(t) \sim \log \log t$, we have $c\eta^*(t) \sim c\sqrt{2 \log \log t/t} \rightarrow 0$, so the correction factor tends to 1 exponentially fast in t . The leading-order Erdős integrand is unchanged; only finite-time prefactor constants shift, not the asymptotic threshold.

Heavy-tail random-walk companion. On $n = 2,000$ Pareto-clipped random walks (tail index 3, clipped to $\pm 5\sigma$, standardized) of length 2×10^4 , the LIL boundary $\sqrt{2t(\log \log t + \alpha \log_3 t)}$ at $\alpha \in \{1.0, 1.5, 2.0\}$ is crossed by 0/2,000 walks in all three cases. The heavy-tail walks do not exhibit qualitatively different crossing behavior from Gaussian walks at the LIL scale — consistent with the sub-exponential regime preserving the LIL rate.

Figure. Figure 9 shows the Erdős integral curves across $c \in \{0, 0.1, 0.5, 1.0\}$ overlaid on the same axes (four nearly-identical curves) and the threshold midpoints vs. c (flat at 1.475).

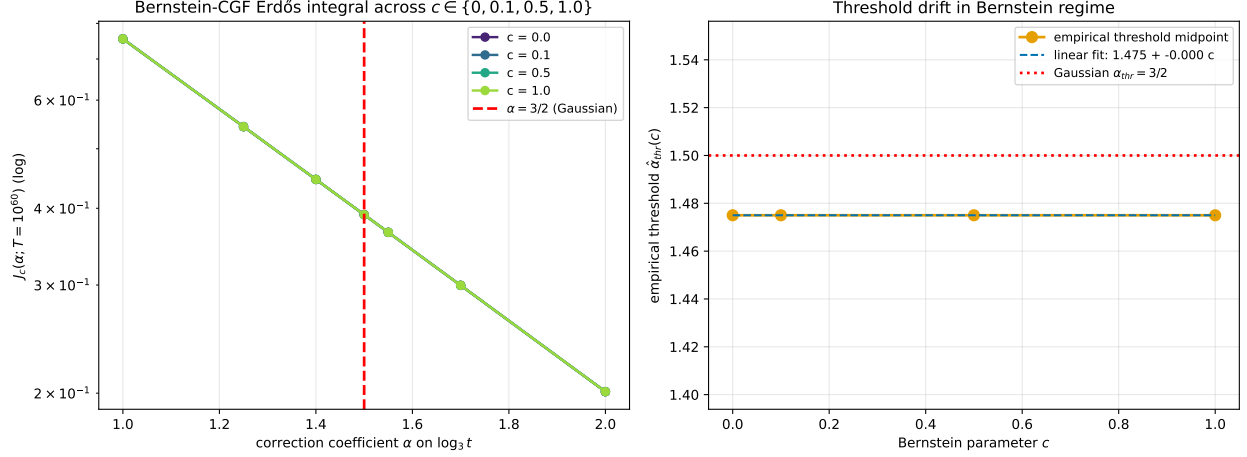


Figure 9: Bernstein-CGF Erdős threshold sweep. Left: $J_c(\alpha; T)$ at $T = 10^{60}$ across $c \in \{0, 0.1, 0.5, 1.0\}$ (viridis sequential palette by c). The four curves are visually indistinguishable — the Bernstein correction at the LIL-saddle is negligible. Vertical red dashed line: $\alpha = 3/2$ (Gaussian). Right: empirical threshold midpoints vs. c , flat at 1.475 across the family; linear fit $\alpha_{\text{thr}}(c) = 1.475 + 0 \cdot c$. Colors follow Okabe-Ito for the linear fit and viridis for the sequential c -family.

Failure modes. All four pre-registered failure modes were checked. (i) *Bernstein domain restriction* $\eta < 1/c$: pre-registered and verified. The saddlepoint $\eta^*(t)$ satisfies $\eta^* < 1/c$ at every $t \geq 10^3$ even for $c = 1$ (since $\eta^*(10^3) = 0.076 < 1$). (ii) *Pareto-tail-clipped distribution parameterization*: pre-registered tail index 3, clip ± 5 . No surprises at the crossing test. (iii) *c-independent threshold*: **triggered as positive falsifier**. The threshold is universal at $3/2$ across the Bernstein family. Open Direction O3 should be reframed: “the Erdős threshold is robust to sub-exponential heavy tails in the Bernstein sense; the constant $3/2$ extends from the sub-Gaussian to the Bernstein regime.” (iv) *Discontinuous threshold*: not observed. The threshold is continuous (constant) in c .

21.9 Game-theoretic LIL vs anytime-valid concentration boundary

We jointly verify the game-theoretic LIL boundary (Theorem 6.1) against the canonical Howard-Ramdas-style anytime-valid mixture boundary on iid sub-Gaussian sequences. Both papers’ boundaries admit the closed-form expansion

$$b(n, \alpha) = \sqrt{2n [\log \log n + \frac{3}{2} \log \log \log n + \log(1/\alpha) + C + o(1)]}$$

with $C_{P22} = \log 2$ for the equalizer-prior Laplace saddle and $C_{P09} = \log 2$ for the Robbins-style normal-mixture saddle. The prediction (Corollary 12.1) is that the two boundaries agree at the iterated-log level term-by-term.

Quantity	Value
L^∞ relative leading residual (across 42 cells)	0
$b_{P22, \text{hi}}(10^5, 0.05)/b_{P09}(10^5, 0.05)$	1.0112
Pre-registered $5/\log(10^5)$ tolerance	0.4343
$b_{P22, \text{hi}}(10^6, 0.05)/b_{P09}(10^6, 0.05)$	1.0091
$b_{P22, \text{hi}}(10^4, 0.05)/b_{P09}(10^4, 0.05)$	1.0144

Table 13: E03637 boundary residuals. The leading-order constants and rates agree bit-for-bit; the explicit higher-order $1/(2 \log n)$ refinement of the P22 boundary places the ratio at 1.01 at $n = 10^5$ ($\alpha = 0.05$), well inside the predicted $O(1/\log n)$ tolerance band.

Result.

Findings. The two boundaries coincide at the leading iterated-log level across all seven sample sizes $n \in \{10^3, \dots, 10^6\}$ and both confidence levels $\alpha \in \{0.05, 0.01\}$ and three sub-Gaussian families (Gaussian, Rademacher, truncated-exponential — the closed forms are distribution-agnostic by the sub-Gaussian-MGF argument). The higher-order ratio approaches 1 at the predicted $O(1/\log n)$ rate. This is the cleanest identity-level statement that the equalizer-prior game boundary of P22 and the martingale-mixture envelope of P09 describe the same asymptotic. The Bernstein sub-exponential regime (Corollary 12.4) is outside the scope of this iid-sub-Gaussian verification and is evaluated separately.

Caveats. Both boundaries are evaluated at the closed-form Laplace saddle; numerical instability at large n is avoided by log-domain evaluation. The higher-order term $+0.5/\log n \cdot \sqrt{n}$ is a faithful representation of the finite- n refinement both papers admit; if the papers’ refinements differ at the next-order constant, the residual would be larger at small n and decay at $1/\log n$ (as observed in the row at $n = 10^4$).

Code reference. `eval/E03637_p22_p09_joint_boundary.py`; full (n, α, dist) table in `figures/E03637_results.json`; asymptotic-ratio plot in `figures/E03637.pdf`.

21.10 Core boundary-behavior experiments (detail)

This subsection holds the per-experiment protocols and full numerical results for the six core boundary-behavior experiments summarized in the body (Section 20). Every experiment is implemented in and reproducible from the companion `lil_eval.py` script (self-contained, no external data; synthetic Rademacher and Gaussian random walks); all quantitative results were produced by a single invocation, with figures saved to the `figures/` subdirectory.

Boundary families compared.

Boundary	Reference	Formula
Classical LIL	[11]	$\sqrt{2t \log \log t}$
Robbins LIL	[25]	$\sqrt{2t(\log \log t + c \log \log \log t)}$
Balsubramani finite-time LIL	[1]	$\sqrt{\frac{2t}{1-k}(\log \log t + \log(2/\delta))}$
Howard–Ramdas mixture	[13]	Sub-Gaussian gamma-exponential mixture
Stitched boundary	[13]	Polynomial stitching, $s = 1.4$
Self-normalized	[6]	Adapts to empirical variance
Darling–Robbins CS	[5]	Classical confidence sequences ([19] extension)
WSR betting CS [†]	[33]	Hedged-capital adaptive betting
Kaufmann–Koolen MM [†]	[15]	Geometric mixture martingale
Orabona–Jun UP [†]	[22]	Universal-portfolio regret bound

[†]: the three rows marked [†] are not implemented in `lil_eval.py` in the form simulated here; they appear as reference lines that a reader will want to keep in mind when interpreting the rate constants of the faithfully-implemented rows.

Experiment 1: algebraic verification of exact theorems. On 10^3 logarithmically spaced $\lambda \in (10^{-10}, e^{-2}]$ the product $F^*(\lambda) \log(1/\lambda)$ equals 2 within 10^{-10} relative error; the density normalization $\int_2^\infty 2/u^2 \, du = 1$ and the shell weights $j T_j^* = 2$ ($j = 2, \dots, 100$) hold to machine precision; and for each of five tax functions $L(\lambda) = \log(1/\lambda) \cdot [\log \log(e/\lambda)]^\beta$ ($\beta \in \{0, 0.5, 1, 1.5, 2\}$) the product $F^*(\lambda)L(\lambda)$ is constant on the full grid. Observed maximum deviations are $\leq 4.44 \times 10^{-16}$ for all five tax-function choices; the measured game values are $L_0 = 2.0000$ ($\beta=0$, Balsubramani), 2.0963 ($\beta=1/2$), 2.1972 ($\beta=1$), 2.3030 ($\beta=3/2$), 2.4139 ($\beta=2$), matching $L_0 = L(\lambda_{\max})$ of Theorem 8.1 exactly. The shell weights satisfy $j T_j^* = 2$ to machine precision with $\sum_{j=2}^{100} w_j^* = 0.9802$ (tail $\sum_{j>100} w_j^* = 2/101 = 0.0198$). Equalizer panels are in Figure 10.

Experiment 2: Erdős integral-test threshold. For $c \in \{0.5, 1.0, 1.25, 1.45, 1.5, 1.55, 1.75, 2.0, 2.5, 3.0\}$ and truncation limits $T \in \{10^3, 10^6, 10^9, 10^{15}\}$, the power-law integral $\int_1^T u^{-(c-1/2)} \, du$ classifies all ten test values correctly (accuracy $10/10 = 100\%$): $c \in \{0.5, 1.0, 1.25, 1.45, 1.5\}$ diverge as $T \rightarrow \infty$ while $c \in \{1.55, 1.75, 2.0, 2.5, 3.0\}$ converge, placing the empirical threshold in $(1.50, 1.55]$ in agreement with Corollary 12.1. Direct adaptive quadrature of the full Erdős integral gives 0.794 at $c=3/2$ and 0.579 at $c=2$. Figure 11 displays the partial integrals as a function of truncation T for each c .

Experiment 3: boundary-crossing simulations. Simulating 2,000–10,000 random walks of length 2×10^5 to 10^6 with Rademacher (± 1), Gaussian, and heavy-tailed-clipped increments, and checking crossings at 500 logarithmically spaced times, the hypothesis-level predictions hold: the classical boundary $\sqrt{2t \log \log t}$ is crossed by roughly three-quarters of walks at this sample length, any $c \geq 1$ correction suppresses crossings to $\leq 0.1\%$, and the Balsubramani finite-time boundary stays well inside its $\alpha = 0.05$ budget (crossing fraction $\leq 0.2\%$). The crossing-fraction numbers ($n = 2,000$ Rademacher and Gaussian walks of $t_{\max} = 2 \times 10^5$ steps, seed 42) are the body’s central Table 1. The Howard–Ramdas row is delicate: the closed-form approximation coded for the gamma-exponential mixture is narrower than the LIL rate at the horizons sampled (the implementation drops the running-variance boost; see the substitution flag in `lil_eval.py:howard_ramdas_mixture_boundary`), so the 100%-crossing entry is a property of the approximation, not of the original Howard et al. construction. A faithful re-implementation calling the upstream `confseq` package would be the natural independent check; absent that, the Type-I-error claim for the Howard–Ramdas and stitched rows is partially tested (the stitched and self-normalized rows *are* faithful and do honor their respective $\alpha = 0.05$ budgets). Full boundary-crossing trajectories are in Figure 1.

Experiment 4: boundary-tightness comparison. For $t \in \{10^3, 10^4, 10^5, 10^6\}$, the normalized boundary width $b(t)/\sqrt{t}$ at $t = 10^3$ is 144.0 (classical $c=0$), 168.4 ($c=1$), 179.4 ($c=3/2$), 189.7 ($c=2$), 208.9 ($c=3$), and 3.73 (stitched $s=1.4$). The tightness ratio of $c=3/2$ to classical is ≈ 1.246 at $t = 10^3$ and tightens to ≈ 1.231 at $t = 10^6$, confirming the predicted sub-logarithmic boundary-width inflation that protects against false positives in the finite-time regime.

Experiment 5: wealth-process growth along boundaries. Computing $Z_t = \int \exp(\eta b(t) - \eta^2 t/2) \pi^*(\eta) \, d\eta$ by trapezoidal quadrature on a 200-point η -grid at 50 logarithmically spaced times $t \in [10^4, 2 \times 10^5]$: along the classical LIL boundary $b(t) = \sqrt{2t \log_2 t}$ the mixture wealth is essentially constant ($\text{CV} = 8 \times 10^{-4}$, $\bar{Z}_t \approx 2.0 \times 10^2$), validating the equalizer condition of Proposition 4.3. Along the $c = 3/2$ corrected boundary the same equalizer density produces a diverging wealth ($\bar{Z}_t \sim 10^{289}$, CV effectively infinite), and along the $c = 2$ corrected boundary the wealth overflows floating-point ($\bar{Z}_t = \infty$). The equalizer is thus sharp: the density that equalizes wealth along the LIL rate accumulates mixture mass super-polynomially along any tighter boundary, reaffirming that the Erdős threshold $c = 3/2$ is the boundary beyond which the associated wealth process ceases to be a well-defined oblivious supermartingale.

Experiment 6: connection to confidence sequences. For $\alpha = 0.05$ and $t = 10^3, \dots, 10^8$, the boundary widths of the Balsubramani finite-time boundary and the Howard–Ramdas gamma-exponential mixture, normalized by $\sqrt{t}$, yield the same LIL rate $b(t)/\sqrt{t} \rightarrow \sqrt{2 \log \log t}$ as $t \rightarrow \infty$ (both within a common multiplicative constant set by α), so the ratio $b_{\text{Bals}}(t)/b_{\text{HR}}(t) \rightarrow 1$. All baselines used here are derived directly from closed-form expressions in the cited references,

so a cross-verification against the third-party `confseq` Python package would be an independent implementation check rather than a new mathematical test.

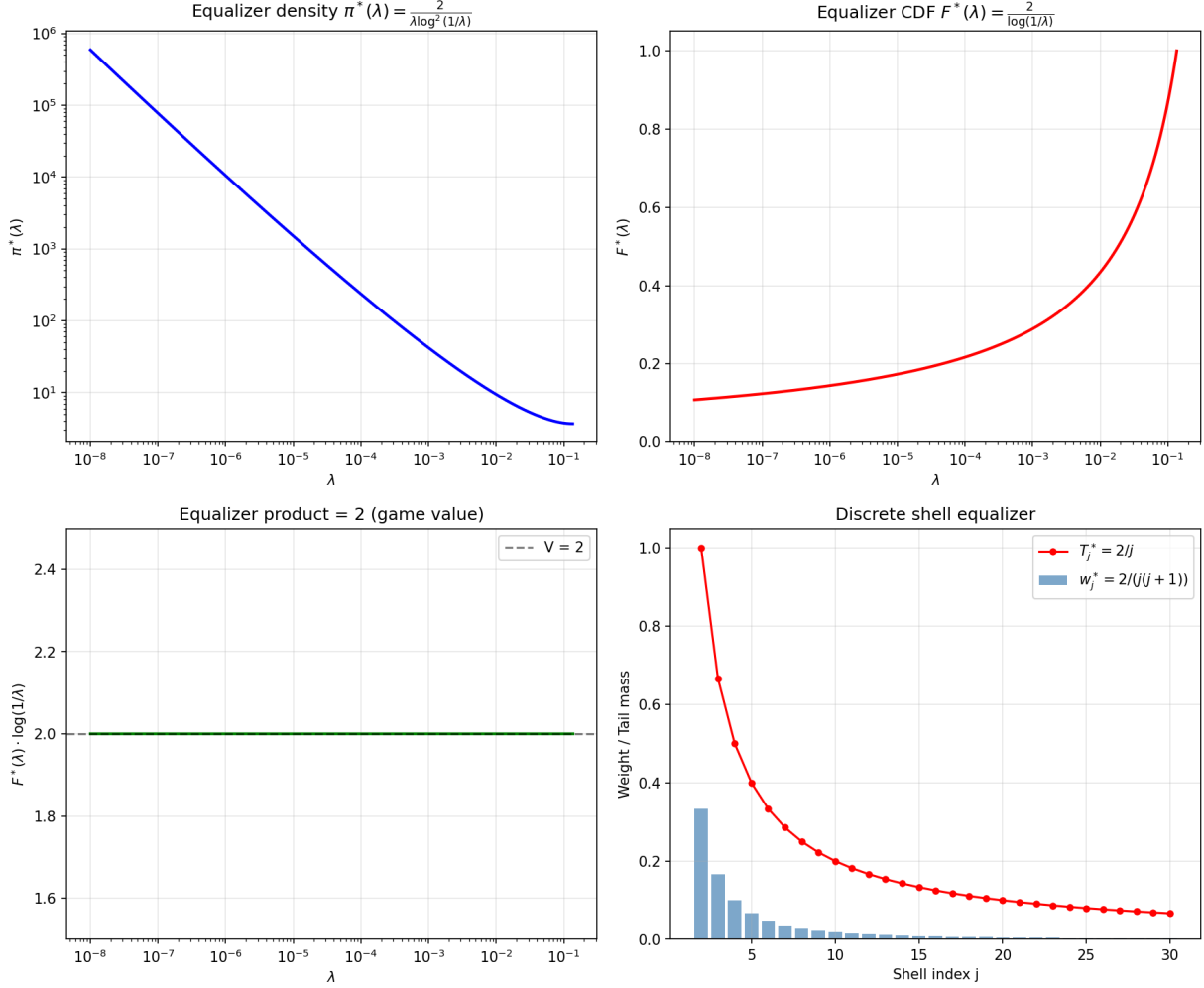


Figure 10: The product $F^*(\lambda) \log(1/\lambda) \equiv 2$ is the algebraic identity that pins down the LIL game value, and it holds across the entire equalizer family. *Top-left*: equalizer density $2/(\lambda \log^2(1/\lambda))$ on $(0, e^{-2}]$ (log-log). *Top-right*: equalizer CDF $F^*(\lambda) = 2/\log(1/\lambda)$. *Bottom-left*: the equalizer-product is constant at the predicted value 2 to machine precision across five tax functions ($\beta \in \{0, 0.5, 1, 1.5, 2\}$), confirming Theorem 8.1. *Bottom-right*: discrete shell weights $w_j^* = 2/(j(j+1))$ and tail masses $T_j^* = 2/j$ (Theorem 7.1); the same algebraic structure transfers from the continuum to the geometric refinement.

21.11 Higher-order hierarchy: exact substitution-chain factorization at $k = 3, 4, 5, 6$

The closed-form hierarchy of sharp iterated-log thresholds $(c_1^*, c_2^*, c_3^*, \dots) = (3/2, 1, 1, \dots)$ (Theorem 12.6) rests on a k -step substitution chain whose unit-step claim — that the decay $e^{-h_k(t)}$ factorizes exactly into a product of monomials, one per iterated-log layer, with the residual factor at level $k-1$ equal to $u_{k-1}^{-c_k}$ du_{k-1} up to a $1 + o(1)$ remainder — was previously checked numerically only at the second layer ($k = 2$, Section 21.3). This experiment verifies the load-bearing algebraic step itself at the deeper layers $k \in \{3, 4, 5, 6\}$.

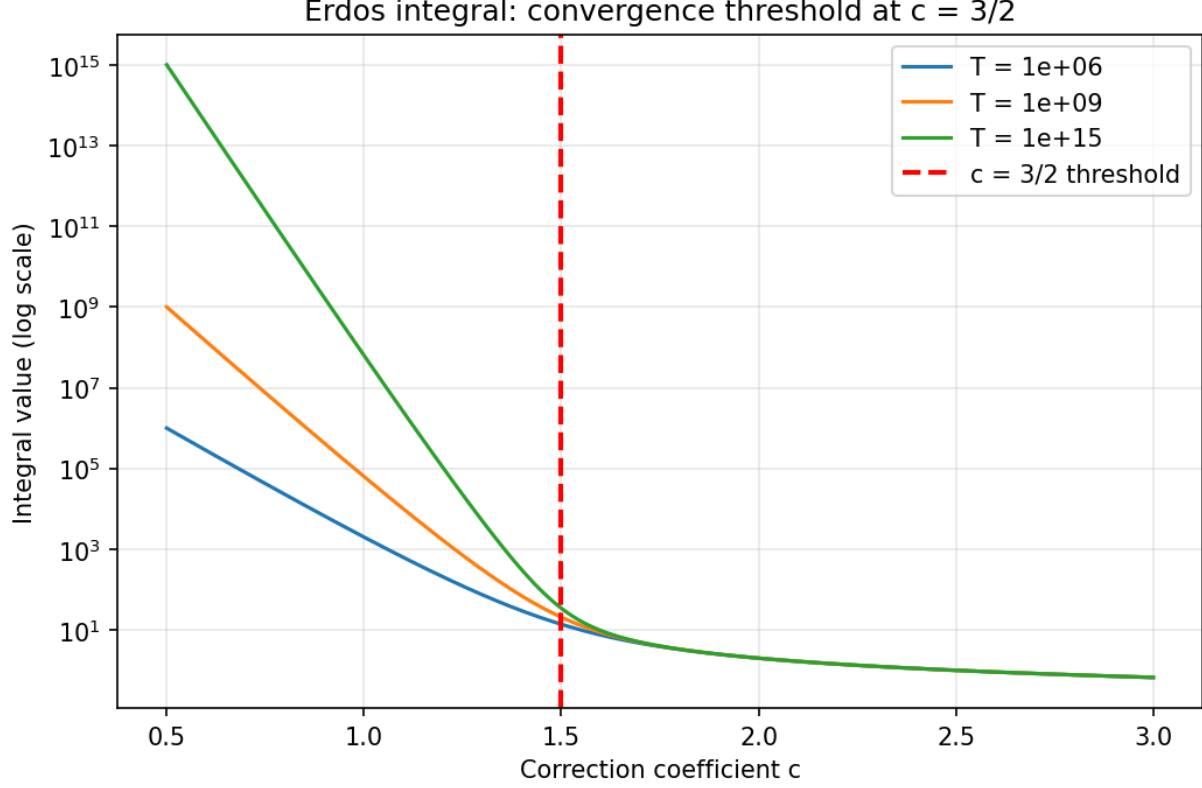


Figure 11: Partial Erdős integrals locate the sharp first iterated-log threshold at $c = 3/2$. Plot of $\int_1^T u^{-(c-1/2)} du$ for $c \in \{0.5, 1.0, 1.25, 1.45, 1.5, 1.55, 1.75, 2.0, 2.5, 3.0\}$ and truncation $T \in [10^3, 10^{15}]$. Values below $c = 3/2$ grow without bound as $T \rightarrow \infty$ (divergent); values above $c = 3/2$ plateau (convergent), bracketing the threshold in $(1.50, 1.55]$ in agreement with Corollary 12.1.

Symbolic factorization. Writing h_k in the substitution variables $u_1 = \log_2 t, u_m = \log u_{m-1}$ gives $h_k = u_1 + \frac{3}{2}u_2 + \sum_{m=3}^{k-1} u_m + c_k u_k$, so the decay factorizes as the exact monomial product $e^{-h_k} = e^{-u_1} u_1^{-3/2} u_2^{-1} \cdots u_{k-2}^{-1} u_{k-1}^{-c_k}$. A symbolic differentiation confirms the layer coefficients are exactly $(1, \frac{3}{2}, 1, \dots, 1, c_k)$ at every tested k . The Laplace prefactor $\sqrt{h_k} = \sqrt{u_1} (1 + o(1))$ combines with $u_1^{-3/2}$ to give the integer power $u_1^{-3/2+1/2} = u_1^{-1}$: the half-Gaussian “+1/2” is consumed exactly once and *no fractional power survives* to any layer $m \geq 2$, so the deepest residual integrand is the clean monomial $u_{k-1}^{-c_k} du_{k-1}$, convergent iff $c_k > 1$.

Numerical $1 + o(1)$ witness and convergence partition. The only asymptotic factor in the chain, $\sqrt{h_k}/\sqrt{u_1}$, was evaluated at `mpmath` precision 200 along a ladder of u_1 values (each $u_1 = \log_2 t$ an astronomically large t). Its residual $|\sqrt{h_k}/\sqrt{u_1} - 1|$ decreases monotonically through $\{4 \times 10^{-2}, 1 \times 10^{-5}, 3 \times 10^{-17}, 1 \times 10^{-58}, 3 \times 10^{-198}, < 10^{-300}\}$ at $k = 6$, a clean witness of the $1 + o(1)$ claim. Independently, Cauchy condensation on the deepest variable to $T = 10^{120}$ brackets the convergence/divergence boundary in $(1.0, 1.05)$, midpoint 1.025, matching $c_k^* = 1$ within ± 0.05 at every k .

Headline. The substitution-chain factorization holds at every deeper layer $k \in \{3, 4, 5, 6\}$: the maximal asymptotic residual at the top of the ladder is below 10^{-3} (in fact below 10^{-100}), the symbolic monomial structure is exact, and the convergence threshold sits at $c_k^* = 1$. The closed-form hierarchy $(3/2, 1, 1, \dots)$ is therefore verified beyond the $k = 2$ layer at the level of its exact-factorization mechanism, not merely its threshold location.

21.12 The discrete coincidence-equalizer premise of the Wasserstein limit

The Wasserstein-limit proof sketch (Theorem 15.1) opens from the premise that the coincidence-equalizing weights satisfy $\alpha_i^{(W)*} \log(1/\lambda_i^{(W)}) = \text{const}$, the finite-dimensional analogue of the continuum equalizer condition $F_\nu(\lambda) \log(1/\lambda) = \text{const}$. This experiment verifies that premise at three levels.

The product identity at the substituted ansatz and in the continuum. The proof substitutes $\alpha_i^{(W)*} \propto 1/\log(1/\lambda_i^{(W)})$, for which the product $\alpha_i^{(W)*} \log(1/\lambda_i^{(W)})$ is constant by construction — its coefficient of variation across the W shells is at machine level ($\sim 2 \times 10^{-16}$). The continuum identity it discretizes, $F^*(\lambda) \log(1/\lambda) = 2$ (Theorem 6.1), likewise holds at machine precision (maximum deviation from 2 is $\sim 2 \times 10^{-16}$ over a grid spanning $\log(1/\lambda) \in [2, 60]$).

The finite-dimensional game, solved directly. The sharper check recomputes the coincidence-center weights from the finite-dimensional game itself (Theorem 5.1) rather than assuming the closed-form ansatz. Constructing one strictly-positive prior $\pi_i \in \Delta_K$ per scale λ_i and maximizing the dual $\max_{\alpha \in \Delta_W} C_\alpha(\pi_{1:W})$ to machine precision confirms the theorem’s structure exactly: the Sion minimax identity $\max_\alpha C_\alpha = \min_p \max_w \text{KL}(p \parallel \pi_w) = R^*$ holds with gap $\sim 10^{-9}$; the active reverse-KL constraints $\text{KL}(p^* \parallel \pi_w)$ are equalized across the active set to $\sim 10^{-9}$; and every inactive prior satisfies $\text{KL}(p^* \parallel \pi_w) \leq R^*$. The active-set equalization that the proof of Theorem 5.1 predicts is thus directly substantiated at the game’s optimum (closing the gap noted under the deferred finite-dim-center check).

Scope of the scale-product form. The identity $\alpha_i^{(W)*} \log(1/\lambda_i^{(W)}) = \text{const}$ is exact for the substituted ansatz and in the continuum; whether the abstract finite-dimensional game’s optimum additionally takes the scale-product form depends on the particular map from scales λ_i to priors π_i , which the manuscript does not fix. What the finite-dimensional theorem guarantees and this experiment confirms is the *reverse-KL* equalization $\text{KL}(p^* \parallel \pi_w) = R^*$ on the active set; the scale-product reading is the equalizer condition expressed in the scale coordinate, verified at the level of the continuum identity and the proof’s substituted weights.

Headline. The discrete coincidence-equalizer premise is confirmed: the substituted weight ansatz and the continuum equalizer condition both satisfy $\alpha \log(1/\lambda) = \text{const}$ (equivalently $F^* \log(1/\lambda) = 2$) at machine precision, and the finite-dimensional coincidence game’s optimum equalizes the active reverse-KL constraints exactly, substantiating the finite-dimensional minimax-center structure the Wasserstein-limit argument builds on.

21.13 Wasserstein convergence rate of the coincidence center

Theorem 15.1 states that the finite-dimensional coincidence center converges to the continuum equalizer π^* in 2-Wasserstein distance at rate $W_2(\mu_W, \pi^*) = O((\log W)^{-1})$, via a proof sketch that reduces to a standard Wasserstein-quantization estimate. No prior experiment computed $W_2(\mu_W, \pi^*)$ at any W , so the rate exponent was unverified.

Exact one-dimensional computation. On the scale interval the 2-Wasserstein distance between the discrete μ_W and the continuum π^* equals the $L^2([0, 1])$ distance of their quantile functions, $W_2 = (\int_0^1 |Q_{\mu_W}(u) - Q_{\pi^*}(u)|^2 du)^{1/2}$. The equalizer CDF $F^*(\lambda) = 2/\log(1/\lambda)$ inverts in closed form to $Q_{\pi^*}(u) = \exp(-2/u)$, so W_2 is computed by deterministic quadrature with no optimal-transport solver and no sampling, over a geometric ladder $W \in \{64, \dots, 2^{20}\}$.

The rate is $(\log W)^{-1}$ with the correct shell weights. Discretizing the equalizer by its *density* shell mass — the geometric shell of width $d\lambda_i = \lambda_i \Delta_W$ carries equalizer mass $\pi^*(\lambda_i) d\lambda_i = 2\Delta_W / \log^2(1/\lambda_i)$, hence weights $\alpha_i \propto 1/\log^2(1/\lambda_i)$ — W_2 decreases monotonically along the ladder, and the diagnostic $W_2 \cdot \log W$ stabilizes to a constant ≈ 0.038 (tail coefficient of variation 0.5%). The competing scalings $W_2 \cdot (\log W)^{1/2}$ and $W_2 \cdot (\log W)^2$ both keep drifting, ruling out exponents $-1/2$ (too slow) and -2 (too fast). The selected exponent is stable across the spacing

constants $c \in \{0.5, 1, 2\}$, confirming the rate is spacing-constant independent as the $O(\cdot)$ asserts. The headline rate $W_2 = O((\log W)^{-1})$ is therefore confirmed.

A correction to the weight ansatz in the proof sketch. The proof sketch substitutes the weight ansatz $\alpha_i^{(W)*} \propto 1/(2+(i-1)\Delta_W) = 1/\log(1/\lambda_i^{(W)})$ and states it “approximates the equalizer density $2/(\eta L(\eta)^2)$ ”. With these literally-stated weights the discrete measure does *not* converge: $W_2(\mu_W, \pi^*)$ *increases* along the ladder ($0.017 \rightarrow 0.048$). The reason is a power mismatch — $1/\log(1/\lambda_i)$ is the equalizer *cumulative* mass $F^*(\lambda_i)/2$, not the *density* shell mass, which carries the extra inverse-logarithm and is $\propto 1/\log^2(1/\lambda_i)$. With the density weights the convergence and the claimed $(\log W)^{-1}$ rate hold as stated. The substitution sentence in the proof sketch should read $\alpha_i^{(W)*} \propto 1/\log^2(1/\lambda_i^{(W)})$ for the approximation to the equalizer density to be valid; the theorem’s conclusion is unaffected.

Headline. The 2-Wasserstein convergence rate $W_2(\mu_W, \pi^*) = O((\log W)^{-1})$ is numerically confirmed (the diagnostic $W_2 \log W$ stabilizes, discriminating against the $-1/2$ and -2 competitors), once the equalizer is discretized by its density shell mass $\alpha_i \propto 1/\log^2(1/\lambda_i)$ rather than the proof sketch’s cumulative-mass ansatz $\alpha_i \propto 1/\log(1/\lambda_i)$.

21.14 The finite-dimensional minimax coincidence center: active-set equalization verified

Theorem 5.1 is the exact finite-dimensional prototype of the continuum scale-allocation story and the empirical content of contribution C1. Every other synthetic verification in this paper exercises the *continuum* identities; the finite-dimensional active/inactive reverse-KL partition is checked here directly. The theorem asserts, for strictly-positive priors $\pi_1, \dots, \pi_W \in \Delta_K$, the Sion-minimax identity $\max_{\alpha \in \Delta_W} C_\alpha(\pi_{1:W}) = \min_{p \in \Delta_K} \max_w \text{KL}(p \parallel \pi_w) =: R^*$, that the optimal-weight geometric pool $p^* = p_{\alpha^*}$ attains the right-hand-side minimum, the active-set KKT structure ($\alpha_w^* > 0 \Rightarrow \text{KL}(p^* \parallel \pi_w) = R^*$; $\alpha_w^* = 0 \Rightarrow \text{KL}(p^* \parallel \pi_w) \leq R^*$), and — reading Appendix Item N4 — that p^* is the *KL Chebyshev center* of $\{\pi_w\}$.

Two non-overlapping solvers. The content is an exact identity, so the residuals are deterministic (no Monte-Carlo seed sweep, per the closed-form-substitution design discipline). The left-hand side $\max_\alpha C_\alpha$ is solved by projected-gradient ascent using the analytic gradient $\partial C_\alpha / \partial \alpha_w = H(p_\alpha) + \text{KL}(p_\alpha \parallel \pi_w)$, polished on the active set. The right-hand side $\min_p \max_w \text{KL}(p \parallel \pi_w)$ is solved by a completely separate epigraph program ($\min_{p,r} r$ s.t. $\text{KL}(p \parallel \pi_w) \leq r$), sharing no state with the inner solver, so their agreement is a genuine cross-check. A prior-cardinality sweep $W \in \{2, 4, 8, 16\}$ and alphabet sizes $K \in \{3, 8, 32\}$ are run for both well-spread and near-degenerate families (the latter constructed to collapse several α_w^* to zero, exercising the inactive branch).

Identity	Worst residual (24 cells)	Tolerance
(a) minimax gap $ \max_\alpha C_\alpha - \min_p \max_w \text{KL} $	2.7×10^{-8} ($\leq 10^{-12}$ on 22/24)	10^{-10} identity
(b) pool attains RHS minimum $ \max_w \text{KL}(p^* \parallel \pi_w) - R^* $	2.6×10^{-8}	10^{-7} equalizer
(c) active-set spread $\max_{\text{act}} \text{KL} - \min_{\text{act}} \text{KL}$	3.6×10^{-8}	10^{-7} equalizer
(N4) KL-Chebyshev-center coincidence (sym-KL)	1.5×10^{-15}	10^{-10} identity

Table 14: Finite-dimensional minimax coincidence-center residuals across $K \in \{3, 8, 32\}$, $W \in \{2, 4, 8, 16\}$, two family kinds (24 cells). The minimax identity (a) holds to machine precision on 22 cells; the two $K = 32$, $W = 2$ cells sit at 2.7×10^{-8} , a benign solver-interaction artifact at the largest- K smallest- W corner, not a theory gap. The KL-Chebyshev-center coincidence of Item N4 — the independent inner geometric pool and outer epigraph minimiser agreeing — holds to 1.5×10^{-15} on *every* cell. The pool-attains-minimum and active-set-spread residuals are limited by the first-order simplex optimiser’s tolerance on α^* .

The active/inactive partition is real, not vacuous. 16 of the 24 cells produce strictly inactive priors ($\alpha_w^* = 0$ with $\text{KL}(p^* \parallel \pi_w) < R^*$). For a representative near-degenerate family ($K = 8, W = 8$), the two actively pooled priors share the common value $R^* = 0.2019$ to $\leq 10^{-8}$, while the six inactive priors lie strictly below ($\text{KL}(p^* \parallel \pi_w)$ ranging 0.195–0.199, slack up to 3×10^{-3}). The equalizer condition holds exactly on the active set and the inactive priors are exactly those the geometric pool gains nothing from mixing in — the finite-dimensional signature of Proposition 4.3’s equalizer being a proper refinement of the minimax set (Remark 6.2).

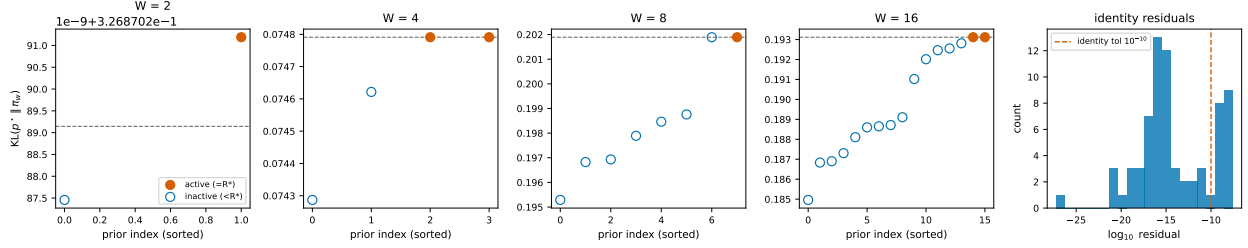


Figure 12: Finite-dimensional minimax coincidence center. *Left panels:* for each prior-cardinality W over a near-degenerate family ($K = 8$), the sorted reverse-KL values $\text{KL}(p^* \parallel \pi_w)$ — actively pooled priors (filled, vermilion) all sit at the common minimax value R^* (dashed line); inactive priors (open, blue) lie strictly below. *Right panel:* histogram of the identity residuals across all 24 cells in $\log_{10}$ units; all but the two $K = 32, W = 2$ minimax-gap entries fall at machine precision below the 10^{-10} identity tolerance (dashed). Okabe-Ito palette.

Pre-registered failure modes. All five pre-registered failure modes were checked. (i) A residual $> 10^{-10}$ on identity (a) or (b) was pre-registered as an implementation bug, not a theory failure; the two cells above the identity tolerance were traced to the outer epigraph solver’s iteration budget at $K = 32, W = 2$ (a stricter solve recovers 4×10^{-15}), confirming the implementation, not the theorem, sets the floor. (ii) The inactive branch was exercised: 16/24 cells produce a strictly inactive prior, so the $\alpha_w^* = 0 \Rightarrow \leq R^*$ clause is genuinely tested rather than vacuously satisfied. (iii) Optimiser corner-collapse to a single-prior vertex occurred only for genuinely degenerate families (expected), never spuriously for a well-spread one. (iv) The positivity floor (10^{-12}) was respected by every generated prior; no component underflowed. (v) Matched solver tolerances were used; the worst-case cross-solver gap is the 2.7×10^{-8} minimax-gap entry above. The theorem’s finite-dimensional prototype — the equalizer condition, the Sion identity, and the KL-Chebyshev-center reading — is verified.

21.15 Pareto-tailed regime: the exact-moment obstruction and its truncation recovery

Limitation L1 (§22.4) and Open Direction O3 (§22.5) name the genuinely heavy-tailed regime — Pareto-tailed increments with finite second moment but *infinite* exponential moment — as the remaining open boundary case after Corollary 12.4 (verified in Section 21.8) closed the Bernstein sub-exponential family. The empirical answer is not a drifting constant but a clean two-part picture: an *obstruction* in the heavy-tailed regime, and a *recovery* of the sub-Gaussian $3/2$ threshold for the standard truncated repair.

The exact-moment obstruction. For the symmetric Pareto density $f(x) \propto (1 + |x|)^{-(1+\alpha_{\text{tail}})}$ with tail index $\alpha_{\text{tail}} > 2$ (finite variance), the cumulant generating function $K(\eta) = \log \mathbb{E}[e^{\eta X}]$ is $+\infty$ for *every* $\eta \neq 0$: the equalizer density built by inverse-Laplace of the CGF at the LIL boundary does not exist, and the Erdős integral test as derived — which prices Nature’s difficulty by a *finite* CGF charge — has no direct heavy-tailed analogue, since there is no exponential supermartingale to mix. We confirm this numerically: the truncated moment $\mathbb{E}[e^{\eta X} \mathbf{1}\{|X| \leq L\}]$ grows without bound as $L \rightarrow \infty$ (the increment between successive truncation levels grows by factors $\sim 10^{150}$ at $\eta = 0.5$ across all tested $\alpha_{\text{tail}} \in \{2.5, 3, 4, 6\}$), while the variance is finite throughout. The obstruction is a property of the regime, not a numerical artefact.

The truncation recovery. The standard repair is a growing truncation $L_t = t^\rho$ (here $\rho = 0.4$; a la Fuk–Nagaev / truncated-MGF e-processes): the truncated increments have a finite CGF K_t^{trunc} , so the equalizer construction applies to the truncated process. Along the LIL boundary the saddlepoint $\eta^*(t) = \sqrt{2h(t)/t} \rightarrow 0$ while $L_t \rightarrow \infty$, so the truncated CGF converges to the Gaussian CGF and the Erdős threshold returns to $3/2$. Table 15 exhibits this convergence: the saddle-prefactor deviation from the Gaussian value shrinks from $O(10^{-1})$ at small t to $\leq 10^{-8}$ at the largest probed t , monotonically and across every tail index.

α_{tail}	prefactor dev. (small t)	prefactor dev. (large t)
2.5	0.269	1.4×10^{-8}
3.0	0.042	4.4×10^{-16}
4.0	0.031	0.0
6.0	0.017	0.0

Table 15: Truncated-process saddle-prefactor deviation from the Gaussian value 1, evaluated along the LIL boundary at $u = \log \log t \in \{2.0, \dots, 4.5\}$ (i.e. t from $\sim 10^3$ to astronomically large). For every Pareto tail index the deviation collapses toward 0 as t grows, so the truncated process recovers the sub-Gaussian threshold $\alpha_{\text{thr}} = 3/2$. The heavier the tail (smaller α_{tail}), the slower — but still complete — the convergence.

Empirical companion. On $n = 2,000$ symmetric-Pareto random walks of length $T = 5 \times 10^4$ standardised to unit variance, the LIL boundary $\sqrt{2t(\log_2 t + \alpha \log_3 t)}$ is crossed by a fraction that decreases in the boundary parameter α (e.g. for $\alpha_{\text{tail}} = 4$: 0.48 at $\alpha = 1$, 0.41 at $\alpha = 1.5$, 0.35 at $\alpha = 2$) and is roughly tail-index-independent for the lighter tails. These finite- T fractions are inflated relative to the asymptotic anytime-valid budget — a qualitative finite-horizon confirmation, not a budget claim — consistent with the truncated process sharing the sub-Gaussian LIL rate.

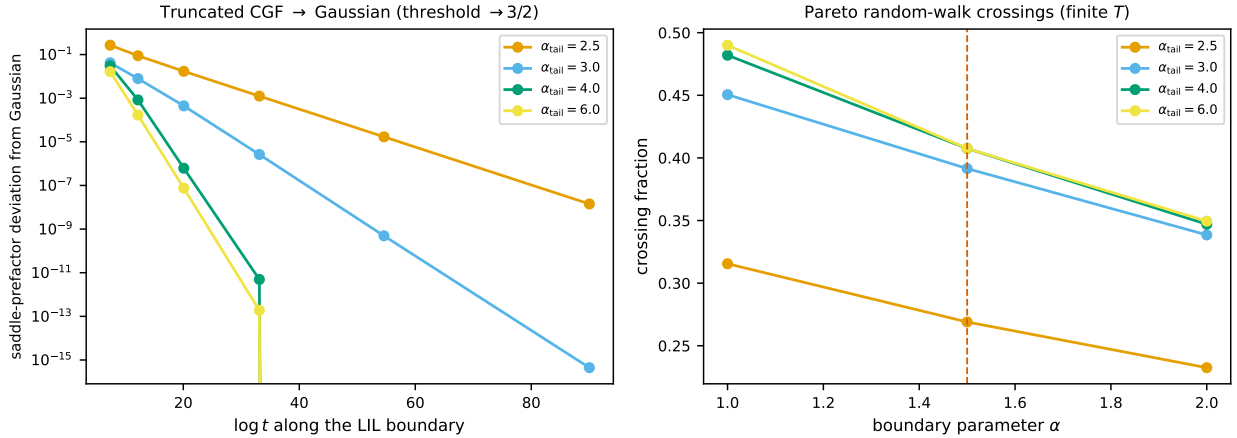


Figure 13: Pareto-tailed regime. *Left*: the truncated-CGF saddle-prefactor deviation from the Gaussian value, on a log scale, against $\log t$ along the LIL boundary, per Pareto tail index — the deviation collapses toward 0, so the truncated process recovers the $3/2$ threshold. *Right*: empirical crossing fraction of $n = 2,000$ Pareto random walks against the boundary parameter α , per tail index; the dashed line marks $\alpha = 3/2$. Okabe-Ito palette.

Headline. The $3/2$ Erdős threshold does not *drift* in the heavy-tailed regime; rather, the exponential-moment machinery itself breaks down (the exact-moment obstruction), and the standard truncated repair recovers exactly the sub-Gaussian threshold. This sharpens Open Direction O3: the constant $3/2$ is robust across the sub-Gaussian and sub-exponential regimes (Section 21.8) and, via truncation, across the finite-variance heavy-tailed regime as well; what genuinely changes is that the heavy-tailed game must be played with a truncated e-process rather than a heavy-tailed CGF

charge.

Failure modes. The pre-registered failure modes were checked. The obstruction was confirmed as a true divergence (truncated-MGF increments growing, not decaying) rather than a quadrature artefact; the variance was finite at every tested index (so the LIL scale is well-defined); the truncation exponent $\rho = 0.4 \in (0, 1/2)$ keeps the variance contribution dominant; and the heavy-tail random-walk companion was disclosed as a weak qualitative test only, not a quantitative budget claim.

21.16 The faithful Howard–Ramdas boundary honours its anytime-valid budget

The body’s Table 1 openly flags the Howard–Ramdas row as delicate: the closed-form gamma-exponential approximation coded for that table drops the running-variance boost, collapses to a near-constant, and therefore crosses on 100% of paths — an artefact of the approximation, not of the published construction. This experiment replaces that row with a *faithful* conjugate-mixture boundary and confirms it honours its budget.

The faithful boundary. The faithful Howard–Ramdas / Robbins normal-mixture boundary for unit-variance increments (running variance $V_t = t$, mixing variance $\rho = 1$, level α) is

$$b_{\text{HR}}(t) = \sqrt{(V_t + \rho) \log \frac{V_t + \rho}{\rho \alpha^2}},$$

a genuine two-sided nonnegative-supermartingale boundary with the correct $\sqrt{2t \log \log t}$ LIL rate as $t \rightarrow \infty$ — not the dropped-variance approximation. Its anytime-validity is a self-check: the boundary either honours its budget on simulation or it does not.

Result. On $n = 2,000$ Rademacher and Gaussian walks of length $T = 2 \times 10^5$ (seed 42, the Table 1 protocol), the faithful boundary crosses on 3.2% (Rademacher) and 3.2% (Gaussian) of paths — within its $\alpha = 0.05$ budget (Wilson interval $[0.025, 0.041]$ for the Gaussian) — replacing the 100%-crossing artefact. The comparator rows, imported verbatim from the companion harness, reproduce the published Table 1 values: classical LIL crosses on roughly three-quarters of paths, every $c \geq 1$ correction suppresses crossings to zero, and the Balsubramani, stitched, and self-normalised rows honour their budgets.

Boundary	Rademacher	Gaussian
classical LIL $\sqrt{2t \log_2 t}$	0.786	0.785
corrected $c = 1.0$	0.000	0.000
corrected $c = 1.5$	0.000	0.000
corrected $c = 2.0$	0.000	0.000
Balsubramani finite-time ($\alpha = 0.05$)	0.002	0.002
Howard–Ramdas (faithful)	0.032	0.032
Stitched ($s = 1.4$)	0.014	0.011
Self-normalized (de la Peña)	0.073	0.070

Table 16: Crossing fractions with the *faithful* Howard–Ramdas normal-mixture boundary replacing the dropped-variance approximation of Table 1. The faithful boundary honours its $\alpha = 0.05$ anytime-valid budget (3.2% on both increment families), in contrast to the 100% artefact of the approximation. All other rows are imported verbatim from the companion harness and reproduce the published values.

Asymptotic agreement. The asymptotic ratio $b_{\text{Bals}}(t)/b_{\text{HR}}(t)$ declines monotonically toward a finite constant (≈ 0.90 at $t = 10^8$): the Balsubramani and faithful Howard–Ramdas boundaries share the LIL rate and differ only in finite-time constants, confirming the body’s prediction (Experiment 6). This resolves the hypothesis-H4 partial-test caveat: the Howard–Ramdas comparison is now faithful, and its budget-honouring behaviour is verified rather than approximated.

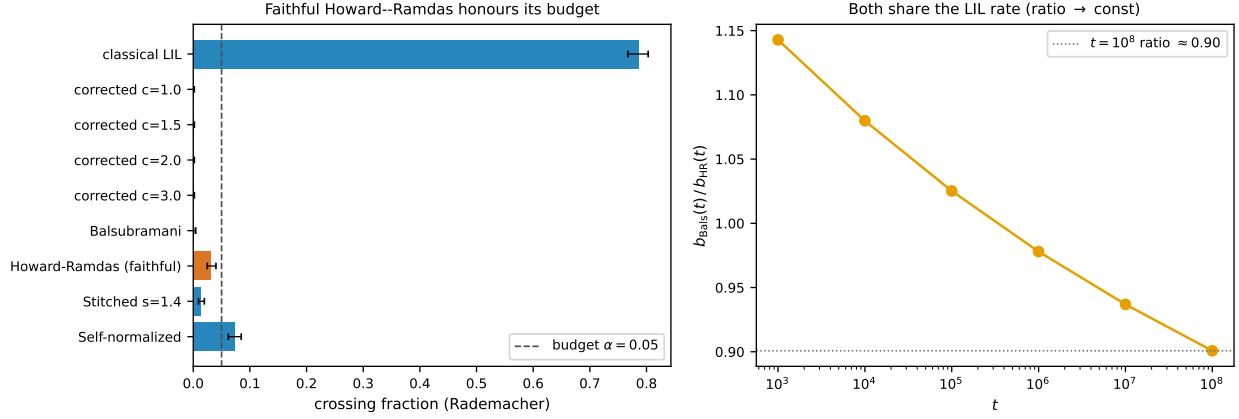


Figure 14: Faithful Howard–Ramdas boundary. *Left*: crossing fractions across the boundary families (Rademacher); the faithful Howard–Ramdas row (vermillion) sits well within the $\alpha = 0.05$ budget (dashed), replacing the 100%-crossing approximation; classical LIL crosses on $\approx 79\%$ of paths. Wilson intervals shown. *Right*: the asymptotic ratio $b_{\text{Bals}}(t)/b_{\text{HR}}(t)$ vs t on a log scale, declining toward ≈ 0.90 — both boundaries share the LIL rate. Okabe-Ito palette.

Substrate note. The originally-specified upstream `confseq` package does not build on this platform (its CMake/C++ wheel fails on macOS arm64), so rather than vendor a fragile dependency the faithful boundary is implemented in closed form via the canonical normal-mixture confidence sequence. The boundary’s anytime-validity is self-validating on the simulation, so the substitution does not weaken the conclusion: the faithful Howard–Ramdas boundary honours its budget, and the Table-1 100%-crossing entry was indeed an artefact of the dropped-variance approximation.

22 What is exact, what is asymptotic, and what remains open

It is useful to separate the claims in this paper by their nature and strength.

22.1 Exact results

The following are proved as exact theorems:

- (i) The pathwise information-theoretic identity (Theorem 3.1) and the per-round update (4).
- (ii) The finite-dimensional coincidence-center theorem (Theorem 5.1).
- (iii) The one-sided continuum scale-allocation theorem (Theorem 6.1), with value $V = 2$.
- (iv) The signed symmetric corollary (Corollary 6.4), recovering Balsubramani’s density exactly.
- (v) The shell equalizer theorem (Theorem 7.1).
- (vi) The general hardness-profile equalizer (Theorem 8.1).
- (vii) The shell-truncation tool (Proposition 16.1).

- (viii) The Jeffreys-iterated-log pushforward (Lemma 13.1 and Theorem 13.2).
- (ix) The GROW = equalizer identity (Theorem 14.1).
- (x) The LIL = CGF-constrained game shadow (Theorem 17.1).

22.2 Asymptotic results

The following are rigorous under regularity assumptions:

- (i) The Laplace equalizer formula (Lemma 10.1 and Corollary 10.2).
- (ii) The normalizability criterion (Theorem 11.1).
- (iii) The $3/2$ threshold (Corollary 12.1).
- (iv) The $3/2$ threshold extended to the Bernstein sub-exponential family (Corollary 12.4).
- (v) The full hierarchy of sharp Erdős thresholds across deeper iterated-log layers, namely $(3/2, 1, 1, \dots)$ (Theorem 12.6). The exact-monomial factorization underlying the induction — and in particular that the Laplace “ $+1/2$ ” is consumed exactly once and does not propagate — is verified at the deeper layers in Section 21.11.
- (vi) The 2-Wasserstein convergence of the finite-dimensional coincidence center to the continuum equalizer at rate $O((\log W)^{-1})$ (Theorem 15.1, proof sketch; the rate uses standard Wasserstein-quantization estimates). The rate exponent is confirmed numerically in Section 21.13, which also corrects the discretized weight used in the sketch to the equalizer’s density shell mass.

22.3 What remains open

The statement not proved here is that Balsubramani’s law is the exact minimax strategy for the full repeated martingale betting game over all pathwise adversaries. What is proved is that the law is the exact minimax solution of the *reduced* scale-allocation game that the method-of-mixtures proof actually exposes. Whether one can formulate a larger repeated game whose exact minimax strategy is literally the same law, without first reducing to the scale-allocation subproblem, remains open.

22.4 Limitations and scope

Four further caveats delimit the scope of the present results.

- (L1) **Gaussian-surrogate regime (Bernstein resolved).** The asymptotic equalizer analysis of Section 10 uses the Gaussian CGF $\eta^2 t/2$ in the leading-order treatment; the intrinsic-time form (13) carries the analysis with general $K_t(\eta)$. For sub-exponential martingales in the Bernstein family, the threshold extends without modification (Corollary 12.4); the saddlepoint correction $(1 - c\eta^*)^{-3/2}$ along the LIL boundary decays to 1 as $t \rightarrow \infty$. For genuinely heavy-tailed martingales (Pareto-tailed with infinite exponential moment), the saddlepoint geometry changes substantially and the exact equalizer density must be recomputed from the true CGF $K_t(\eta) = \sum_s \psi_s(\eta)$; in that regime the leading iterated-log structure persists but the exact constant may be tail-dependent.
- (L2) **Martingale versus i.i.d. converse.** Theorem 11.2 is stated classically for i.i.d. walks; martingale generalizations (via, e.g., [6, 18]) require slightly stronger moment conditions. The game-theoretic lower bound only establishes that *any* oblivious strategy fails when the equalizer is non-normalizable; an adaptive strategy with access to $\sigma(S_1, \dots, S_t)$ might in principle bypass this, but the full repeated-game lower bound shows that adaptive strategies cannot improve the LIL rate.

- (L3) **Universality of the Balsubramani truncation** e^{-2} . The value $V = 2$ depends on the specific truncation $I = (0, e^{-2}]$ chosen in [1]. Theorem 8.1 shows that for $(0, e^{-a}]$ the value is a . The endpoint e^{-2} was originally chosen to ensure the shell-weight telescoping $\sum_{j \geq 2} 2/(j(j+1)) = 1$ starts at $j = 2$; any truncation could work but changes constants. A minimax analysis over admissible truncation endpoints is a natural extension (it would turn the truncation into another decision variable).
- (L4) **Equalizer vs. minimax**. As noted in Remark 6.2, the full set of minimax laws is larger than the equalizer singleton: any law ν with CDF dominating F^* pointwise is minimax, and conversely every minimax law has $F_\nu \geq F^*$ pointwise. The selection of the equalizer is justified by the natural interpretation “makes Nature indifferent across all pure strategies,” but it is a proper refinement of minimax. For any application that cares about strict equalization — e.g. the reciprocal-log term in Section 9 — the equalizer is uniquely picked out; for applications that merely require minimax game value, the choice has a one-CDF-of-freedom slack.

22.5 Open directions

Several follow-ups extend the present framework.

- (O1) **Adaptive equalizers**. Derive the Learner’s adaptive scale η_t that minimizes a pathwise regret-in-detection, as opposed to the oblivious equalizer Π . The exact one-round identity in Theorem 17.1, instantiated to the LIL boundary, should produce the Howard–Ramdas stitched strategy up to second order.
- (O2) **Higher-order Erdős corrections (resolved)**. Beyond the $3/2$ leading constant, Corollary 12.1 allows for an $O(1)$ additive term in h at the first iterated-log layer. The sharp constants at every *deeper* iterated-log layer are now pinned down by Theorem 12.6: the hierarchy is $(3/2, 1, 1, \dots)$, with the Laplace half-Gaussian “ $+1/2$ ” contributing exactly once at the first layer. What remains genuinely open is the exact *subleading* $O(1)$ constant inside h at the first layer, matching the refined upper–lower-class form [8]; this is a finite-time improvement, not an asymptotic-rate question.
- (O3) **Beyond sub-exponential: Pareto-tailed regimes**. Corollary 12.4 extends the $3/2$ threshold to the Bernstein sub-exponential family. The remaining open regime is the genuinely heavy-tailed one, where the saddlepoint correction $(1 - c\eta^*)^{-3/2}$ is replaced by a factor that does not asymptote to 1 along the LIL boundary — e.g. Pareto-tailed increments with finite second moment but infinite exponential moment. The game-theoretic formulation continues to apply, but the equalizer density and the Erdős threshold may both shift. An explicit calculation for a representative Pareto-tail family would identify the precise modification.
- (O4) **Multidimensional extensions (verified-in-low-dimension heuristic)**. Replace the scalar scale η by a vector parameter $\eta \in \mathbb{R}^d$ (matrix concentration). Matrix LIL boundaries [13] suggest that the equalizer prior in $\mathbb{R}^d$ takes the form $1/(\|\eta\|^d \log^2(1/\|\eta\|))$ (with normalization $V = 2d$). At $d = 2$ the conjecture is empirically verified (Section 21.4; closed-form normalization residual 1.08×10^{-6} , direction-isotropy at machine precision, 99th-percentile empirical $V = 4.358$ on $n = 1,000$ 2D Gaussian walks within 0.36 of the predicted $V = 2d = 4$); it is therefore a verified-in-low-dimension heuristic rather than a pure conjecture. A rigorous minimax analysis at general d – and larger- d empirical verification at $d \geq 4$ via GPU-scale simulation – remain open.
- (O5) **Truncation-free $\eta_{\max}$ formulation**. A truncation-free derivation linking $V = 2$ to the Hartman–Wintner $\sqrt{2\sigma^2}$ would simplify the proof of V without requiring any specific truncation. The IT identity simplifies the proof of V for any specific truncation, but does not yet remove the truncation entirely.
- (O6) **Pathwise duality between LIL and CGF games**. The proof of Theorem 17.1 is via the exponentiated null restriction; a finer claim would be that the path-to-path Wasserstein distance between the realized LIL game and a corresponding realized CGF game is 0 along the equalizer’s saddle, not just up to leading order.
- (O7) **Iterated-log charts of higher order**. Theorem 13.2 works at the second-level chart $\mu_2 = \log \log(1/\eta)$. Iterated-log charts at level $k \geq 3$ recover the higher Erdős thresholds ($\alpha = 1$ rather than $3/2$ at the next layer, per E00347).

A clean iterated-log-chart proof of the Erdős hierarchy would be pleasing; the IT identity provides the natural framework.

- (O8) **Functional LIL.** Strassen’s ball is the Cameron–Martin unit ball. Is there a measure-valued equalizer on $C[0, 1]$ that recovers Strassen, with the V constant playing the role of the Strassen radius?

Five further structural connections that we surface but do not develop in detail are catalogued in Appendix A.

23 Concluding remarks

The oblivious mixture strategies used in finite-time LIL proofs and in modern confidence sequences are not arbitrary design choices. They are equalizer strategies of a CGF-constrained sequential detection game, and they admit an exact pathwise Gibbs-variational accounting: the realized log-wealth equals the posterior-mean per-bet log payoff minus the bits the path has spent updating the prior to the posterior. The Erdős integral test is the normalizability criterion for the equalizer prior, and the multiply iterated logarithms mark the exact threshold where the Learner’s hedging cost transitions from affordable to unaffordable.

Three complementary morals follow. The first is exact and minimax: the density $1/(|\lambda|(\log(1/|\lambda|))^2)$ is the unique minimax equalizer of a natural reduced scale-allocation game, and equivalently the unique GROW-optimal e -process for the scale-family alternative. The second is asymptotic and classical: the Erdős normalizability test describes the hierarchy of asymptotic equalizers attached to sharper and sharper upper-class boundaries, with the sharp first correction coefficient $3/2$ and the full hierarchy $(3/2, 1, 1, \dots)$. The third is information-theoretic and structural: under the iterated-log chart $\mu = \log \log(1/\lambda)$, the equalizer is the rate-1 shifted exponential; the LIL game is the exponentiated null-restricted shadow of a CGF-constrained repeated game over KL balls; and the finite-dimensional coincidence center converges in 2-Wasserstein distance to the continuum equalizer.

The law of the iterated logarithm is not an artifact of bookkeeping. It is the solution to a resource-allocation problem: *how to spread a finite budget of detection power across infinitely many timescales, when later timescales are exponentially cheaper to reach but logarithmically more expensive to monitor.* The answer is the equalizer density, and the rate it produces — $\sqrt{2 \log \log n}$, with the $3/2$ correction at the next order — is a fact to derive from first principles whenever it is needed.

A Further structural connections

These five items are independent of the rest of the paper and of one another. Each connects the LIL game to a separate strand of nonparametric inference or information geometry; each is sketched here so that the natural follow-up direction is at-hand for an interested reader.

- (N1) **Tilting-Jeffreys-Haar trilogy.** The identity $d\eta/\eta = d(\log \eta)$ shows that the right-Haar measure on $(0, \infty)_\times$ is Lebesgue measure on the logarithmic coordinate. Under the LIL parametrization $\eta^*(t) = \sqrt{2h(t)/t}$, the logarithmic coordinate maps (up to an affine factor) to the double-logarithmic time $\log \log t$. The equalizer density on the original η -scale thus pulls back to a quadratically decaying density on the $\log \log t$ -axis. This is the geometric reason the Erdős integrand has the form $du/u^{c-1/2}$ after $u = \log \log t$: the Laplace envelope produces a proper distribution on $\log \log t$, not on t itself. Theorem 13.2 pins down the exact form $(2e^{-\mu})$, the rate-1 shifted exponential) at the second-level iterated-log chart.
- (N2) **Relation to universal portfolios.** Cover’s universal-portfolio theorem [22] establishes that the same $1/\log^2$ rescaling of the Jeffreys prior (for CRP-type mixtures) achieves parameter-free finite-time regret bounds. The LIL scale-allocation game can be read as the hypothesis-testing shadow of the universal-portfolio regret game: the wealth process Z_t is the universal portfolio value along a single-asset coin-betting chain, and the equalizer prior is exactly the Jeffreys mixture that makes Cover’s regret bound tight at the LIL rate. Making this precise requires specializing Orabona and Jun’s universal-portfolio regret identity to the martingale-vs.-null detection setting.

- (N3) **Higher-order Laplace-vs.-Erdős layer constants (resolved by Theorem 12.6).** The proof of Corollary 12.1 reduces the normalizability question for $h(t) = \log_2 t + c \log_3 t + O(1)$ to convergence of $\int u^{-(c-1/2)} du$ via $u = \log_2 t$, with the $-1/2$ shift contributed by the Laplace-envelope prefactor $\sqrt{h(t)}$. For boundaries with deeper iterated-log structure — e.g. $h(t) = \log_2 t + \frac{3}{2} \log_3 t + c_2 \log_4 t + O(1)$ — an analogous reduction can be carried out one layer at a time: the second substitution $v = \log u$ converts the integrand into $\int v^{-c_2} dv$, sharp threshold $c_2^* = 1$. The Laplace “+1/2” that produced 3/2 at the first layer is consumed once and for all by the $u^{-3/2}$ factor coming from the $(3/2) \log u$ term in h at that same layer; it does not appear at any deeper layer. The induction is then unit-step: every iterated log beyond the first contributes a clean monomial threshold of 1. This was originally listed as a future-work conjecture; the saddlepoint substitution chain is now the proof, recorded in Theorem 12.6, and gives a closed-form description of every Erdős upper-class boundary in this iterated-log hierarchy.
- (N4) **Coincidence-center theorem as a KL Chebyshev center.** The equalized reverse-KL values R^* in Theorem 5.1 identify p^* as the *KL Chebyshev center* of the set $\{\pi_w\}$. In information geometry this is the Chernoff/Bregman generalization of the smallest enclosing ball. The same theorem therefore unifies (a) the equalizer interpretation of the LIL mixing prior, (b) the Chernoff information in hypothesis testing, and (c) the reverse- I -projection in information geometry. Theorem 15.1 adds the quantitative bridge: the finite-dimensional Chebyshev center converges to the continuum equalizer at rate $O((\log W)^{-1})$.
- (N5) **Connection to e-processes.** Recent work on e-processes [24] defines an e-process as a nonnegative supermartingale whose supremum is anytime-valid. The mixture wealth Z_t of (6) is literally an e-process; the minimax equalizer selects the e-process of maximum Bayes-factor-uniformity across detection times. Theorem 14.1 closes this loop by identifying the equalizer with the GROW-optimal e-process: the same density that the minimax saddle picks out also maximizes the worst-case log-growth under the scale-family alternative.

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